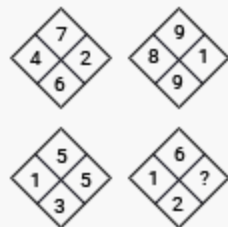


Missing Number

Introduction

Questions, which ask to insert the missing number, come under this category. In most cases one of the following will be given.

- a. A diagram or a set of diagrams



- b. Or a matrix

13	54	?
7	45	32
27	144	68

- c. Or any random figure e.g. star

Each of these has some characters. These characters can be numbers or letters or a combination of number and letters that follow a particular pattern. We need to crack the reasoning or logic behind this pattern so as to find the missing term.

Most common patterns

Different types of patterns could be there, some of the patterns are as following

1. It could be sum of two numbers divided by a constant
2. It could be average of numbers
3. It could be in the form of alphabets, where alphabets are increased by constant or increased by square of numbers or increased by prime numbers
4. It could be the difference of product of two diametrically opposite numbers.
5. It could be the difference of sum of adjacent numbers.
6. The difference of the numbers could be in the pattern $13 \pm 1, 23 \pm 1$, and so on.

7. The difference of the numbers could be in the pattern 12 ± 1 , 22 ± 1 , and so on.
8. Numbers could be 12, 22, 32 and so on or 13, 23, 33... and so on.
9. The difference could be prime numbers or difference could be square of prime numbers.
10. The difference could be in the form of $N^2 \pm N$ or $N^3 \pm N$.
11. The difference could be in the form of $\times N + N$ or $\times 2 + 1$, $\times 2 + 2$, $\times 2 + 3$... and so on.
12. The difference could be in the form of $\times 2 \pm 1$ alternatively.
13. Numbers could be $\times 1$, $\times 2$, $\times 3$ and so on.

Different types of Number Series:

The most common patterns followed by number series are:

■ Series consisting of Perfect Squares:

A series based on Perfect squares is most of the times based on the perfect squares of the numbers in a specific order & generally one of the numbers is missing in this type of series.

Example: 324, 361, 400, 441, ?

Sol: $324 = 18^2$, $361 = 19^2$, $400 = 20^2$, $441 = 21^2$, $484 = 22^2$

■ Perfect Cube Series:

It is based on the cubes of numbers in a particular order and one of the numbers is missing in the series.

Example: 512, 729, 1000, ?

Sol: 8^3 , 9^3 , 10^3 , 11^3

■ Geometric Series:

It is based on either descending or ascending order of numbers and each successive number is obtained by dividing or multiplying the previous number by a specific number.

Example: 4, 36, 324, 2916?

Sol: $4 \times 9 = 36$, $36 \times 9 = 324$, $324 \times 9 = 2916$, $2916 \times 9 = 26244$.

■ Arithmetic Series:

It consists of a series in which the next term is obtained by adding/subtracting a constant number to its previous term. Example: 4, 9, 14, 19, 24, 29, 34 in which the number to be added to get the new number is 5. Now, we get an arithmetic sequence 2, 3, 4, 5.

■ Two-stage Type Series:

In a two step Arithmetic series, the differences of consecutive numbers themselves form an arithmetic series.

Example: 1, 3, 6, 10, 15.....

Sol: $3 - 1 = 2$, $6 - 3 = 3$, $10 - 6 = 4$, $15 - 10 = 5$

Now, we get an arithmetic sequence 2, 3, 4, 5

■ Mixed Series:

This particular type of series may have more than one pattern arranged in a single series or it may have been created according to any of the unorthodox rules.

Example: 10, 22, 46, 94, 190, ?

Sol:

$$10 \times 2 = 20 + 2 = 22,$$

$$22 \times 2 = 44 + 2 = 46,$$

$$46 \times 2 = 92 + 2 = 94,$$

$$94 \times 2 = 188 + 2 = 190,$$

$$190 \times 2 = 380 + 2 = 382.$$

So the missing number is 382.

■ Arithmetico –Geometric Series :

As the name suggests, Arithmetico –Geometric series is formed by a peculiar combination of Arithmetic and Geometric series. An important property of Arithmetico-Geometric series is that the differences of consecutive terms are in Geometric Sequence.

Example: 1, 4, 8, 11, 22, 25, ?

Sol : Series Type +3 , X2 (i.e Arithmetic and Geometric Mixing)

$$1 + 3 = 4, 4 \times 2 = 8, 8 + 3 = 11, 11 \times 2 = 22, 22 + 3 = 25, 25 \times 2 = 50$$

Geometrico - Arithmetic Series is the reverse of Arithmetico - Geometric Series. The differences of suggestive terms are in Arithmetic Series.

Example: 1, 2, 6, 36, 44, 440, ?

Sol : Series Type - X 2, + 4, X 6, +8 , X 10

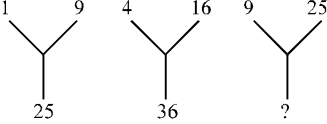
$$1 \times 2 = 2, 2 + 4 = 6, 6 \times 6 = 36, 36 + 8 = 44, 44 \times 10 = 440, 440 + 12 = 452$$

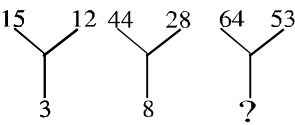


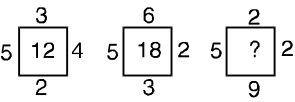
FINDING THE MISSING NUMBER

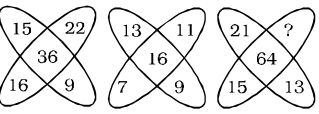
TYPE-I

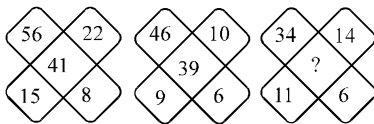
Directions : Find the missing number from the given responses in each of the following questions.

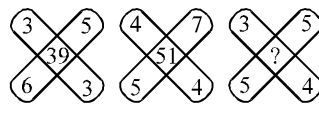
1. 
 (1) 47 (2) 49
 (3) 50 (4) 57
 (SSC Combined Graduate Level Prelim Exam. 04.07.1999 (First Sitting))

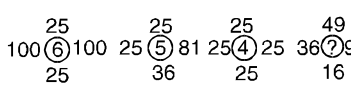
2. 
 (1) 30 (2) 13
 (3) 70 (4) 118
 (SSC Combined Graduate Level Prelim Exam. 04.07.1999 (Second Sitting))

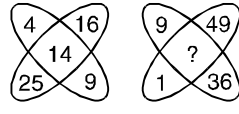
3. 
 (1) 15 (2) 18
 (3) 17 (4) 16
 (SSC Combined Graduate Level Prelim Exam. 27.02.2000 (First Sitting))

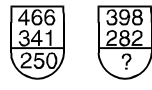
4. 
 (1) 23 (2) 19
 (3) 20 (4) 22
 (SSC Combined Graduate Level Prelim Exam. 24.02.2002 (First Sitting))

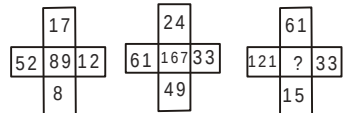
5. 
 (1) 25 (2) 52
 (3) 12 (4) 48
 (SSC Combined Graduate Level Prelim Exam. 24.02.2002 (Second Sitting))

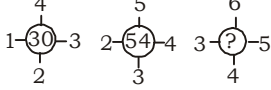
6. 
 (1) 35 (2) 37
 (3) 45 (4) 47
 (SSC Combined Graduate Level Prelim Exam. 24.02.2002 (Middle Zone))

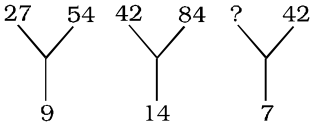
7. 
 (1) 3 (2) 2
 (3) 5 (4) 4
 (SSC Combined Graduate Level Prelim Exam. 27.02.2000 (Second Sitting))

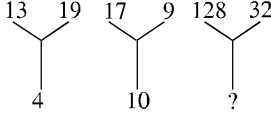
8. 
 (1) 25 (2) 15
 (3) 17 (4) 18
 (SSC Combined Graduate Level Prelim Exam. 27.02.2000 (Second Sitting))

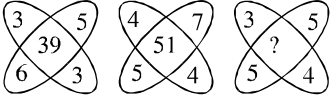
9. 
 (1) 298 (2) 232
 (3) 350 (4) 268
 (SSC Combined Graduate Level Prelim Exam. 27.02.2000 (Second Sitting))

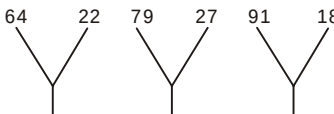
10. 
 (1) 240 (2) 230
 (3) 232 (4) 251
 (SSC Combined Matric Level (PRE) Exam. 13.05.2001 (1st Sitting))

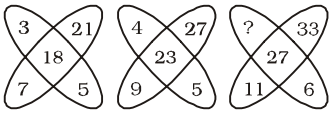
11. 
 (1) 68 (2) 65
 (3) 86 (4) 52
 (SSC Combined Matric Level (PRE) Exam. 05.05.2002 (IInd Sitting) (Eastern Zone, Guwahati))

12. 
 (1) 12 (2) 21
 (3) 24 (4) 35
 (SSC CPO Sub-Inspector Exam. 12.01.2003)

13. 
 (1) 10 (2) 15
 (3) 20 (4) 25
 (SSC CPO Sub-Inspector Exam. 07.09.2003)

14. 
 (1) 35 (2) 37
 (3) 45 (4) 47
 (SSC CPO Sub-Inspector Exam. 07.09.2003)

15. 
 (1) 62 (2) 37
 (3) 73 (4) 19
 (SSC CPO Sub-Inspector Exam. 05.09.2004)

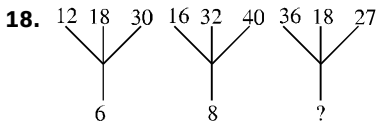
16. 
 (1) 7 (2) 6
 (3) 5 (4) 4
 (SSC CPO Sub-Inspector Exam. 05.09.2004)

FINDING THE MISSING NUMBER



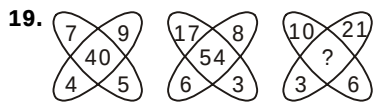
- (1) 04 (2) 01
(3) 02 (4) 03

(SSC CPO Sub-Inspector Exam. 26.05.2005)



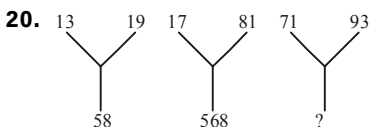
- (1) 6 (2) 9
(3) 12 (4) 18

(SSC CPO Sub-Inspector Exam. 26.05.2005)



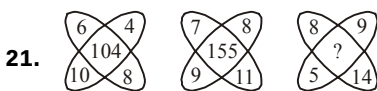
- (1) 60 (2) 62
(3) 64 (4) 66

(SSC Combined Graduate Level Prelim Exam. 13.11.2005 (Second Sitting))



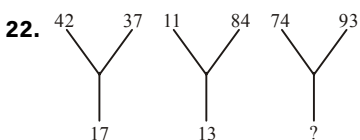
- (1) 94 (2) 96
(3) 98 (4) 100

(SSC Combined Graduate Level Prelim Exam. 04.02.2007 (First Sitting))



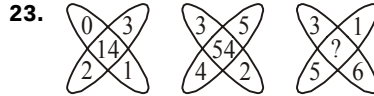
- (1) 152 (2) 164
(3) 140 (4) 142

(SSC Combined Graduate Level Prelim Exam. 04.02.2007 (First Sitting))



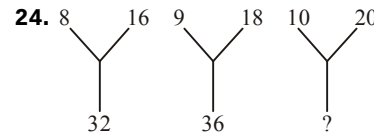
- (1) 40 (2) 42
(3) 44 (4) 46

(SSC Combined Graduate Level Prelim Exam. 04.02.2007 (Second Sitting))



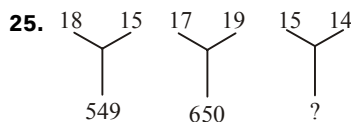
- (1) 35 (2) 72
(3) 94 (4) 71

(SSC Combined Graduate Level Prelim Exam. 04.02.2007 (Second Sitting))



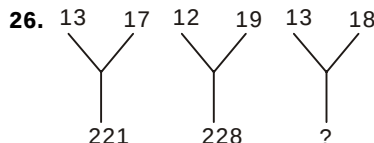
- (1) 30 (2) 40
(3) 24 (4) 32

(SSC CPO Sub-Inspector Exam. 16.12.2007)



- (1) 210 (2) 225
(3) 196 (4) 421

(SSC CPO Sub-Inspector Exam. 09.11.2008)



- (1) 31 (2) 229
(3) 234 (4) 312

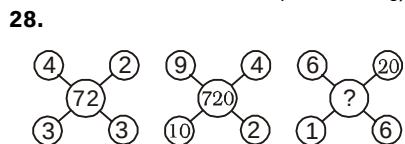
(SSC CPO Sub-Inspector Exam. 06.09.2009)

27. What is the number missing from the third target ?

5	9	15
16	29	?
49	89	147

- (1) 45 (2) 48
(3) 51 (4) 54

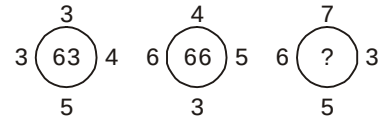
(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010 (First Sitting))



- (1) 72 (2) 720
(3) 7200 (4) 38

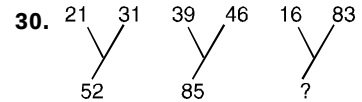
(SSC CPO (SI, ASIn & Intelligence Officer) Exam. 28.08.2011 (Paper-I))

29.



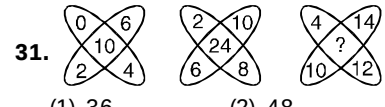
- (1) 57 (2) 53
(3) 105 (4) 111

(SSC CPO (SI, ASIn & Intelligence Officer) Exam. 28.08.2011 (Paper-I))



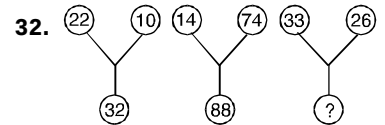
- (1) 92 (2) 72
(3) 62 (4) 99

(SSC Combined Matric Level (PRE) Exam. 24.10.1999 (1st Sitting))



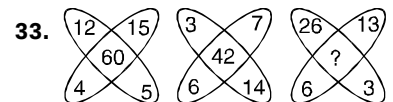
- (1) 36 (2) 48
(3) 38 (4) 30

(SSC Combined Matric Level (PRE) Exam. 24.10.1999 (1st Sitting))



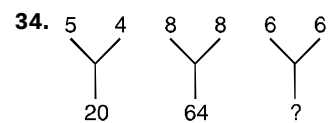
- (1) 48 (2) 58
(3) 59 (4) 95

(SSC Combined Matric Level (PRE) Exam. 24.10.1999 (IInd Sitting))



- (1) 19 (2) 29
(3) 78 (4) 48

(SSC Combined Matric Level (PRE) Exam. 24.10.1999 (IInd Sitting))



- (1) 25 (2) 12
(3) 36 (4) 0

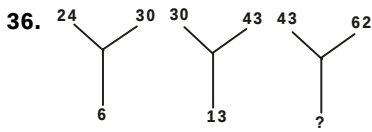
(SSC Combined Matric Level (PRE) Exam. 21.05.2000 (1st Sitting) (East Zone))



- (1) 196 (2) 184
(3) 169 (4) 200

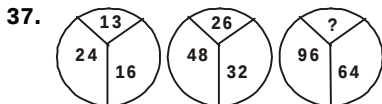
(SSC CML (PRE) Exam. 21.05.2000 (1st Sitting) (East Zone))

FINDING THE MISSING NUMBER



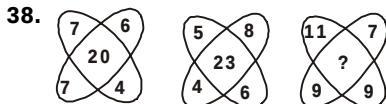
- (1) 12 (2) 21
(3) 19 (4) 9

(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (1st Sitting)
(Raipur, Madhya Pradesh)



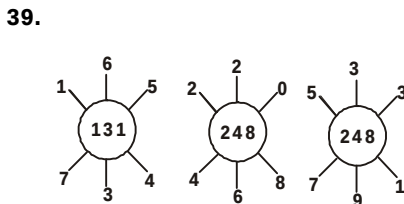
- (1) 160 (2) 25
(3) 32 (4) 52

(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (1st Sitting)
(Raipur, Madhya Pradesh)



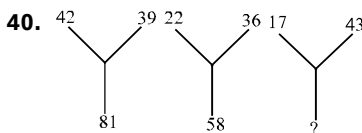
- (1) 77 (2) 36
(3) 99 (4) 63

(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (1st Sitting)
(Raipur, Madhya Pradesh)



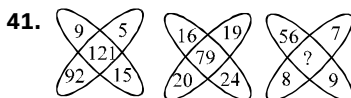
- (1) 132 (2) 320
(3) 274 (4) 262

(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (1st Sitting)
(Raipur, Madhya Pradesh)



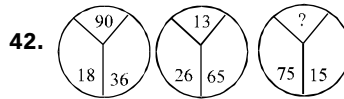
- (1) 26 (2) 60
(3) 52 (4) 80

(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting)
(East Zone)



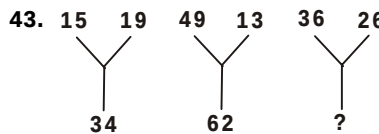
- (1) 98 (2) 80
(3) 89 (4) 18

(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting)
(East Zone)



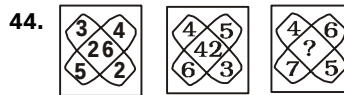
- (1) 30 (2) 75
(3) 45 (4) 60

(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting)
(East Zone)



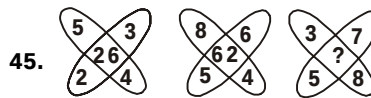
- (1) 10 (2) 62
(3) 4 (4) 124

(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (IInd Sitting)
(Middle Zone, Allahabad)



- (1) 64 (2) 62
(3) 60 (4) 58

(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (IInd Sitting)
(Middle Zone, Allahabad)



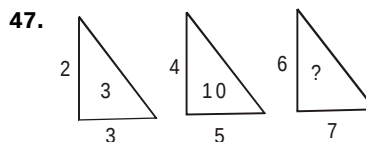
- (1) 71 (2) 59
(3) 62 (4) 55

(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (1st Sitting)



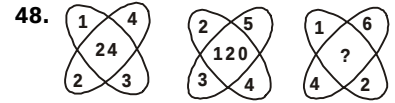
- (1) 0 (2) 10
(3) 2.5 (4) 20

(SSC Combined Matric Level (PRE) Exam.
13.05.2001 (1st Sitting)



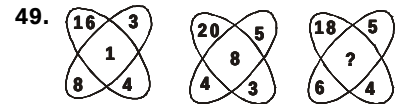
- (1) 21 (2) 32
(3) 22 (4) 24

(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (IInd Sitting)



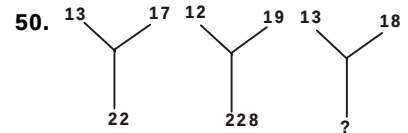
- (1) 40 (2) 140
(3) 48 (4) 36

(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (IInd Sitting)



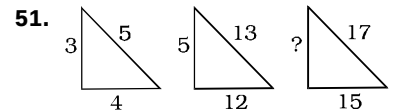
- (1) 3 (2) 10
(3) 15 (4) 60

(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (IInd Sitting)



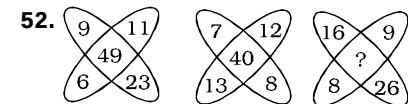
- (1) 229 (2) 234
(3) 312 (4) 246

(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (IInd Sitting)



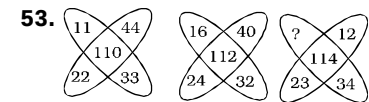
- (1) 2 (2) 8
(3) 64 (4) 6

(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting)
(East Zone)



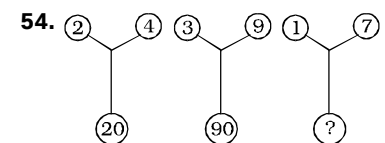
- (1) 52 (2) 42
(3) 59 (4) 17

(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting)
(East Zone)



- (1) 37 (2) 45
(3) 35 (4) 46

(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting)
(East Zone)



FINDING THE MISSING NUMBER

- (1) 20 (2) 25
(3) 50 (4) 75

(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting
(East Zone)

55.
$$\begin{array}{ccccc} 16 & 64 & 256 & & \\ 1 & 10 & 9 & 25 & 27 & 100 & 26 & ? & 144 \\ 4 & & 16 & & 64 & & & & \end{array}$$

- (1) 37 (2) 47
(3) 06 (4) 42

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(Eastern Zone, Guwahati)

56.
$$\begin{array}{ccccc} 4 & 3 & 11 & 9 & 15 & 6 \\ & \diagdown & \diagup & \diagdown & \diagup & \diagdown \\ & 144 & 9801 & ? & & \end{array}$$

- (1) 2250 (2) 8100
(3) 11036 (4) 1216

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(North Zone, Delhi)

57.

$$\begin{array}{ccc} 5 & 9 & 8 \\ 7 & 13 & 4 \\ 3 & 5 & 4 \\ 4 & 8 & 9 \\ 3 & 3 & 3 \end{array}$$

- (1) 12 (2) 15
(3) 18 (4) 14

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(North Zone, Delhi)

58.

$$\begin{array}{ccc} 3 & 5 & ? \\ 24 & 13 & 7 \\ 9 & 7 & 10 \\ 17 & 50 & 47 \\ 9 & 11 & 21 \end{array}$$

- (1) 3 (2) 5
(3) 9 (4) 10

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(North Zone, Delhi)

59.

$$\begin{array}{ccc} 5 & 3 & 7 \\ 63 & 41 & ? \\ 6 & 8 & 5 \\ 2 & 7 & 6 \\ 3 & 9 & 8 \end{array}$$

- (1) 82 (2) 83
(3) 86 (4) 26

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(North Zone, Delhi)

60.

$$\begin{array}{ccc} 27 & 42 & 27 \\ 4 & 13 & 11 \\ 3 & 16 & 7 \\ 13 & 65 & 8 \\ 9 & 72 & 9 \end{array}$$

- (1) 9 (2) 18
(3) 12 (4) 6

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(North Zone, Delhi)

61.

$$\begin{array}{ccc} 64 & 125 & 216 \\ 1 & 10 & 27 \\ 8 & 27 & 64 \\ 27 & 64 & 125 \end{array}$$

- (1) 18 (2) 2
(3) 9 (4) 17

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(North Zone, Delhi)

62.

$$\begin{array}{ccc} 8 & 12 & 9 \\ 6 & 5 & 4 \\ 7 & 7 & 14 \\ 4 & 9 & 8 \\ 10 & ? & 16 \end{array}$$

- (1) 7 (2) 8
(3) 6 (4) 9

(SSC Combined Matric Level (Pre)
Exam. 05.05.2002 (IInd Sitting)
(North Zone Delhi)

63.

$$\begin{array}{ccc} 4 & 5 & 6 \\ 1 & 30 & 3 \\ 2 & 42 & 4 \\ 3 & 3 & 5 \\ 2 & 4 & 4 \end{array}$$

- (1) 54 (2) 45
(3) 35 (4) 53

(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (Ist Sitting)

64.

$$\begin{array}{ccc} 4 & 5 & 6 \\ 1 & 4 & 3 \\ 2 & 4 & 3 \\ 3 & 4 & 5 \\ 2 & 4 & 4 \end{array}$$

- (1) 2 (2) 4
(3) 5 (4) 6

(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (IInd Sitting)

65.

$$\begin{array}{ccc} 4 & 24 & 15 \\ 6 & 9 & 9 \\ 6 & 3 & 6 \end{array}$$

- (1) 8 (2) 7
(3) 10 (4) 12

(SSC Combined Matric Level (Pre)
Exam. 16.06.2002 (Re-Exam)

66. Which number will replace the question mark?

$$\begin{array}{ccc} 50 & 31 & 43 \\ 9 & 7 & ? \\ 31 & 21 & 11 \end{array}$$

- (1) 6 (2) 5
(3) 8 (4) 9

(SSC Combined Matric Level (Pre)
Exam. 30.07.2006 (Ist Sitting)
(East Zone)

67. Insert the missing number :

$$\begin{array}{ccc} 2 & 3 & 6 \\ 3 & 33 & 2 \\ 4 & 5 & 4 \end{array}$$

- (1) 94 (2) 86
(3) 82 (4) 78

(SSC Combined Matric Level (Pre) Exam.
30.07.2006 (IInd Sitting) (Central Zone)

68.

$$\begin{array}{ccc} 7 & 9 & 8 \\ 19 & ? & 28 \\ 4 & 3 & 5 \end{array}$$

- (1) 27 (2) 21
(3) 28 (4) 17

(SSC Combined Matric Level (Pre)
Exam. 30.03.2008 (Ist Sitting)

69.

$$\begin{array}{ccc} 12 & 13 & 27 \\ 25 & 51 & 64 \end{array}$$

- (1) 35 (2) 36
(3) 37 (4) 38

(SSC Combined Matric Level (Pre)
Exam. 30.03.2008 (Ist Sitting)

70.

$$\begin{array}{ccc} 45 & 20 & 40 \\ 25 & 30 & 25 \\ 30 & 30 & 65 \end{array}$$

- (1) 36 (2) 33
(3) 45 (4) 60

(SSC Data Entry Operator
Exam. 31.08.2008)

71.

$$\begin{array}{ccc} 46 & 22 & 69 \\ 24 & 27 & ? \end{array}$$

- (1) 40 (2) 41
(3) 31 (4) 51

(SSC Data Entry Operator
Exam. 02.08.2009)

72.

$$\begin{array}{ccc} 4 & 3 & 11 \\ 144 & 9801 & ? \end{array}$$

- (1) 2250 (2) 8100
(3) 11036 (4) 1216

(SSC Stenographer Grade 'C' & 'D'
Exam. 09.01.2011)

FINDING THE MISSING NUMBER

73.

48	12	64	16	?	15
25	5	81	9	49	7

- (1) 75 (2) 60
(3) 30 (4) 90

(SSC Assistant Grade-III
Exam. 11.11.2012 (IInd Sitting))

74.

12	6	5
27	20	18
16	13	12
10	?	11
13	9	4

- (1) 9 (2) 12
(3) 8 (4) 13

(SSC Assistant Grade-III
Exam. 11.11.2012 (IInd Sitting))

75.

27	9
2	7
1	4

35	7
3	3
2	?

36	4
4	3
?	2

- (1) 3 (2) 4
(3) 5 (4) 6

(SSC Multi-Tasking Staff Exam.
10.03.2013, 1st Sitting : Patna)

76. Select the missing number from the given responses.

6	9	4	8	5	8
15	6	12	6	12	11

- (1) 12 (2) 10
(3) 8 (4) 6

(SSC Multi-Tasking Staff
Exam. 10.03.2013)

77. Find the missing number.

3	6	2
4	27	5
2	3	9

- (1) 37 (2) 45
(3) 47 (4) 57

(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)

78.

2	3	5	1	4	3
30	5	-5	-1	?	-1

- (1) 7 (2) -12
(3) 12 (4) 9

(SSC Constable (GD)
Exam. 12.05.2013)

Directions % In each of the following questions, select the missing number from the given responses.

79.

3	9	2	8	4	7
81	5	64	6	49	5

- (1) 1 (2) 8
(3) 6 (4) 16

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 22.06.2014)

80.

4	6	8	5	11	9
66	142	?	6	7	

- (1) 121 (2) 82
(3) 131 (4) 74

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 22.06.2014)

81.

81	729	64	512	49	?
					
9		8		7	

- (1) 444 (2) 515
(3) 343 (4) 373

(SSC CHSL (10+2) DEO & LDC
Exam. 09.11.2014 & Bihar SSC
2nd CGL (Pre) Exam. 23.02.2015)

82.

2	4	3	1	5	4
265	16	1	81	256	?

- (1) 125 (2) 25
(3) 625 (4) 1225

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, Patna Region :
1st Sitting)

83.

5	4	3	8	9	4
20	9	24	11	?	13

- (1) 36 (2) 117
(3) 52 (4) 26

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014)

84.

28	4	25	5	?	3
7	11	5	10	8	11

- (1) 22 (2) 24
(3) 25 (4) 28

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014)

85. If $\frac{7 \times 8}{9} = 3$; $\frac{8 \times 9}{144} = 12$;

$\frac{10 \times 11}{169} = 13$; then $\frac{12 \times 13}{441} = ?$

- (1) 196 (2) 225
(3) 144 (4) 21

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, 1st Sitting
TF No. 333 LO 2)

Directions : In each of the following questions, select the missing number from the given responses.

86.

3	2	1	5	3	4
35	126	?			

- (1) 90 (2) 91
(3) 103 (4) 75

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, 1st Sitting
TF No. 333 LO 2)

87.

4	36	13	13	8	32
					
12	?	16			

- (1) 8 (2) 13
(3) 4 (4) 12

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, IInd Sitting
TF No. 545 QP 6)

88.

8	12	14
6	54	7
4	7	9

- (1) 53 (2) 68
(3) 71 (4) 76

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 21.06.2015
(1st Sitting) TF No. 8037731)

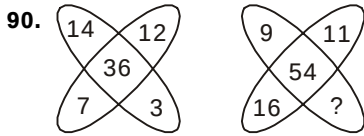
89.

3	2	2	-1	6	5
4	24	-2	4	0	?

- (1) 1 (2) 30
(3) 11 (4) 0

(SSC CGL Tier-I Exam, 09.08.2015
(1st Sitting) TF No. 1443088)

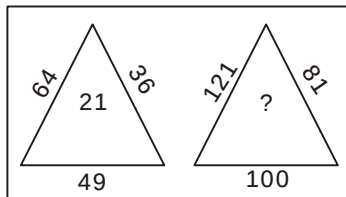
FINDING THE MISSING NUMBER



- (1) 16 (2) 12
(3) 17 (4) 18

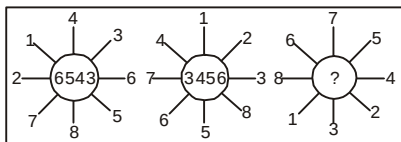
(SSC CGL Tier-I Exam, 09.08.2015
(Ist Sitting) TF No. 1443088)

91.



- (1) 10 (2) 20
(3) 30 (4) 40

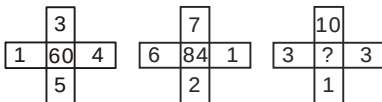
92.



- (1) 6543 (2) 5634
(3) 5364 (4) 3564

(SSC CGL Tier-I Exam, 09.08.2015
(IInd Sitting) TF No. 4239378)

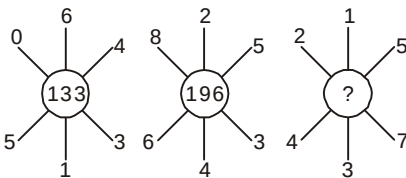
93.



- (1) 90 (2) 12
(3) 48 (4) 16

(SSC CGL Tier-I Exam, 16.08.2015
(Ist Sitting) TF No. 3196279)

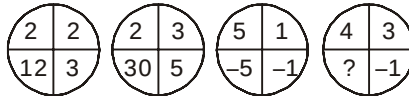
94.



- (1) 535 (2) 451
(3) 154 (4) 702

(SSC CGL Tier-I Exam, 16.08.2015
(Ist Sitting) TF No. 3196279)

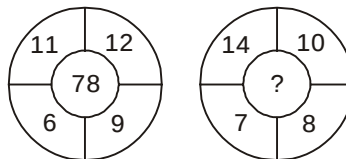
95.



- (1) -12 (2) 9
(3) 7 (4) 12

(SSC CGL Tier-I Exam, 16.08.2015
(Ist Sitting) TF No. 3196279)

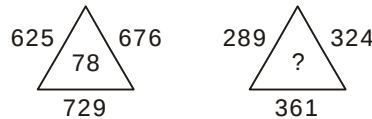
96.



- (1) 84 (2) 104
(3) 94 (4) 74

(SSC CGL Tier-I Exam, 16.08.2015
(IInd Sitting) TF No. 2176783)

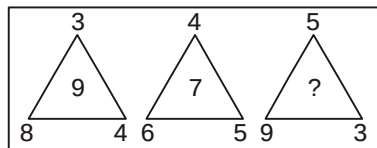
97.



- (1) 34 (2) 54
(3) 44 (4) 64

(SSC CGL Tier-I
Re-Exam, 30.08.2015)

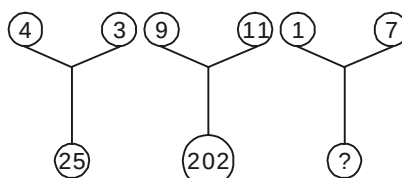
98. Find the missing number from the given responses.



- (1) 8 (2) 9
(3) 6 (4) 7

(SSC Constable (GD)
Exam, 04.10.2015, IInd Sitting)

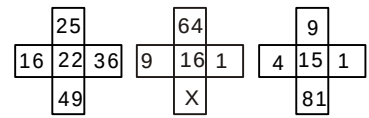
99.



- (1) 100 (2) 75
(3) 25 (4) 50

(SSC (10+2) LDC/DEO/PA/SA
Exam, 01.11.2015 TF No. 1098066)

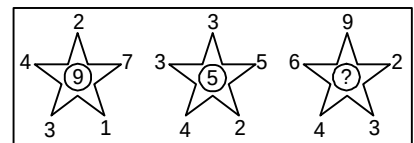
100.



- (1) 2 (2) 32
(3) 4 (4) 16

(SSC CHSL (10+2) LDC, DEO & PA/SA
Exam, 01.11.2015, IInd Sitting)

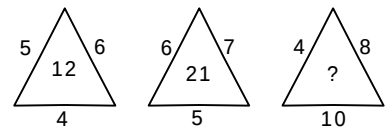
101.



- (1) 4 (2) 7
(3) 10 (4) 11

(SSC CHSL (10+2) LDC, DEO & PA/SA
Exam, 01.11.2015, IInd Sitting)

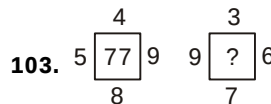
102.



- (1) 32 (2) 22
(3) 320 (4) 14

(SSC CHSL (10+2) LDC, DEO
& PA/SA Exam, 15.11.2015
(Ist Sitting) TF No. 6636838)

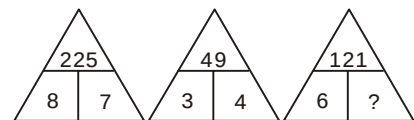
103.



- (1) 77 (2) 79
(3) 73 (4) 75

(SSC CHSL (10+2) LDC, DEO
& PA/SA Exam, 15.11.2015
(IInd Sitting) TF No. 7203752)

104.

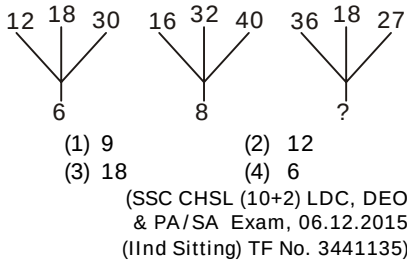


- (1) 21 (2) 20
(3) 5 (4) 4

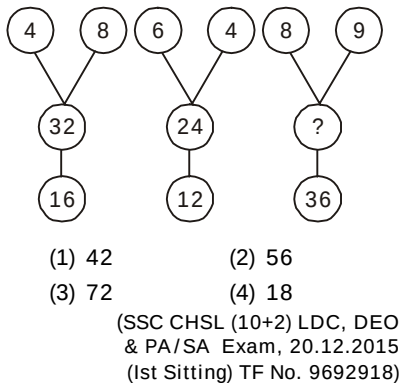
(SSC CHSL (10+2) LDC, DEO
& PA/SA Exam, 15.11.2015
(IInd Sitting) TF No. 7203752)

FINDING THE MISSING NUMBER

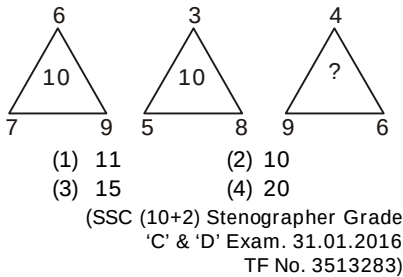
105.



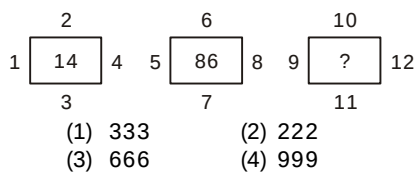
106.



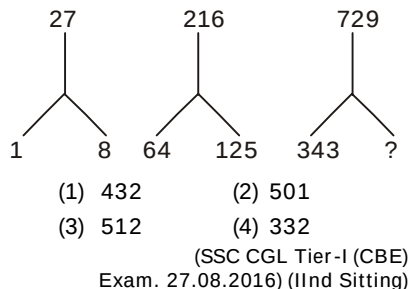
107.



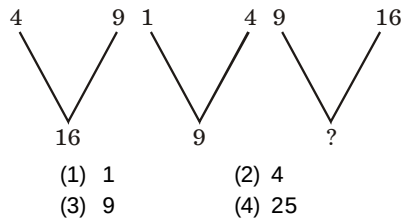
108. Select the missing number from the given responses.



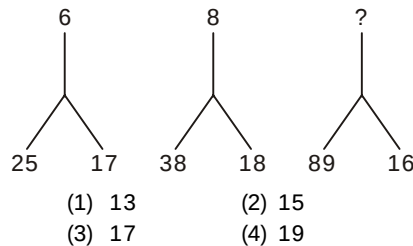
109. Select the missing number from the given responses:



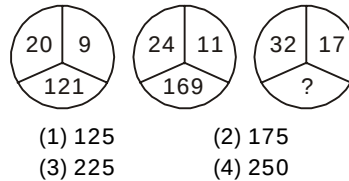
110. Select the missing number from the given responses.



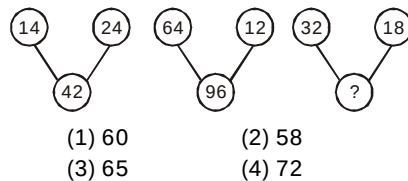
111. Find the missing number from the given alternatives :



112. Find the missing number from the given alternatives :

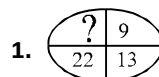


113. Select the missing number from the given responses :



TYPE-II

Directions : Find the missing number from the given responses in each of the following questions.



1. (1) 40 (2) 38
(3) 39 (4) 44

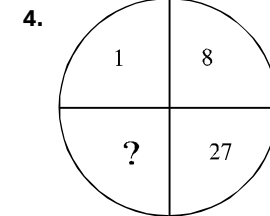
(SSC Combined Graduate Level Prelim Exam, 04.07.1999 (First Sitting))

2. 3 ? 5
5 4 7
4 4 4
60 96 140
(1) 4 (2) 6
(3) 8 (4) 9

(SSC Combined Graduate Level Prelim Exam, 04.07.1999 (First Sitting))

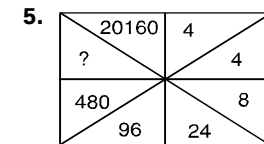
3. 4 9 2
3 5 7
8 1 ?
(1) 9 (2) 6
(3) 15 (4) 14

(SSC Combined Graduate Level Prelim Exam, 04.07.1999 (Second Sitting))



- (1) 41 (2) 64
(3) 35 (4) 61

(SSC Combined Graduate Level Prelim Exam, 04.07.1999 (Second Sitting))



- (1) 860 (2) 1140
(3) 2880 (4) 3240

(SSC Combined Graduate Level Prelim Exam, 27.02.2000 (First Sitting))

6. The trend results are shown at the end of each column. Find out the figure against the missing number.

9	8	7
6	7	8
5	4	?
270	224	336

- (1) 9 (2) 6
(3) 7 (4) 5

(SSC Combined Graduate Level Prelim Exam, 27.02.2000 (First Sitting))

FINDING THE MISSING NUMBER

7. In the question given below the numbers given at the top follow a certain specific pattern. Study out the pattern and find out the missing number.

9	4	20
8	5	12
7	6	?

- (1) 2 (2) 4
(3) 6 (4) 9

(SSC Combined Graduate Level
Prelim Exam. 27.02.2000
(First Sitting))

8. a = 23 (529) (1024),
b = 21 (441) (144),
c = 19 (361) (?)
(1) 1441 (2) 3529
(3) 9361 (4) 8281

(SSC Combined Graduate Level
Prelim Exam. 27.02.2000
(Second Sitting))

9.

0	-1	-2
1	0	-1
2	?	0

- (1) 1 (2) -1
(3) -2 (4) 4

(SSC Combined Graduate Level
Prelim Exam. 27.02.2000
(Second Sitting))

10.

5	6	3
25	42	21
2	10	20
7	17	?

- (1) 72 (2) 26
(3) 27 (4) 73

(SSC Combined Graduate Level
Prelim Exam. 27.02.2000
(Second Sitting))

11.

5	4	9
6	3	?
7	2	4
65	20	45
(1) 04	(2) 02	
(3) 03	(4) 01	

(SSC Combined Graduate Level
Prelim Exam. 24.02.2002
(First Sitting))

12.

9	11	13
3	4	7
3	4	5
81	176	?

- (1) 169 (2) 143
(3) 455 (4) 545

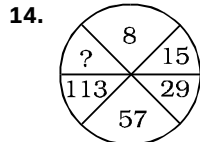
(SSC Combined Graduate Level Prelim
Exam. 24.02.2002 (Second Sitting))

13.

9	6	8
5	8	4
7	4	?
11	2	7

- (1) 4 (2) 7
(3) 3 (4) 6

(SSC Combined Graduate Level
Prelim Exam. 24.02.2002
(Middle Zone))



- (1) 223 (2) 224
(3) 225 (4) 227

(SSC CPO Sub-Inspector
Exam. 12.01.2003)

15.

4	12	11	5
6	7	10	3
8	9	10	7
7	5	?	4

- (1) 12 (2) 14
(3) 13 (4) 8

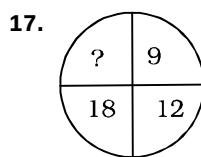
(SSC CPO Sub-Inspector
Exam. 12.01.2003)

16.

6	18	15
3	2	5
4	3	?
—	—	—
8	27	9

- (1) 2 (2) 11
(3) 3 (4) 6

(SSC CPO Sub-Inspector
Exam. 12.01.2003)



- (1) 11 (2) 25
(3) 10 (4) 27

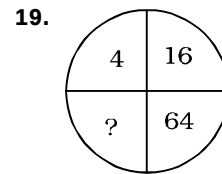
(SSC Combined Graduate Level
Prelim Exam. 11.05.2003
(First Sitting))

18.

5	5	2
2	4	1
8	3	10
40	30	?

- (1) 10 (2) 12
(3) 13 (4) 20

(SSC Combined Graduate Level
Prelim Exam. 11.05.2003
(First Sitting))



- (1) 16 (2) 52
(3) 112 (4) 256

(SSC Combined Graduate Level Prelim
Exam. 11.05.2003 (Second Sitting))

20.

2	12	?
50	300	550
10	60	110
124	744	1364

- (1) 22 (2) 33
(3) 44 (4) 55

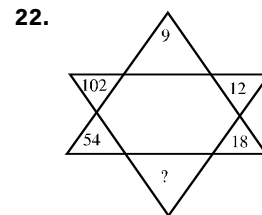
(SSC Combined Graduate Level Prelim
Exam. 11.05.2003 (Second Sitting))

21.

8	7	10	12
13	12	15	17
10	9	?	14

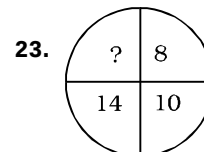
- (1) 12 (2) 10
(3) 21 (4) 25

(SSC CPO Sub-Inspector
Exam. 07.09.2003)



- (1) 40 (2) 48
(3) 30 (4) 24

(SSC CPO Sub-Inspector
Exam. 07.09.2003)



- (1) 20 (2) 16
(3) 12 (4) 18

(SSC Combined Graduate Level
Prelim Exam. 08.02.2004
(First Sitting))

24.

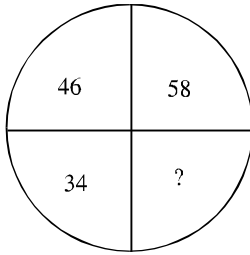
2	7	4
5	2	3
1	?	6
10	42	72

- (1) 2 (2) 4
(3) 5 (4) 3

(SSC Combined Graduate Level Prelim
Exam. 08.02.2004 (First Sitting))

FINDING THE MISSING NUMBER

25.



- (1) 92 (2) 72
(3) 22 (4) 68

(SSC Combined Graduate Level Prelim Exam. 08.02.2004 (Second Sitting))

26.

1	4	?
4	2	5
2	2	3
49	64	169

- (1) 3 (2) 6
(3) 5 (4) 4

(SSC Combined Graduate Level Prelim Exam. 08.02.2004 (Second Sitting))

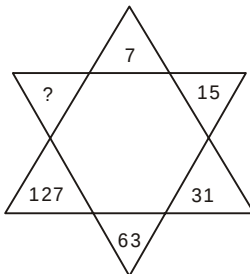
27.

9	3	7
12	2	9
13	5	?
1404	30	504

- (1) 5 (2) 8
(3) 15 (4) 56

(SSC CPO Sub-Inspector Exam.05.09.2004)

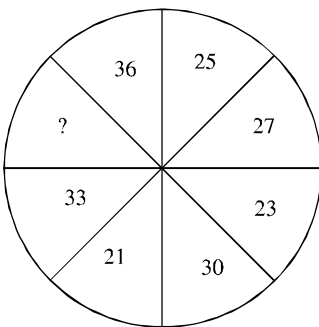
28.



- (1) 190 (2) 255
(3) 221 (4) 236

(SSC CPO Sub-Inspector Exam.05.09.2004)

29.



- (1) 35 (2) 32
(3) 22 (4) 19

(SSC CPO Sub-Inspector Exam. 26.05.2005)

30. 7 21 15

49 441 225

98 882 450

140 1302 ?

- (1) 6750 (2) 690
(3) 1380 (4) 660

(SSC CPO Sub-Inspector Exam. 26.05.2005)

31. 11 6 8

17 12 ?

25 34 19

19 28 11

- (1) 13 (2) 15
(3) 16 (4) 9

(SSC Statistical Investigators Grade-IV Exam. 31.07.2005)

32. 3 4 2

2 3 1

1 2 ?

6 24 0

- (1) 3 (2) 4
(3) 0 (4) 5

(SSC Statistical Investigators Grade-IV Exam. 31.07.2005)

33. 3 4 5

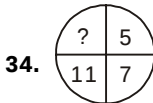
4 5 3

4 3 ?

48 60 105

- (1) 2 (2) 6
(3) 7 (4) 9

(SSC Combined Graduate Level Prelim Exam. 13.11.2005 (Second Sitting))



34.

- (1) 1 (2) 9
(3) 12 (4) 17

(SSC Combined Graduate Level Prelim Exam. 13.11.2005 (Second Sitting))

35. 25 49 81

5 7 ?

15 13 11

20 20 20

- (1) 9 (2) 3
(3) 61 (4) 31

(SSC Combined Graduate Level Prelim Exam.04.02.2007 (First Sitting))

36.

12	6	8
3	2	?
9	10	5
13	13	13

- (1) 01 (2) 0
(3) 10 (4) 11

(SSC Combined Graduate Level Prelim Exam. 04.02.2007 (Second Sitting))

37. The given equations follow the same rule. Find the missing number according to it.

836 (316) 112

213 () 420

(1) 368 (2) 220

(3) 211 (4) 468

(SSC CPO Sub-Inspector Exam. 16.12.2007)

38.

18 15 11

6 5 8

3 4 ?

324 300 528

(1) 7 (2) 5

(3) 6 (4) 4

(SSC CPO Sub-Inspector Exam. 16.12.2007)

39.

5	20	6	9
4	8	15	3
9	25	7	9
22	7	8	?

- (1) 7 (2) 8
(3) 9 (4) 10

(SSC Combined Graduate Level Prelim Exam. 27.07.2008 (First Sitting))

40. 341 (16) 521

613 (25) 816

452 (?) 326

(1) 27 (2) 22

(3) 30 (4) 41

(SSC Combined Graduate Level Prelim Exam. 27.07.2008 (First Sitting))

41. Find the missing number from the given responses :

4	9	17	6
20	5	8	9
7	23	9	9
?	9	4	19

- (1) 7 (2) 9
(3) 8 (4) 6

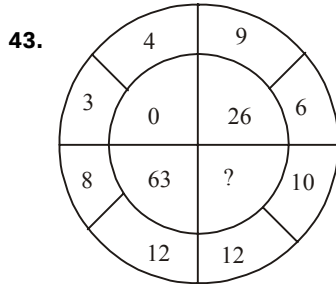
(SSC Combined Graduate Level Prelim Exam. 27.07.2008 (Second- Sitting))

FINDING THE MISSING NUMBER

42.
$$\begin{array}{r} 5 \quad 8 \quad 13 \\ 6 \quad 9 \quad 15 \\ 4 \quad 12 \quad ? \\ \hline 60 \quad 432 \quad 1560 \end{array}$$

- (1) 16 (2) 32
(3) 29 (4) 120

(SSC CPO Sub-Inspector Exam.09.11.2008)



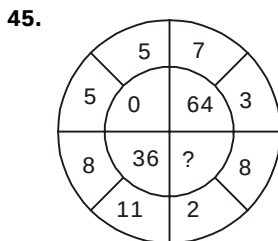
- (1) 12 (2) 7
(3) 16 (4) 14

(SSC CPO Sub-Inspector Exam.09.11.2008)

44.
$$\begin{array}{r} 3 \quad 6 \quad 7 \\ 9 \quad 18 \quad 21 \\ 27 \quad 54 \quad ? \\ 81 \quad 162 \quad 189 \end{array}$$

- (1) 22 (2) 63
(3) 190 (4) 55

(SSC CPO Sub-Inspector Exam.06.09.2009)



- (1) 0 (2) 125
(3) 100 (4) 144

(SSC CPO Sub-Inspector Exam.06.09.2009 & Bihar SSC 2nd CGL (Pre) Exam. 16.02.2015)

46. Find the missing number from the given responses.
173 (24) 526
431 (18) 325
253 (?) 471

- (1) 22 (2) 42
(3) 30 (4) 06

(SSC Combined Graduate Level Tier-1 Exam. 16.05.2010 (First Sitting))

47. Find the missing number from the given responses :

5	6	12
4	3	4
2	3	?
18	27	96

- (1) 4 (2) 5
(3) 3 (4) 6

(SSC Combined Graduate Level Tier-1 Exam.16.05.2010 (Second Sitting))

48. Select the missing number from the given alternatives.

$$\begin{array}{r} 3 \quad 7 \quad 5 \\ 4 \quad 2 \quad ? \\ 5 \quad 4 \quad 6 \\ 60 \quad 56 \quad 90 \end{array}$$

- (1) 9 (2) 3
(3) 4 (4) 8

(SSC SAS Exam. 26.06.2010 (Paper-I))

49. Find the missing number from the given responses.

8	9	10
5	4	3
28	?	16
12	25	14

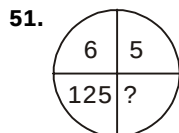
- (1) 28 (2) 11
(3) 32 (4) 18

(SSC CISF ASI Exam. 29.08.2010 (Paper-I))

50.
$$\begin{array}{r} 1 \quad 3 \quad 7 \\ 2 \quad 4 \quad 4 \\ 4 \quad 5 \quad 9 \\ 3 \quad 2 \quad 3 \\ 50 \quad 70 \quad ? \end{array}$$

- (1) 23 (2) 115
(3) 118 (4) 220

(SSC CPO Sub-Inspector Exam. 12.12.2010 (Paper-I))



- (1) 127 (2) 31
(3) 217 (4) 328

(SSC Combined Graduate Level Prelim Exam. 19.06.2011 (First Sitting))

52.

2	1	2
21	22	?
1	2	5
20	23	43

- (1) 40 (2) 48
(3) 50 (4) 36

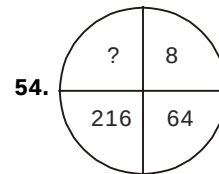
(SSC Combined Graduate Level Prelim Exam. 19.06.2011 (First Sitting))

53.

7	3	2
4	9	6
2	1	5
69	91	?

- (1) 58 (2) 51
(3) 65 (4) 64

(SSC Combined Graduate Level Prelim Exam. 19.06.2011 (Second Sitting))



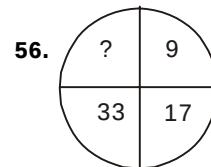
- (1) 343 (2) 512
(2) 729 (4) 1000

(SSC Combined Graduate Level Prelim Exam. 19.06.2011 (Second Sitting))

55.
$$\begin{array}{r} 2 \quad 7 \quad 9 \\ 7 \quad 3 \quad 4 \\ 9 \quad 8 \quad ? \\ 126 \quad 168 \quad 216 \end{array}$$

- (1) 8 (2) 3
(3) 6 (4) 36

(SSC Combined Graduate Level Tier-1 Exam. 26.06.2011 (First Sitting))



- (1) 60 (2) 68
(3) 55 (4) 65

(SSC Combined Graduate Level Tier-1 Exam. 26.06.2011 (First Sitting))

57.

169	64	81	30
625	?	49	50
1296	576	100	70

| FINDING THE MISSING NUMBER |

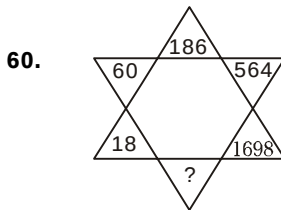
- (1) 324 (2) 289
(3) 441 (4) 361
(SSC Combined Graduate Level
Tier-1 Exam. 26.06.2011
(Second Sitting))

58.

1	2	3
4	5	6
7	8	9
27	38	?

- (1) 49 (2) 50
(3) 51 (4) 52
(SSC Combined Graduate Level
Tier-1 Exam. 26.06.2011
(Second Sitting))

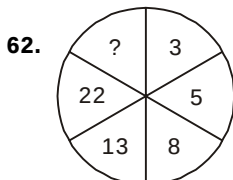
- 59.** 6 9 12
36 81 144
24 63 ?
(1) 120 (2) 80
(3) 94 (4) 102
(SSC CPO (SI, ASIn & Intelligence Officer)
Exam. 28.08.2011 (Paper-I))



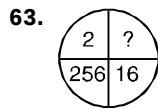
- (1) 5052 (2) 5100
(3) 5094 (4) 4860
(SSC CPO (SI, ASIn & Intelligence Officer)
Exam. 28.08.2011 (Paper-I))

- 61.** Find out the numbers that would fit in the second row and third row middle and last blank spaces (?) respectively.

- 18 23 16
17 19 ?
22 ? ?
(1) 26, 24, 25 (2) 15, 21, 20
(3) 21, 15, 20 (4) 25, 24, 26
(SSC CHSL (10+2) DEO & LDC
Exam. 02.11.2014, Patna Region :
1st Sitting)



- (1) 1 (2) 26
(3) 39 (4) 45
(SSC CPO (SI, ASIn & Intelligence Officer)
Exam. 28.08.2011 (Paper-I))



- (1) 8 (2) 4
(3) 32 (4) 16
(SSC Combined Matric Level (PRE)
Exam. 24.10.1999 (1st Sitting))

64.

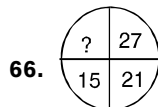
16	28	29
13	12	16
14	10	15
15	30	?

- (1) 60 (2) 30
(3) 2 (4) 45
(SSC Combined Matric Level (PRE)
Exam. 24.10.1999 (1st Sitting))

65.

5	6	5
8	9	7
10	7	?
400	378	315

- (1) 9 (2) 7
(3) 5 (4) 3
(SSC Combined Matric Level (PRE)
Exam. 24.10.1999 (1st Sitting))



- (1) 8 (2) 7
(3) 9 (4) 10
(SSC Combined Matric Level (PRE)
Exam. 24.10.1999 (IInd Sitting))

67.

6	15	20
8	4	5
3	5	20
51	65	?

- (1) 56 (2) 120
(3) 51 (4) 12
(SSC Combined Matric Level (PRE)
Exam. 24.10.1999 (IInd Sitting))

68.

9	11	13
3	4	7
3	4	?
81	176	455

- (1) 3 (2) 5
(3) 7 (4) 9
(SSC Combined Matric Level (PRE)
Exam. 24.10.1999 (IInd Sitting))

69.

1	3	2
8	6	7
9	5	?
72	90	56

- (1) 22 (2) 4
(3) 37 (4) 15
(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (1st Sitting)
(East Zone))

70.

4	1	25
9	4	36
16	9	49
25	16	?

- (1) 81 (2) 64
(3) 144 (4) 100
(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (1st Sitting)
(East Zone))

- 71.** 4 5 2
5 4 2
6 5 ?
120 100 24
(1) 14 (2) 6
(3) 160 (4) 58
(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (1st Sitting)
(East Zone))

72.

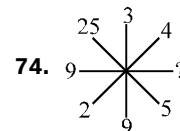
7	4	9
14	8	?
28	16	36
56	32	72

- (1) 81 (2) 27
(3) 16 (4) 18
(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting))

73.

5	6	7
4	5	?
3	4	5
60	120	140

- (1) 5 (2) 6
(3) 7 (4) 4
(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting))



- (1) 81 (2) 64
(3) 32 (4) 20
(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting))

- 75.** 408 (169) 395
129 (?) 122
(1) 49 (2) 39
(3) 59 (4) 48
(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting))

FINDING THE MISSING NUMBER

76.

3	4	9
2	5	2
6	7	?
36	140	18

- (1) 7 (2) 1
(3) 19 (4) 12

(SSC Combined Matric Level (PRE)
Exam. 21.05.2000 (IInd Sitting)
(Middle Zone, Allahabad)

77.

5	7	8
4	6	6
2	3	?
10	14	12

(1) 2 (2) 4
(3) 61 (4) 3

(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (Ist Sitting)

78.

4	3	6
2	5	4
3	7	?
24	105	120

- (1) 5 (2) 4
(3) 6 (4) 7

(SSC Combined Matric Level (PRE)
Exam. 13.05.2001 (IInd Sitting)

79.

8	6	4
8	10	12
4	2	?
256	120	192

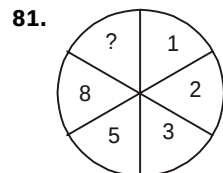
- (1) 8 (2) 4
(3) 2 (4) 0

(SSC Combined Matric Level (PRE)
Exam. 27.05.2001 (IInd Sitting)
(East Zone)

80. 16 (27) 43, 29 (?) 56,
47 (22) 25

- (1) 27 (2) 17
(3) 37 (4) 07

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(Eastern Zone, Guwahati)



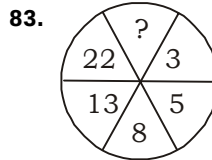
- (1) 10 (2) 15
(3) 13 (4) 12

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(Eastern Zone, Guwahati)

82. If $19 \times 6 = 38$
 $32 \times 12 = 128$
 $17 \times 9 = 51$
 $9 \times 3 = ?$

- (1) 9 (2) 18
(3) 27 (4) 35

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(Eastern Zone, Guwahati)



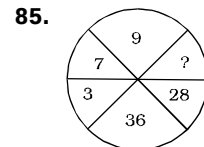
- (1) 29 (2) 39
(3) 37 (4) 49

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (IInd Sitting)
(Eastern Zone, Guwahati)

84. 6 7 8
36 49 64
42 56 ?

- (1) 32 (2) 48
(3) 53 (4) 72

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (IInd Sitting)
(Eastern Zone, Guwahati)



- (1) 16 (2) 12
(3) 8 (4) 6

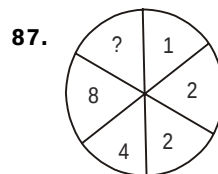
(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(North Zone, Delhi)

86.

26	18	10
11	9	7
5	4	1
10	5	?

- (1) 4 (2) 2
(3) 5 (4) 6

(SSC Combined Matric Level (PRE)
Exam. 05.05.2002 (Ist Sitting)
(North Zone, Delhi)



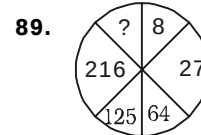
- (1) 32 (2) 06
(3) 12 (4) 20

(SSC Combined Matric Level (Pre)
Exam. 05.05.2002 (IInd Sitting)
(North Zone Delhi)

88. 7 9 5 11
4 15 12 7
13 8 11 ?

- (1) 20 (2) 10
(3) 30 (4) 70

(SSC Combined Matric Level (Pre)
Exam. 05.05.2002 (IInd Sitting)
(North Zone Delhi)



- (1) 729 (2) 343
(3) 305 (4) 4

(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (Ist Sitting)

90.

3	1	4
5	4	7
2	8	?
38	81	74

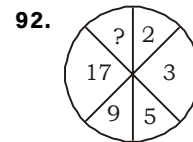
- (1) 9 (2) 6
(3) 3 (4) 7

(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (Ist Sitting)

91. a = 12 (175) 15, b = 14 (219)
16, c = 17 (?) 14

- (1) 223 (2) 233
(3) 224 (4) 230

(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (IInd Sitting)



- (1) 23 (2) 30
(3) 33 (4) 29

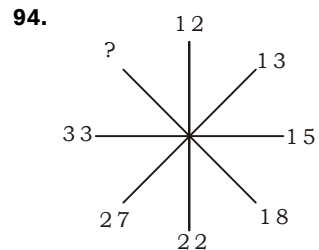
(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (IInd Sitting)

93.

24	144	384
6	36	?
2	12	32
1	6	16

- (1) 85 (2) 80
(3) 96 (4) 91

(SSC Combined Matric Level (Pre)
Exam. 12.05.2002 (IInd Sitting)



| FINDING THE MISSING NUMBER |

- (1) 36 (2) 38
(3) 40 (4) 42

SSC Combined Matric Level (Pre)
Exam. 16.06.2002 (Re-Exam)

- 95.** F J N
M Q U
O S ?

- (1) U (2) W
(3) X (4) Y

SSC Combined Matric Level (Pre)
Exam. 16.06.2002 (Re-Exam)

- 96.** 6 8 7
36 64 49
24 48 ?

- (1) 64 (2) 54
(3) 42 (4) 35

SSC Combined Matric Level (Pre)
Exam. 16.06.2002 (Re-Exam)

- 97.** 5 2 4
4 4 7
2 5 3
18 30 ?

- (1) 42 (2) 43
(3) 32 (4) 33

SSC Combined Matric Level (Pre)
Exam. 16.06.2002 (Re-Exam)

- 98.**

1	$\frac{1}{2}$	$\frac{3}{2}$
2	$\frac{2}{3}$	$\frac{8}{3}$
3	?	$\frac{19}{5}$

- (1) $\frac{1}{2}$ (2) $\frac{2}{3}$

- (3) $\frac{3}{4}$ (4) $\frac{4}{5}$

SSC Combined Matric Level (Pre)
Exam. 16.06.2002 (Re-Exam)

- 99.** Find the missing number in the matrix.

10	17	8
5	3	15
6	14	?
42	68	92

- (1) 23 (2) 10
(3) 25 (4) 46

SSC Combined Matric Level (Pre)
Exam. 30.07.2006 (1st Sitting)
(East Zone)

- 100.** Find the missing number in the matrix :

3	10	?
6	2	6
2	3	1
121	225	100

- (1) 9 (2) 3
(3) 10 (4) 5

SSC Combined Matric Level (Pre) Exam.
30.07.2006 (IInd Sitting) (Central Zone)

- 101.** 5 8 9
7 6 6
9 7 ?
21 21 21

- (1) 7 (2) 6
(3) 5 (4) 4

SSC Combined Matric Level (Pre)
Exam. 30.03.2008 (1st Sitting)

- 102.** 7 8 6
6 5 9
12 13 ?
504 520 486

- (1) 7 (2) 12
(3) 8 (4) 9

SSC Combined Matric Level (Pre)
Exam. 30.03.2008 (1st Sitting)

- 103.**

?	45
15	21

- (1) 8 (2) 10
(3) 14 (4) 16

SSC Combined Matric Level (Pre)
Exam. 30.03.2008 (1st Sitting)

- 104.**

12	14	16
5	9	6
10	16	?
50	110	84

- (1) 11 (2) 12
(3) 14 (4) 18

SSC Data Entry Operator
Exam. 31.08.2008

- 105.** 144 (132) 121
64 (?) 100
(1) 70 (2) 80
(3) 85 (4) 90

SSC Data Entry Operator
Exam. 31.08.2008

- Directions :** Select the missing number from the given alternatives.

SSC Data Entry Operator
Exam. 02.08.2009

- 106.**

6	8	12
12	9	4
10	5	?
720	360	336

- (1) 9 (2) 12
(3) 7 (4) 14

SSC Data Entry Operator
Exam. 31.08.2008

- 107.**

6	7	5
7	8	6
8	9	?
62	79	47

- (1) 4 (2) 7
(3) 8 (4) 9

(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 27.11.2010)

- 108.**

9	4	5
12	16	15
15	20	25
180	80	?

- (1) 125 (2) 75
(3) 20 (4) 25

(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 27.11.2010)

- 109.**

5	4	3
6	5	4
7	6	5
23	14	?

- (1) 23 (2) 17
(3) 7 (4) 12

(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 28.11.2010 (1st sitting))

- 110.**

5	8	7
2	4	6
6	2	3
60	64	?

- (1) 68 (2) 49
(3) 39 (4) 126

(SSC Higher Secondary Level
Data Entry Operator & LDC
Exam. 28.11.2010 (IInd sitting))

- 111.**

6	5	7
7	8	4
11	12	?
462	480	224

- (1) 7 (2) 8
(3) 6 (4) 9

(SSC Stenographer Grade 'C' & 'D'
Exam. 09.01.2011)

FINDING THE MISSING NUMBER

112. Find out the missing number.

8 9 9
6 7 8
9 11 ?
39 52 59

- (1) 10 (2) 11
(3) 12 (4) 13

(SSC Multi-Tasking (Non-Technical)
Staff Exam. 20.02.2011)

113. Find out the missing number.

8 9 7
4 5 6
2 4 ?
64 180 294

- (1) 6 (2) 7
(3) 8 (4) 9

(SSC Multi-Tasking (Non-Technical)
Staff Exam. 27.02.2011)

Directions : Select the missing number from the given responses.

114.

3	15	5
5	35	7
9	—	5

- (1) 54 (2) 50
(3) 49 (4) 45

(SSC CISF Constable (GD)
Exam. 05.06.2011)

115.

36	64	100
6	8	10
12	16	?

- (1) 10 (2) 20
(3) 22 (4) 110

(SSC Stenographer (Grade 'C' & 'D')
Exam. 16.10.2011)

116.

2 7 8
7 5 3
3 8 ?
42 280 120

- (1) 4 (2) 5
(3) 6 (4) 7

(SSC Stenographer (Grade 'C' & 'D')
Exam. 16.10.2011)

117.

14 22 37
04 06 03
05 03 05
61 135 ?

- (1) 120 (2) 112
(3) 116 (4) 555

SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011 (Ist
Sitting (North Zone))

118. 64 25 81

36 49 16
14 12 ?

- (1) 13 (2) 26
(3) 97 (4) 15

SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(Ist Sitting (North Zone))

119. 27 9 3

4 16 64
512 ? 8

- (1) 64 (2) 2
(3) 16 (4) 8

SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(IInd Sitting (North Zone))

120. 5 3 7 1

7 5 9 3
4 4 4 4
3 2 ? 1

- (1) 6 (2) 5
(3) 4 (4) 2

SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(IInd Sitting (North Zone))

121. 8 72 9

6 66 11
9 ? 6
(1) 64 (2) 60
(3) 52 (4) 54

SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(Ist Sitting (East Zone))

122. 14 25 42

2 4 6
3 3 ?
4 7 9

- (1) 3 (2) 4
(3) 6 (4) 7

SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(Ist Sitting (East Zone))

123. 5 4 41

7 3 58
10 2 ?
(1) 34 (2) 12
(3) 99 (4) 104

SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(IInd Sitting (East Zone))

124.

5	6	3	8
6	5	4	7
7	4	5	6
210	120	60	?

- (1) 336 (2) 330
(3) 270 (4) 390

SSC (10+2) Level Data Entry
Operator & LDC Exam. 04.12.2011
(IInd Sitting (East Zone))

125. 16 4 4

81 3 27
? 25 5
(1) 97 (2) 12
(3) 125 (4) 30

SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(Ist Sitting (Delhi Zone))

126.

6	4	5
3	6	5
2	3	4
36	72	?

- (1) 100 (2) 175
(3) 125 (4) 120

SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(Ist Sitting (Delhi Zone))

127. 5 4 9 6

7 5 2 4
1 3 4 5
35 60 72 ?

- (1) 116 (2) 136
(3) 120 (4) 126

SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(IInd Sitting (Delhi Zone))

128. 64 36 2

81 25 4
144 16 ?

- (1) 6 (2) 8
(3) 3 (4) 16

SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(IInd Sitting (Delhi Zone))

129. 10 85 8

7 54 7
8 ? 9

- (1) 72 (2) 77
(3) 74 (4) 79

SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(Ist Sitting (East Zone))

130. 12 13 16

3 5 6
8 8 ?
288 520 384

- (1) 4 (2) 5
(3) 3 (4) 6

SSC (10+2) Level Data Entry
Operator & LDC Exam. 11.12.2011
(Ist Sitting (East Zone))

FINDING THE MISSING NUMBER

131.

4	10	6
12	96	48
16	152	?

- (1) 110 (2) 104
(3) 112 (4) 124

SSC (10+2) Level Data Entry Operator & LDC Exam. 11.12.2011 (IInd Sitting (East Zone))

132.

9	5	6
7	6	7
4	8	?
252	240	210

- (1) 4 (2) 5
(3) 6 (4) 3

SSC (10+2) Level Data Entry Operator & LDC Exam. 11.12.2011 (IInd Sitting (East Zone))

133. Find the missing number from the given responses.

- 13 6 4
6 8 6
1 6 ?
(1) 7 (2) 10
(3) 8 (4) 13

SSC Constable (GD) & Rifleman (GD) Exam. 22.04.2012 (Ist Sitting)

134. Find the missing number from the given responses.

- 16 12 13
32 24 26
48 ? 39
(1) 28 (2) 36
(3) 42 (4) 32

SSC Constable (GD) & Rifleman (GD) Exam. 22.04.1912 (IInd Sitting)

135.

3	15	4
7	38	5
3	?	5

- (1) 18 (2) 15
(3) 19 (4) 20

(SSC Level Data Entry Operator & LDC Exam.21.10.2012 (IInd Sitting))

136.

9	10	11
5	7	8
19	22	?

- (1) 41 (2) 25
(3) 24 (4) 19

(SSC Level Data Entry Operator & LDC Exam.21.10.2012 (IInd Sitting))

137.

7	9	6
8	6	7
3	5	?

- 168 270 126
(1) 5 (2) 4
(3) 3 (4) 6

(SSC Level Data Entry Operator & LDC Exam.21.10.2012 (IInd Sitting))

138.

15	11	7
7	9	12
8	6	?
840	594	420

- (1) 70 (2) 60
(3) 5 (4) 9

(SSC Level Data Entry Operator & LDC Exam.21.10.2012 (IInd Sitting))

139.

1	3	4
5	2	8
26	13	?

- (1) 12 (2) 39
(3) 3 (4) 80

(SSC Level Data Entry Operator & LDC Exam.28.10.2012 (Ist Sitting))

140.

10	11	12
105	126	?

- (1) 144 (2) 149
(3) 3 (4) 2

(SSC Level Data Entry Operator & LDC Exam.28.10.2012 (Ist Sitting))

141.

7	23	21
1	3	?
2	5	7
3	4	3

- (1) 1 (2) 0
(3) 2 (4) 3

(SSC Level Data Entry Operator & LDC Exam.28.10.2012 (Ist Sitting))

142.

$\frac{11}{2}$	$\frac{5}{8}$	$\frac{7}{3}$	?
----------------	---------------	---------------	---

- (1) $\frac{3}{27}$ (2) $\frac{2}{27}$

- (3) $\frac{1}{27}$ (4) $\frac{3}{9}$

(SSC Level Data Entry Operator & LDC Exam.28.10.2012 (Ist Sitting))

143.

14	16	18
26	32	18
10	12	?

- (1) 13 (2) 11
(3) 14 (4) 9

(SSC Level Data Entry Operator & LDC Exam.28.10.2012 (Ist Sitting))

144.

9	9	17	17	16	8
5	5	4	4	?	8
(1) 8	(2) 16	(3) 64	(4) 4		

(SSC Level Data Entry Operator & LDC Exam.28.10.2012 (Ist Sitting))

145.

7	8	10	11	13	14
29	9	38	12	?	15
(1) 42	(2) 47	(3) 44	(4) 12		

(SSC Level Data Entry Operator & LDC Exam.28.10.2012 (Ist Sitting))

146.

7	4	5
2	3	?
5	9	6
70	108	240

- (1) 7 (2) 8
(3) 6 (4) 9

(SSC Level Data Entry Operator & LDC Exam.28.10.2012 (Ist Sitting))

147.

5	4	3
6	7	8
4	2	?
34	30	30

- (1) 6 (2) 5
(3) 10 (4) 3

(SSC Level Data Entry Operator & LDC Exam.04.11.2012 (IInd Sitting))

148.

5	2	7
?	3	1
4	5	2
15	7	13

- (1) 1 (2) 5
(3) 9 (4) 7

(SSC Assistant Grade-III Exam.11.11.2012 (IInd Sitting))

149.

4	8	10	320
2	?	3	180
9	6	4	216
(1) 22	(2) 30	(3) 28	(4) 20

(FCI Assistant Grade-II Exam. 22.01.2012 Paper-I)

150.

18	11	19
12	13	16
36	4	?

(1) 36 (2) 9

(3) 35 (4) 7

(FCI Assistant Grade-II Exam. 22.01.2012 Paper-I)

151-

8	15	22
29	?	43
50	57	64
(1) 34	(2) 50	
(3) 33	(4) 36	

(FCI Assistant Grade-III Exam. 25.02.2012 (Paper-I))

152-

7	9	3
8	3	5
2	5	?
112	135	900

- (1) 6 (2) 70
(3) 60 (4) 65

(FCI Assistant Grade-III Exam. 25.02.2012 (Paper-I)) North Zone (Ist Sitting)

| FINDING THE MISSING NUMBER |

- 153.** 7 9 6
 4 5 7
 2 3 ?
 9 11 9

(1) 4 (2) 5
 (3) 7 (4) 8

FCI Assistant Grade-III
Exam. 05.02.2012 (Paper-I)
East Zone (IInd Sitting)

- 154.** 9 8 7
 18 16 14
 36 32 ?

(1) 28 (2) 21
 (3) 98 (4) 42

FCI Assistant Grade-III
Exam. 05.02.2012 (Paper-I)
East Zone (IInd Sitting)

Directions (155-156) : In each of the following questions, select the missing number from the given responses.

(SSC (10+2) Level Data Entry
Operator & LDC Exam.
04.11.2012, 1st Sitting)

- 155.** ? 3
 9 5
 (1) 14 (2) 15
 (3) 12 (4) 13

- 156.** 7 6 15
 10 ? 12
 35 12 90
 (1) 9 (2) 4
 (3) 25 (4) 11

Direction (157) % In each of the following questions, select the missing number from the given responses.

(SSC CHSL (10+2) DEO & LDC
Exam. 09.11.2014)

- 157.**
 18 11 6 12
 9 38 6 19 32 9 26 44 3 9 ? 20
 17 11 15 8
 (1) 9 (2) 40
 (3) 7 (4) 36

- 158.** If $2 + 3 + 5 = 30$, $3 + 4 + 6 = 72$,
 $5 + 6 + 2 = 60$, then $5 + 4 + 0 = ?$
 (1) 40 (2) 30
 (3) 0 (4) None

(SSC Multi-Tasking Staff Exam.
10.03.2013, 1st Sitting : Patna)

- 159.** If $20 \times 5 = 4$ and $48 \times 12 = 4$,
 then $77 \times 11 = ?$

(1) 7 (2) 4
 (3) 847 (4) 10

(SSC Multi-Tasking Staff
Exam. 10.03.2013)

- 160.** If $53 \div 31 = 2$, $45 \div 27 = 1$,
 $69 \div 32 = 3$, then $97 \div 26 = ?$

(1) 1 (2) 2
 (3) 3 (4) 4

(SSC Multi-Tasking Staff Exam.
17.03.2013, Kolkata Region)

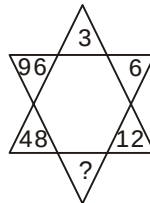
- 161.** The three equations follow the same numerical operation. Find the missing number according to it.

$178, 817; 534 = 453; 294 = ?$

(1) 429 (2) 492
 (3) 924 (4) 942

(SSC Multi-Tasking Staff Exam.
17.03.2013, Kolkata Region)

- 162.** Select the missing number from the given responses



(1) 22 (2) 18
 (3) 26 (4) 24

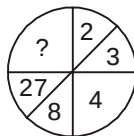
(SSC Multi-Tasking Staff
Exam. 17.03.2013, 1st Sitting)

- 163.** If $4 \times 2 \times 6 = 1626$, $3 \times 7 \times 4 = 974$, then $5 \times 6 \times 8 = ?$

(1) 2568 (2) 5664
 (3) 6456 (4) 3658

(SSC Multi-Tasking Staff
Exam. 17.03.2013, IInd Sitting)

- 164.** Select the missing number from the given responses.



(1) 49 (2) 45
 (3) 64 (4) 56

(SSC Multi-Tasking Staff
Exam. 17.03.2013, IInd Sitting)

Directions (165-166) % In each of the following questions, some equations are solved on the basis of certain system. Find out the correct answer for the unsolved equation on that basis.

(SSC Multi-Tasking Staff
Exam. 24.03.2013, 1st Sitting)

- 165.** $72 + 37 = 6328$; $54 + 13 = 4504$;
 $61 + 53 = ?$

(1) 4524 (2) 5244
 (3) 5424 (4) 5214

- 166.** $1 \times 2 \times 4 = 212$, $5 \times 6 \times 8 = 654$,
 $3 \times 7 \times 2 = ?$

(1) 173 (2) 713
 (3) 731 (4) 317

- 167.** 6 15 20
 8 4 5
 3 5 20
 51 65 ?

(1) 120 (2) 51
 (3) 12 (4) 56

(SSC Graduate Level Tier-I
Exam. 21.04.2013, 1st Sitting)

- 168.**

16	25	9
36	64	81
10	13	?

(1) 12 (2) 13
 (3) 14 (4) 11

(SSC Graduate Level Tier-I
Exam. 21.04.2013, 1st Sitting)

- 169.**

2	9	11	7
8	5	13	-3
7	?	10	(-4)
6	4	10	?

(1) 3 and 2 (2) (-3) and 2
 (3) 3 and (-2) (4) (-3) and (-2)

(SSC Graduate Level Tier-I
Exam. 21.04.2013, 1st Sitting)

- 170.** If $55 + 66 = 33$ and $22 + 99 = 33$, what is $44 + 88 = ?$

(1) 36 (2) 38
 (3) 40 (4) 33

(SSC Graduate Level Tier-I
Exam. 21.04.2013, 1st Sitting)

- 171.** 16 49 64
 25 36 81
 9 13 ?

(1) 22 (2) 17
 (2) 14 (4) 21

(SSC Graduate Level Tier-I
Exam. 21.04.2013, 1st Sitting)

- 172.** 3 4 6
 5 7 3
 1 2 7
 35 69 ?

(1) 94 (2) 84
 (3) 42 (4) 82

(SSC Graduate Level Tier-I
Exam. 21.04.2013, 1st Sitting)

FINDING THE MISSING NUMBER

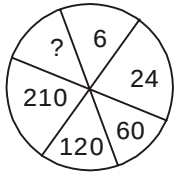
- 173.** 19 18 34 32 44 41
2 4 ?
(1) 6 (2) 9
(3) 4 (4) 3
(SSC Graduate Level Tier-I
Exam. 21.04.2013, Ist Sitting)
- 174.** 4 3 2
36 2 100 7 ? 5
(1) 49 (2) 64
(3) 81 (4) 71
(SSC Graduate Level Tier-I
Exam. 21.04.2013, Ist Sitting)
- 175.** 49 81 64
4 49 9
25 16 36
10 ? 11
(1) 6 (2) 4
(3) 9 (4) 16
(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)
- 176.** 2 14 21 28
3 21 28 35
4 ? 35 ?
(1) 35 and 49 (2) 28 and 42
(3) 21 and 42 (4) 49 and 28
(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)
- 177.** 2 3 4
24 39 ?
20 30 40
(1) 44 (2) 49
(3) 50 (4) 56
(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)
- 178.** 7 3 2
4 9 6
2 1 5
39 65 ?
(1) 91 (2) 68
(3) 56 (4) 104
(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)
- 179.** 81 64 16
4 9 49
36 16 25
108 96 ?
(1) 230 (2) 140
(3) 120 (4) 410
(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)
- 180.** 25 5 5
30 5 6
35 ? 5
(1) 5 (2) 4
(3) 6 (4) 7
(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)

- 181.** 24 51 67
2 4 6
5 7 5
53 211 ?
(1) 135 (2) 235
(3) 347 (4) 407
(SSC Graduate Level Tier-I
Exam. 21.04.2013, IInd Sitting)
- 182.** Find the correct answer for the unsolved equation :
 $5 \times 6 \times 3 = 356$, $1 \times 0 \times 5 = 510$,
 $5 \times 6 \times 7 = ?$
(1) 567 (2) 657
(3) 210 (4) 756
(SSC Constable (GD)
Exam. 12.05.2013)
- 183.** 1 4 9 16
25 36 49 ?
81 100 ? 144
(1) 64 & 121 (2) 20 & 100
(3) 121 & 46 (4) 95 & 150
(SSC Constable (GD)
Exam. 12.05.2013 Ist Sitting)
- 184.** Find the correct set of numbers from the given alternatives.
(4, 3, 2), (16, 9, 4),
(256, 81, 16), (65536, ?, ?)
(1) 6651, 286 (2) 6561, 256
(3) 1486, 97 (4) 190, 20
(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)
- 185.** 16 32 48 64
17 34 51 68
18 36 54 ?
(1) 77 (2) 72
(3) 74 (4) 76
(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)
- 186.** 4 5 10
36 2 49 2 ? 2
(1) 144 (2) 55
(3) 100 (4) 64
(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)
- 187.** 5 1 25
6 2 18
10 4 25
3 3 ?
(1) 10 (2) 9
(3) 3 (4) 4
(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)
- 188.** If $29 \times 48 = 576$, $35 \times 16 = 90$,
 $22 \times 46 = 96$, then $42 \times 17 = ?$
(1) 56 (2) 286
(3) 48 (4) 64
(SSC Graduate Level Tier-I
Exam. 19.05.2013, IInd Sitting)

- 189.** 54 30 112 42 ? 28
24 70 38
(1) 176 (2) 166
(3) 116 (4) 66
(SSC Graduate Level Tier-I
Exam. 19.05.2013, IInd Sitting)
- 190.** 216 209 202
522 515 508
633 626 ?
(1) 620 (2) 608
(3) 602 (4) 619
(SSC Graduate Level Tier-I
Exam. 19.05.2013, IInd Sitting)
- 191.** 15 12 44 28 64 53
3 8 ?
(1) 30 (2) 13
(3) 70 (4) 118
(SSC Graduate Level Tier-I
Exam. 19.05.2013, IInd Sitting)
- 192.** 3 5 8 7
4 6 4 6
5 2 2 3
58 58 62 ?
(1) 124 (2) 126
(3) 122 (4) 128
(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)
- 193.** 4 3 2
6 9 10
9 27 ?
(1) 54 (2) 30
(3) 20 (4) 50
(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)
- 194.** 84 81 88
14 12 18 9 ? 11
(1) 16 (2) 10
(3) 12 (4) 14
(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)
- 195.** If $235 = 38$ and $452 = 45$, then
 $345 = ?$
(1) 49 (2) 66
(3) 72 (4) 50
(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)
- 196.** $2 \times 3 = 49$, $5 \times 6 = 2536$, $1 \times 9 = 181$, $4 \times 7 = ?$
(1) 1628 (2) 1649
(3) 2549 (4) 1219
(SSC Graduate Level Tier-I
Exam. 19.05.2013, Ist Sitting)
- Directions (197-198) :** In each of the following questions, select the missing number from the given responses.
(SSC CAPFs SI & CISF ASI
Exam. 23.06.2013)

FINDING THE MISSING NUMBER

197.



- (1) 330 (2) 336
(3) 428 (4) 420

198. 12 15 36
03 04 05
04 06 04
40 66 ?

- (1) 104 (2) 320
(3) 25 (4) 84

199. If $52 + 26 + 38 = 46$
and $24 + 36 + 52 = 36$,
then $79 + 55 + 28 = ?$

- (1) 104 (2) 200
(3) 198 (4) 176

(SSC Cabinet Secretariat RO
(ECO), DFO (T) & DFO (GD) Tier-I
Exam. 23.06.2013)

200. If $3454 = 5$ and
 $2332 = 5$, then
 $5245 = ?$

- (1) 5 (2) 10
(3) 7 (4) 8

(SSC Cabinet Secretariat RO
(ECO), DFO (T) & DFO (GD) Tier-I
Exam. 23.06.2013)

201. Select the missing number from
the given matrix :

10	12	14
03	04	04
06	06	06
36	54	?

- (1) 60 (2) 61
(3) 62 (4) 63

(SSC Cabinet Secretariat RO
(ECO), DFO (T) & DFO (GD) Tier-I
Exam. 23.06.2013)

202. Select the missing number from
the given responses:

7	14	19
6	12	17
5	10	?

- (1) 20 (2) 18
(3) 15 (4) 16

(SSC Multi-Tasking (Non-Tech.)
Staff Exam. 16.02.2014)

203. Select the missing number from
the given responses.

92	70	48
64	53	42
52	45	?

- (1) 36 (2) 40
(3) 38 (4) 42

(SSC Multi-Tasking Staff
(Patna) Exam. 16.02.2014)

204. Select the missing number from
the given responses.

25	49	16
36	81	64
11	16	?

- (1) 12 (2) 20
(3) 18 (4) 13

(SSC Multi-Tasking (Non-Tech.) Staff
Exam. 23.02.2014, IIInd Sitting)

Directions (205 – 207) : In each
of the following questions, select the
missing number from the given re-
sponses.

(SSC CGL Tier-I
Re-Exam-2013, 27.04.2014)

205.

3	5	2
6	7	8
9	10	4
?	65	30
(1) 50	(2) 51	
(3) 52	(4) 49	

206.

3	4	13
8	8	56
5	3	?
(1) 4	(2) 6	
(3) 8	(4) 2	

207.

25	17	41
32	40	11
26	?	31
(1) 25	(2) 34	
(3) 38	(4) 26	

Directions (208-210) : In each
of the following questions, select the
missing number from the given re-
sponses.

(SSC CGL Tier-I
Re-Exam-2013, 27.04.2014)

208.

24	20	37
31	25	?
26	36	19
(1) 26	(2) 30	
(3) 25	(4) 23	

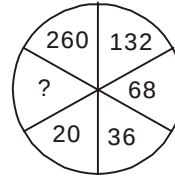
209.

6	5	4
7	6	5
5	7	6
37	23	?
(1) 10	(2) 12	
(3) 13	(4) 14	

210.

8	12	13
6	5	10
2	8	?
4	15	18
(1) 15	(2) 5	
(3) 6	(4) 3	

211. Find the missing number from the
given responses.



- (1) 12 (2) 10
(3) 9 (4) 8

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 22.06.2014)

Directions ¶ In each of the fol-
lowing questions, select the missing
number from the given responses.

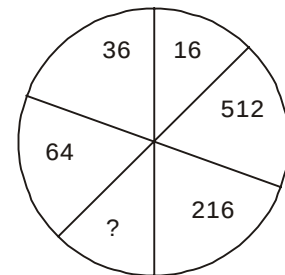
212.

7	9	8
2	4	3
5	7	6
16	32	?

- (1) 17 (2) 23
(3) 47 (4) 73

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 22.06.2014)

213. Find the missing number from the
given responses.



- (1) 128 (2) 64
(3) 72 (4) 108

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 22.06.2014)

214. 81 625 2401

3	5	7
27	125	?
105	745	2737

- (1) 287 (2) 336
(3) 385 (4) 343

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam. 22.06.2014)

Directions (215 – 217) : In each
of the following questions, find the
missing number from the given re-
sponses.

(SSC GL Tier-I Re-Exam. (2013)
20.07.2014, Ist Sitting)

| FINDING THE MISSING NUMBER |

215. 72 44 68
91 ? 86
43 66 37
(1) 33 (2) 22
(3) 11 (4) 55

216. 8 2 9
3 9 6
6 4 9
30 22 ?
(1) 63 (2) 24
(3) 52 (4) 12

217. 8 3 12
2 3 6
4 3 3
4 3 ?
(1) 5 (2) 6
(3) 7 (4) 15

Directions (218 – 220) : In each of the following questions, select the missing number from the given responses.

(SSC GL Tier-I Re-Exam. (2013)
20.07.2014, IInd Sitting)

218. 4 7 9
8 6 8
3 7 9
35 49 ?
(1) 63 (2) 89
(3) 81 (4) 64

219. 7 6 8
5 4 9
3 2 1
83 56 ?
(1) 146 (2) 128
(3) 136 (4) 148

220. 15 225 30
7 70 20
3 ? 8
(1) 70 (2) 12
(3) 16 (4) 24

Directions (221-223) : In each of the following questions, select the missing number from the given responses :

(SSC GL Tier-I Exam.
19.10.2014, Ist Sitting)

221. 96 ? 168
32 48 56
16 24 28
(1) 52 (2) 144
(3) 64 (4) 38

222. 7 8 5
6 9 ?
2 3 6
84 216 900
(1) 90 (2) 70
(2) 65 (4) 30

223. 2 3 4 ?
3 1 4 2
4 2 2 6
5 5 2 2
196 121 144 225

(1) 4 (2) 5
(3) 6 (4) 3

Directions (224-226) : In each of the following questions, select the missing number from the given responses.

(SSC GL Tier-I Exam. 19.10.2014)

224. 6 8 7
36 64 49
24 48 35
18 24 ?
(1) 17 (2) 18
(3) 19 (4) 21

225. 5 2 3
3 6 5
4 7 2
60 84 ?
(1) 10 (2) 25
(3) 30 (4) 40

226. 14 10 22
2 3 4
7 3 11
4 10 ?
(1) 2 (2) 4
(3) 6 (4) 8

Directions (227-229) % In each of the following questions, select the missing number from the given responses.

(SSC GL Tier-I Exam. 26.10.2014)

227. 8 4 7
6 9 9
48 36 ?
16 12 21
(1) 24 (2) 16
(3) 42 (4) 63

228. 3 8 7
9 6 3
2 ? 12
(1) 4 (2) 3
(3) 7 (4) 6

229. 2 4 3 2
9 7 6 5
? 33 27 21
(1) 77 (2) 35
(3) 69 (4) 80

Directions (230-232) : In each of the following questions, find the missing number from the given responses.

(SSC CHSL (10+2) DEO & LDC
Exam. 02.11.2014, Patna Region :
Ist Sitting)

230. 18 20 8
16 20 18 22 9 ?
(1) 6 (2) 7
(3) 10 (4) 11

231. 2 4 5 3
3 3 3 6 ? 6
1 2 1 6
(1) 5 (2) 2
(3) 3 (4) 1

Directions (232-234) : In each of the following questions, select the missing number from the given responses.

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, Patna Region :
Ist Sitting)

232. 1 3 7
2 4 4
4 5 9
3 2 3
50 70 ?
(1) 23 (2) 115
(3) 118 (4) 220

233. 13 9 24
11 ? 6
16 20 10
(1) 11 (2) 20
(3) 19 (4) 16

234. 9 30 21
6 ? 14
12 40 28
(1) 20 (2) 33
(3) 37 (4) 70
(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014)

Directions % In each of the following questions, select the missing number from the given responses.

(SSC CGL Tier-I Exam. 19.10.2014
TF No. 022 MH 3)

235. 7 6 6
8 6 ?
3 4 5
168 144 120
(1) 8 (2) 10
(3) 5 (4) 4

(SSC CGL Tier-I Exam. 19.10.2014
TF No. 022 MH 3)

236. 8 5 6
3 7 5
1 4 2
74 90 ?
(1) 65 (2) 85
(3) 52 (4) 76
(SSC CGL Tier-I Exam. 19.10.2014
TF No. 022 MH 3)

237. 22 46 24
27 58 31
32 68 ?
(1) 46 (2) 36
(3) 32 (4) 38
(SSC CGL Tier-I Exam. 19.10.2014
TF No. 022 MH 3)

FINDING THE MISSING NUMBER

238. Find the missing number.

$$3 * 4 = 64$$

$$2 * 3 = 9$$

$$3 * 2 = 8$$

$$9 * 2 = ?$$

(1) 216 (2) 512

(3) 128 (4) 1024

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, Ist Sitting
TF No. 333 LO 2)

239. 9 5 7

4 7 ?

7 8 3

252 280 126

(1) 8 (2) 3

(3) 2 (4) 6

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, Ist Sitting
TF No. 333 LO 2)

240. 14 19 12

13 15 ?

18 ? 16

(1) 11] 17 (2) 17] 11

(3) 17] 19 (4) 19] 17

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, Ist Sitting
TF No. 333 LO 2)

241. 5 7 9

4 8 2

8 6 ?

160 336 108

(1) 4 (2) 7

(3) 6 (4) 8

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, IInd Sitting
TF No. 545 QP 6)

242. 6 7 4

5 3 5

7 ? 6

3 3 6

(1) 7 (2) 3

(3) 8 (4) 5

(SSC CHSL (10+2) DEO & LDC
Exam. 16.11.2014, IInd Sitting
TF No. 545 QP 6)

243. 9 6 ?

8 5 6

7 4 3

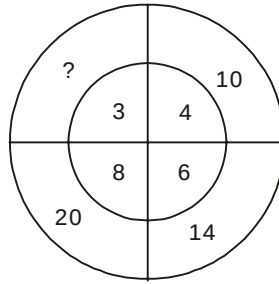
65 26 39

(1) 7 (2) 8

(3) 10 (4) 9

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam, 21.06.2015
(Ist Sitting) TF No. 8037731)

244.



(1) 19 (2) 18

(3) 24 (4) 12

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam, 21.06.2015
(Ist Sitting) TF No. 8037731)

245. 15 25 30

60 100 ?

240 400 480

(1) 125 (2) 110

(3) 120 (4) 126

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam, 21.06.2015
IInd Sitting)

246. 6 7 8

36 49 64

18 28 ?

(1) 53 (2) 48

(3) 32 (4) 40

(SSC CAPFs SI, CISF ASI & Delhi
Police SI Exam, 21.06.2015
IInd Sitting)

247. 7 5 3

8 4 9

2 8 ?

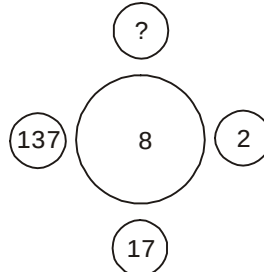
112 160 162

(1) 12 (2) 4

(3) 6 (4) 8

(SSC CGL Tier-I Exam, 09.08.2015
(Ist Sitting) TF No. 1443088)

248.



(1) 9107 (2) 97

(3) 907 (4) 1097

(SSC CGL Tier-I Exam, 09.08.2015
(Ist Sitting) TF No. 1443088)

249. 9 11 13

13 15 17

10 12 14

14 16 18

11 13 ?

(1) 21 (2) 22

(3) 14 (4) 15

(SSC CGL Tier-I Exam, 09.08.2015

(Ist Sitting) TF No. 1443088)

250. 18 21 24

3 9 3

6 4 8

21 26 ?

(1) 24 (2) 27

(3) 29 (4) 22

(SSC CGL Tier-I Exam, 09.08.2015

(IInd Sitting) TF No. 4239378)

251. I 25 15 40 8

II 65 25 90 [?]

III 45 15 60 12

(1) 18 (2) 12

(3) 24 (4) 6

(SSC CGL Tier-I Exam, 09.08.2015

(IInd Sitting) TF No. 4239378)

252. 2 4 2

3 9 3

4 16 4

8 64 ?

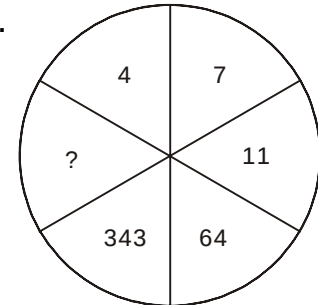
(1) 16 (2) 8

(3) 24 (4) 9

(SSC CGL Tier-I Exam, 09.08.2015

(IInd Sitting) TF No. 4239378)

253.



(1) 1321 (2) 1231

(3) 1332 (4) 1331

(SSC CGL Tier-I Exam, 16.08.2015

(Ist Sitting) TF No. 3196279)

254. I. 40 32 72 12

II. 30 24 54 9

III. 54 ? 90 15

(1) 46 (2) 49

(3) 48 (4) 36

(SSC CGL Tier-I Exam, 16.08.2015

(Ist Sitting) TF No. 3196279)

FINDING THE MISSING NUMBER

255.

1	216	343
8	125	512
27	64	?
35	401	1575

- (1) 575 (2) 729
(3) 340 (4) 615

(SSC CGL Tier-I Exam, 16.08.2015
(IInd Sitting) TF No. 2176783)

256.

2	3	8
4	5	10
6	7	12
32	50	?

- (1) 128 (2) 92
(3) 200 (4) 30

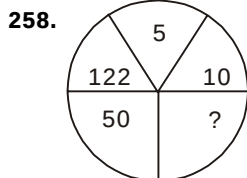
(SSC CGL Tier-I Exam, 16.08.2015
(IInd Sitting) TF No. 2176783)

257.

4	3	2	8	32
5	3	1	9	24
7	3	3	7	70
2	9	4	12	?

- (1) 84 (2) 120
(3) 27 (4) 60

(SSC CGL Tier-I Exam, 16.08.2015
(IInd Sitting) TF No. 2176783)



- (1) 23 (2) 27
(3) 25 (4) 26

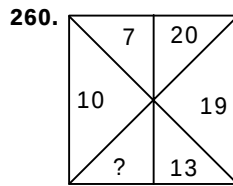
(SSC CGL Tier-I Exam, 16.08.2015
(IInd Sitting) TF No. 2176783)

259.

?	2
128	4
64	8
32	16

- (1) 240 (2) 246
(3) 250 (4) 256

(SSC CGL Tier-I
Re-Exam, 30.08.2015)



- (1) 29 (2) 39
(3) 31 (4) 41

(SSC CGL Tier-I
Re-Exam, 30.08.2015)

261.

2	3	4
4	6	8
6	9	12
36	81	?

- (1) 24 (2) 100
(3) 144 (4) 64

(SSC CGL Tier-I
Re-Exam, 30.08.2015)

262. Observing the Trend find the missing number.

5	4	3
6	7	8
4	2	?
34	30	30

- (1) 6 (2) 3
(3) 5 (4) 10

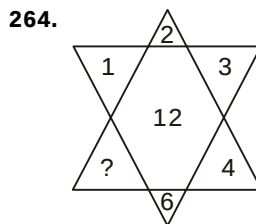
(SSC Constable (GD)
Exam, 04.10.2015, 1st Sitting)

263.

3	21	7
4	?	3
2	16	8

- (1) 12 (2) 5
(3) 10 (4) 1

(SSC (10+2) LDC/DEO/PA/SA
Exam. 01.11.2015 TF No. 1098066)



- (1) 8 (2) 1
(3) 12 (4) 6

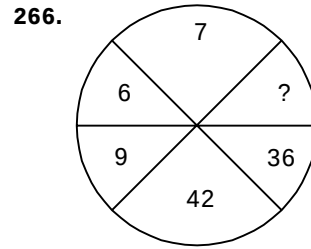
(SSC (10+2) LDC/DEO/PA/SA
Exam. 01.11.2015 TF No. 1098066)

265.

8	13	10
7	12	9
10	15	?

- (1) 5 (2) 19
(3) 12 (4) 8

(SSC (10+2) LDC/DEO/PA/SA
Exam. 01.11.2015 TF No. 1098066)



- (1) 24 (2) 54
(3) 78 (4) 34

(SSC CHSL (10+2) LDC, DEO & PA/SA
Exam, 01.11.2015, IInd Sitting)

267.

5	6	7	8
10	18	21	40
7	9	10	?

- (1) 11 (2) 20
(3) 13 (4) 15

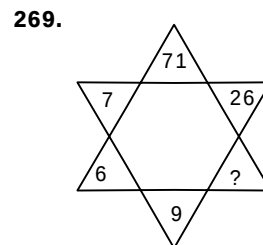
(SSC CHSL (10+2) LDC, DEO & PA/SA
Exam, 01.11.2015, IInd Sitting)

268.

65	77	87
21	?	21
44	55	66

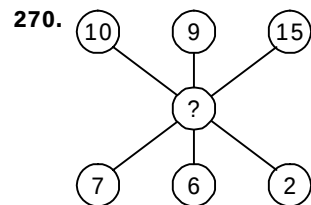
- (1) 23 (2) 21
(3) 22 (4) 20

(SSC CHSL (10+2) LDC, DEO
& PA/SA Exam, 15.11.2015
(1st Sitting) TF No. 6636838)



- (1) 35 (2) 49
(3) 39 (4) 42

(SSC CHSL (10+2) LDC, DEO
& PA/SA Exam, 15.11.2015
(1st Sitting) TF No. 6636838)



- (1) 2 (2) 5
(3) 4 (4) 3

(SSC CHSL (10+2) LDC, DEO
& PA/SA Exam, 15.11.2015
(1st Sitting) TF No. 6636838)

FINDING THE MISSING NUMBER

271.

874		
1	3	5
2	4	6
3	1	9
1	7	?

- (1) 4 (2) 6
(3) 8 (4) 2

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 15.11.2015 (IInd Sitting) TF No. 7203752)

272.

4	8	16	32
5	15	?	135
6	24	96	384

- (1) 32 (2) 45
(3) 80 (4) 30

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 15.11.2015 (IInd Sitting) TF No. 7203752)

Directions (273-275) : In each of the following questions, select the missing number from the given responses.

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 06.12.2015 (Ist Sitting) TF No. 1375232)

273.

5	4	9
6	3	?
7	2	4
65	20	45

- (1) 1 (2) 4
(3) 3 (4) 2

274.

43	48	41
42	44	?
47	?	?

- (1) 49, 45, 46 (2) 45, 49, 46
(3) 40, 48, 46 (4) 46, 40, 45

275.

?	6
58	19

- (1) 175 (2) 147
(3) 171 (4) 417

276.

?	7
42	14
21	18

- (1) 42 (2) 45
(3) 58 (4) 54

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 06.12.2015 (IInd Sitting) TF No. 3441135)

277.

?	9
33	17

- (1) 68 (2) 65
(3) 55 (4) 60

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 06.12.2015 (IInd Sitting) TF No. 3441135)

278.

121	8
144	?
64	11

- (1) 9 (2) 14
(3) 12 (4) 8

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 20.12.2015 (Ist Sitting) TF No. 9692918)

279.

3	4	5
2	3	4
1	2	3
14	29	?

- (1) 30 (2) 40
(3) 32 (4) 50

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 20.12.2015 (Ist Sitting) TF No. 9692918)

280.

113	118	?
112	?	116
?	110	115

- (1) 109, 111, 117
(2) 114, 111, 117
(3) 111, 114, 117
(4) 117, 109, 111

(SSC CHSL (10+2) LDC, DEO & PA/SA Exam, 20.12.2015 (Ist Sitting) TF No. 9692918)

281.

?	8	30
80	?	90
3	?	?

- (1) 40 (2) 10
(3) 20 (4) 15

(SSC (10+2) Stenographer Grade 'C' & 'D' Exam. 31.01.2016 TF No. 3513283)

282.

594	198
?	66

- (1) 22 (2) 33
(3) 11 (4) 12

(SSC (10+2) Stenographer Grade 'C' & 'D' Exam. 31.01.2016 TF No. 3513283)

283.

27	7	14
3	4	5
75	165	?
(1) 185	(2) 285	
(3) 165	(4) 425	

(SSC (10+2) Stenographer Grade 'C' & 'D' Exam. 31.01.2016 TF No. 3513283)

Directions (284-285) : In each of the following questions, select the missing number from the given responses.

(SSC (10+2) Stenographer Grade 'C' & 'D' Exam. 31.07.2016)

284.

1	4	9	16
1	2	3	4
2	4	6	?
(1) 7	(2) 5		
(3) 4	(4) 8		

285.

121	100	?	169
4	8	9	7
7	2	5	6
(1) 114	(2) 196		
(3) 214	(4) 81		

286. Select the missing number from the given responses

43	25	?
21	40	35
35	34	28

- (1) 36 (2) 46
(3) 40 (4) 26

(SSC CGL Tier-I (CBE) Exam.10.09.2016)

| FINDING THE MISSING NUMBER |

- 287.** Select the missing number from the given responses :

7	14	4
4	12	9
6	24	?

- (1) 14 (2) 16
(3) 21 (4) 22

(SSC CGL Tier-I (CBE)
Exam.11.09.2016) (Ist Sitting)

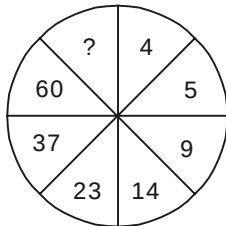
- 288.** Select the missing number from the given responses.

9	8	5
6	6	7
2	?	7
3	6	5

- (1) 9 (2) 4
(3) 6 (4) 8

(SSC CPO SI, ASI Online
Exam.05.06.2016) (IInd Sitting)

- 289.** Select the missing number from the given responses



- (1) 87 (2) 97
(3) 90 (4) 82

(SSC CPO SI, ASI Online
Exam.05.06.2016) (IInd Sitting)

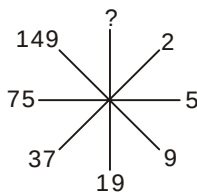
- 290.** Select the missing number from the given responses.

14	26	17
2	2	?
3	6	5
4	4	3

- (1) 1 (2) 6
(3) 4 (4) 2

(SSC CPO SI, ASI Online
Exam.05.06.2016) (IInd Sitting)

- 291.** Select the missing number in the pattern from the given responses.



- (1) 298 (2) 299
(3) 499 (4) 199

(SSC CPO Exam. 06.06.2016)
(Ist Sitting)

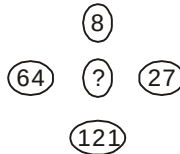
- 292.** Select the missing number in the pattern from the given responses.

1	3	28
3	7	58
2	5	133
4	9	?

- (1) 97 (2) 88
(3) 91 (4) 106

(SSC CPO Exam. 06.06.2016)
(Ist Sitting)

- 293.** Select the missing number in the pattern from the given responses.



- (1) 22 (2) 28
(3) 20 (4) 88

(SSC CPO Exam. 06.06.2016)
(Ist Sitting)

- 294.** Select the missing number from the given responses:

6	5	26
4	7	32
?	9	44

- (1) 8 (2) 31
(3) 32 (4) 36

(SSC CHSL (10+2) Tier-I (CBE)
Exam. 08.09.2016) (Ist Sitting)

- 295.** Select the missing number from the given alternatives

9	11	13
3	4	7
3	4	5
81	176	?

- (1) 143 (2) 169
(3) 455 (4) 545

(SSC CGL Tier-I (CBE)
Exam. 09.09.2016) (Ist Sitting)

- Directions (296-299) :** In each of the following questions, select the missing number from the given responses.

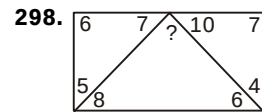
(SSC CAPFs (CPO) SI & ASI, DP
Exam. 20.03.2016) (IInd Sitting)

- 296.** 25 36 64
81 9 4
16 49 100
18 ? 20

- (1) 14 (2) 22
(3) 16 (4) 19

- 297.** 6 5 3 10
2 8 ? 4
4 6 3 8
5 9 15 3

- (1) 7 (2) 6
(3) 4 (4) 5



- (1) 8 (2) 7
(3) 5 (4) 6

- 299.** 6 11 25
8 6 16
12 5 ?

- (1) 18 (2) 16
(3) 12 (4) 22

- 300.** Select the missing number from the given responses.

- 8 5 4
7 6 8
12 20 12
44 10 ?

- (1) 40 (2) 30
(3) 20 (4) 35

(SSC CAPFs (CPO) SI & ASI,
Delhi Police Exam. 05.06.2016)
(Ist Sitting)

- 301.** Select the missing number from the given responses.

- 121 81 49
100 64 36
15 16 25
? 31 41

- (1) 11 (2) 289
(3) 24 (4) 15

(SSC CPO SI & ASI, Online
Exam. 06.06.2016) (IInd Sitting)

- 302.** Select the missing number from the given responses.

- 2 3 4
8 6 9
6 3 5
10 ? 29

- (1) 12 (2) 14
(3) 16 (4) 9

(SSC CPO SI & ASI, Online
Exam. 06.06.2016) (IInd Sitting)

FINDING THE MISSING NUMBER

- 303.** In this question, select the missing number from the given alternatives.

6	15	20
8	4	5
3	5	20
51	65	?

- (1) 56 (2) 120
(3) 151 (4) 154

(SSC CGL Tier-I (CBE)

Exam. 27.08.2016) (Ist Sitting)

- 304.** Select the missing number from the given responses

21	56	70
45	87	84
115	180	?

- (1) 110 (2) 120
(3) 130 (4) 140

(SSC CGL Tier-I (CBE)

Exam. 28.08.2016) (IInd Sitting)

- 305.** Select the missing number from the given responses.

24	20	36
15	11	18
55	40	?

- (1) 45 (2) 65
(3) 70 (4) 80

(SSC CGL Tier-I (CBE)

Exam. 29.08.2016) (IInd Sitting)

- 306.** Select the missing numbers from the given alternatives

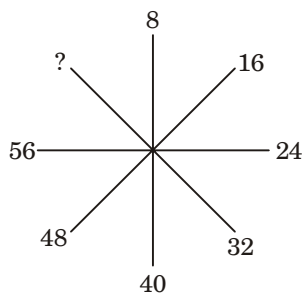
8	32	4
7	?	5
2	6	3

- (1) 30 (2) 35
(3) 20 (4) 25

(SSC CGL Tier-I (CBE)

Exam. 01.09.2016) (Ist Sitting)

- 307.** Find the missing number from the given alternatives.



- (1) 60 (2) 62
(3) 64 (4) 66

(SSC CGL Tier-I (CBE)

Exam. 02.09.2016) (Ist Sitting)

- 308.** Select the missing number from the given responses :

3	5	9
4	6	?
3	7	2
36	210	36

- (1) 4 (2) 3
(3) 2 (4) 1

(SSC CGL Tier-I (CBE)

Exam. 03.09.2016) (IInd Sitting)

- 309.** Select the missing number from the given responses:

7	56	8
9	45	?
11	99	9

- (1) 6 (2) 9
(3) 5 (4) 4

(SSC CGL Tier-I (CBE)

Exam. 04.09.2016) (Ist Sitting)

- 310.** Select the missing numbers from the given alternatives :

12	8	4
8	4	12
4	12	8
100	44	?

- (1) 56 (2) 48
(3) 38 (4) 36

(SSC CGL Tier-I (CBE)

Exam. 06.09.2016) (Ist Sitting)

- 311.** Select the missing numbers from the given alternatives :

17	11	19
12	13	16
25	4	?

- (1) 9 (2) 16
(3) 25 (4) 36

(SSC CGL Tier-I (CBE)

Exam. 07.09.2016) (Ist Sitting)

- 312.** In the following question, a series is given with one term missing. Choose the correct alternative from the given ones that will complete the series.

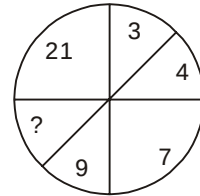
1	2	3
11	7	5
120	45	?

- (1) 19 (2) 16
(3) 15 (4) 17

(SSC CGL Tier-I (CBE)

Exam. 30.08.2016) (IInd Sitting)

- 313.** Find the missing number.



- (1) 15 (2) 12
(3) 11 (4) 16

(SSC CGL Tier-I (CBE)

Exam. 01.09.2016) (Ist Sitting)

- 314.** Select the missing number from the given alternatives :

28	20	7
84	35	12
45	?	9

- (1) 19 (2) 22
(3) 26 (4) 25

(SSC CGL Tier-I (CBE)

Exam. 02.09.2016) (Ist Sitting)

- 315.** Select the missing number from the given alternatives :

4	3	6
3	5	4
2	2	3
14	16	?

- (1) 12 (2) 18
(3) 20 (4) 30

(SSC CGL Tier-I (CBE)

Exam. 28.08.2016) (Ist Sitting)

- 316.** Find the missing number from the given responses :

7	3	2
4	9	6
2	1	5
69	91	?

- (1) 58 (2) 65
(3) 64 (4) 51

(SSC CGL Tier-I (CBE)

Exam. 31.08.2016) (IInd Sitting)

- 317.** Find the missing number from the given alternatives :

8	7	6
7	6	5
6	5	4
90	65	?

| FINDING THE MISSING NUMBER |

- (1) 54 (2) 44
(3) 34 (4) 26

(SSC CGL Tier-I (CBE)

Exam. 01.09.2016) (IInd Sitting)

318. Find the missing number from the given responses :

6	15	20
8	4	5
3	5	20
51	65	?

- (1) 12 (2) 51
(3) 56 (4) 120

(SSC CGL Tier-I (CBE)

Exam. 02.09.2016) (IInd Sitting)

319. Find the missing term in the following question :

?	8
64	27

- (1) 3 (2) 9
(3) 1 (4) 7

(SSC CGL Tier-I (CBE)

Exam. 03.09.2016) (IIInd Sitting)

320. Select the missing number from the given alternatives :

8	21	34
27	?	53

- (1) 35 (2) 40
(3) 17 (4) 23

(SSC CGL Tier-I (CBE)

Exam. 04.09.2016) (IIInd Sitting)

321. Find the missing number from the given alternatives

86	480	38
72	510	21
36	?	12

- (1) 120 (2) 360
(3) 100 (4) 240

(SSC CGL Tier-I (CBE)

Exam. 06.09.2016) (IIInd Sitting)

322. Find the missing number in the following question :

8	6	7
9	13	11
3	7	?

- (1) 5 (2) 6
(3) 7 (4) 8

(SSC CGL Tier-I (CBE)

Exam. 07.09.2016) (IIInd Sitting)

323. Find the missing term in the following question :

3	4	9
2	5	2
6	7	?
36	140	18

- (1) 1 (2) 7
(3) 12 (4) 19

(SSC CGL Tier-I (CBE)

Exam. 08.09.2016) (IIInd Sitting)

324. Select the missing number from the given matrix :

5	2	4
4	4	7
2	5	3
18	30	?

- (1) 43 (2) 42
(3) 33 (4) 32

(SSC CGL Tier-I (CBE)

Exam. 09.09.2016) (IIInd Sitting)

325. Select the missing number from the given alternatives :

?	120	150
110	100	90
80	60	40

- (1) 60 (2) 70
(3) 80 (4) 90

(SSC CGL Tier-I (CBE)

Exam. 10.09.2016) (IIInd Sitting)

326. Find the missing number from the given alternatives :

7	10	5
16	40	8
15	?	9

- (1) 75 (2) 45
(3) 20 (4) 30

(SSC CGL Tier-I (CBE)

Exam. 11.09.2016) (IIInd Sitting)

327. Select the missing number from the given alternatives :

7	14	4
4	12	9
6	24	?

- (1) 19 (2) 16
(3) 21 (4) 20

(SSC CGL Tier-I (CBE)

Exam. 03.09.2016) (IIInd Sitting)

328. Find the missing number from the given alternatives :

2	4	0
1	2	4
3	1	3
36	?	91

- (1) 73 (2) 68
(3) 101 (4) 55

(SSC CGL Tier-I (CBE)

Exam. 04.09.2016) (IIInd Sitting)

329. Select the missing number from the given responses :

2	4	2	4
5	5	3	7
8	6	4	?

- (1) 2 (2) 3
(3) 10 (4) 11

(SSC CGL Tier-I (CBE)

Exam. 06.09.2016) (IIInd Sitting)

330. Select the missing number from the given alternatives :

8	13	72	117
21	7	189	63
4	6	36	?

- (1) 65 (2) 54
(3) 72 (4) 110

(SSC CGL Tier-I (CBE)

Exam. 07.09.2016) (IIInd Sitting)

331. Find the missing term in the following question :

6	15	20
8	4	5
3	5	20
51	65	?

- (1) 56 (2) 12
(3) 120 (4) 51

(SSC CGL Tier-I (CBE)

Exam. 08.09.2016) (IIInd Sitting)

332. Select the missing number from the given alternatives :

7	9	8
8	9	?
4	9	6
60	90	70

- (1) 9 (2) 8
(3) 7 (4) 6

(SSC CGL Tier-I (CBE)

Exam. 09.09.2016) (IIInd Sitting)

FINDING THE MISSING NUMBER

- 333.** Select the missing number from the given responses :

5	5	15
6	8	6
8	12	?
240	480	450

- (1) 15 (2) 12
(3) 8 (4) 5

(SSC CGL Tier-I (CBE)

Exam. 10.09.2016) (IInd Sitting)

- 334.** Find the missing term in the following question :

10	6	9
18	14	17
28	34	?
20	26	32

- (1) 25 (2) 37
(3) 40 (4) 57

(SSC CGL Tier-I (CBE)

Exam. 11.09.2016) (IInd Sitting)

- 335.** Select the missing numbers from the given alternatives :

4	8	16
32	64	128
?	512	1024

- (1) 468 (2) 456
(3) 268 (4) 256

(SSC CGL Tier-I (CBE)

Exam. 27.10.2016) (Ist Sitting)

- 336.** Select the missing number from the given alternatives :

6	18	15
3	2	5
4	3	?
8	27	9

- (1) 11 (2) 6
(3) 3 (4) 2

(SSC CGL Tier-I (CBE)

Exam. 27.10.2016) (IInd Sitting)

- 337.** In the following question, select the missing number from the given series :

24	30	19
5	7	18
9	1	?

- (1) 4 (2) 1
(3) 5 (4) 6

(SSC CHSL (10+2) Tier-I (CBE)

Exam. 15.01.2017) (IInd Sitting)

- 338.** In the following question, select the missing number from the given alternatives.

6	8	12
7	9	14
85	145	?

- (1) 175 (2) 450
(3) 340 (4) 740

(SSC CHSL (10+2) Tier-I (CBE)

Exam. 16.01.2017) (IInd Sitting)

- 339.** Find the missing number from the given alternatives :

7	10	5
16	40	8
15	?	9

- (1) 75 (2) 45
(3) 20 (4) 30

(SSC CGL Tier-I (CBE)

Exam. 11.09.2016) (Ist Sitting)

- 340.** Find the missing number from the given responses.

8	7	9
4	5	6
9	6	?
288	210	162

- (1) 4 (2) 8
(3) 7 (4) 3

(SSC Multi-Tasking Staff

Exam. 30.04.2017 Ist Sitting)

ANSWERS

TYPE-I

1. (2)	2. (2)	3. (2)	4. (1)
5. (1)	6. (2)	7. (1)	8. (3)
9. (2)	10. (2)	11. (3)	12. (2)
13. (3)	14. (2)	15. (3)	16. (2)
17. (3)	18. (2)	19. (2)	20. (4)
21. (4)	22. (2)	23. (4)	24. (2)
25. (4)	26. (3)	27. (2)	28. (2)
29. (4)	30. (4)	31. (3)	32. (3)
33. (3)	34. (3)	35. (4)	36. (3)
37. (4)	38. (2)	39. (4)	40. (2)
41. (2)	42. (1)	43. (2)	44. (2)
45. (2)	46. (3)	47. (1)	48. (3)
49. (1)	50. (2)	51. (2)	52. (3)
53. (2)	54. (3)	55. (4)	56. (2)

57. (1)	58. (3)	59. (1)	60. (3)
61. (1)	62. (1)	63. (1)	64. (2)
65. (3)	66. (2)	67. (2)	68. (2)
69. (3)	70. (2)	71. (2)	72. (2)
73. (2)	74. (2)	75. (3)	76. (4)
77. (3)	78. (2)	79. (3)	80. (3)
81. (3)	82. (3)	83. (1)	84. (2)
85. (4)	86. (2)	87. (2)	88. (2)
89. (4)	90. (4)	91. (3)	92. (1)
93. (1)	94. (3)	95. (1)	96. (1)
97. (2)	98. (4)	99. (4)	100. (4)
101. (3)	102. (1)	103. (4)	104. (3)
105. (1)	106. (3)	107. (1)	108. (2)
109. (3)	110. (4)	111. (2)	112. (3)
113. (4)			

TYPE-II

1. (2)	2. (2)	3. (2)	4. (2)
5. (3)	6. (2)	7. (2)	8. (4)
9. (1)	10. (3)	11. (4)	12. (3)
13. (3)	14. (3)	15. (4)	16. (3)
17. (4)	18. (1)	19. (4)	20. (1)
21. (1)	22. (3)	23. (1)	24. (4)
25. (3)	26. (3)	27. (2)	28. (2)
29. (4)	30. (4)	31. (3)	32. (3)
33. (3)	34. (4)	35. (1)	36. (1)
37. (3)	38. (3)	39. (2)	40. (2)
41. (3)	42. (1)	43. (4)	44. (2)
45. (4)	46. (1)	47. (4)	48. (2)
49. (2)	50. (2)	51. (3)	52. (1)
53. (3)	54. (2)	55. (3)	56. (4)
57. (1)	58. (3)	59. (1)	60. (2)
61. (2)	62. (3)	63. (2)	64. (2)
65. (1)	66. (3)	67. (2)	68. (2)
69. (2)	70. (2)	71. (2)	72. (4)
73. (4)	74. (1)	75. (1)	76. (2)
77. (2)	78. (1)	79. (2)	80. (1)
81. (3)	82. (1)	83. (2)	84. (4)
85. (2)	86. (2)	87. (1)	88. (2)
89. (2)	90. (3)	91. (2)	92. (3)
93. (3)	94. (3)	95. (2)	96. (4)
97. (4)	98. (4)	99. (1)	100. (2)
101. (2)	102. (4)	103. (3)	104. (2)

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105. (2)	106. (3)	107. (2)	108. (2)
109. (3)	110. (4)	111. (2)	112. (4)
113. (2)	114. (4)	115. (2)	116. (2)
117. (3)	118. (1)	119. (1)	120. (3)
121. (4)	122. (2)	123. (4)	124. (1)
125. (3)	126. (1)	127. (3)	128. (2)
129. (2)	130. (1)	131. (2)	132. (2)
133. (2)	134. (2)	135. (1)	136. (3)
137. (3)	138. (3)	139. (4)	140. (2)
141. (2)	142. (1)	143. (4)	144. (4)
145. (2)	146. (2)	147. (1)	148. (2)
149. (2)	150. (2)	151. (4)	152. (3)
153. (1)	154. (1)	155. (2)	156. (2)
157. (1)	158. (3)	159. (1)	160. (2)
161. (1)	162. (4)	163. (1)	164. (3)
165. (2)	166. (3)	167. (1)	168. (1)
169. (3)	170. (1)	171. (2)	172. (1)
173. (1)	174. (1)	175. (1)	176. (2)
177. (4)	178. (1)	179. (2)	180. (4)
181. (4)	182. (4)	183. (1)	184. (2)
185. (2)	186. (1)	187. (3)	188. (1)
189. (4)	190. (4)	191. (2)	192. (1)
193. (4)	194. (1)	195. (4)	196. (2)
197. (2)	198. (4)	199. (1)	200. (1)
201. (3)	202. (3)	203. (3)	204. (1)
205. (2)	206. (3)	207. (4)	208. (3)
209. (4)	210. (1)	211. (1)	212. (2)
213. (2)	214. (4)	215. (4)	216. (1)
217. (2)	218. (3)	219. (1)	220. (2)
221. (2)	222. (4)	223. (2)	224. (4)
225. (3)	226. (4)	227. (4)	228. (1)
229. (1)	230. (2)	231. (3)	232. (2)
233. (1)	234. (1)	235. (4)	236. (1)
237. (2)	238. (2)	239. (4)	240. (1)
241. (3)	242. (3)	243. (1)	244. (3)
245. (3)	246. (4)	247. (3)	248. (4)
249. (4)	250. (3)	251. (1)	252. (2)
253. (4)	254. (4)	255. (2)	256. (3)
257. (1)	258. (4)	259. (4)	260. (2)
261. (3)	262. (1)	263. (1)	264. (3)
265. (3)	266. (2)	267. (3)	268. (3)
269. (1)	270. (1)	271. (1)	272. (2)
273. (1)	274. (4)	275. (1)	276. (4)

277. (2)	278. (3)	279. (4)	280. (3)
281. (2)	282. (1)	283. (2)	284. (4)
285. (2)	286. (1)	287. (2)	288. (3)
289. (2)	290. (4)	291. (2)	292. (1)
293. (3)	294. (1)	295. (3)	296. (3)
297. (3)	298. (2)	299. (4)	300. (3)
301. (2)	302. (4)	303. (2)	304. (3)
305. (2)	306. (2)	307. (3)	308. (3)
309. (3)	310. (1)	311. (1)	312. (2)
313. (2)	314. (4)	315. (4)	316. (2)
317. (2)	318. (4)	319. (3)	320. (2)
321. (4)	322. (1)	323. (1)	324. (3)
325. (4)	326. (4)	327. (2)	328. (1)
329. (3)	330. (2)	331. (3)	332. (2)
333. (4)	334. (3)	335. (4)	336. (3)
337. (2)	338. (3)	339. (4)	340. (4)

EXPLANATIONS

TYPE-I

$$\begin{array}{c}
 1 \quad 9 \rightarrow (3)^2 \\
 \swarrow \quad \searrow \\
 25 \rightarrow (5)^2
 \end{array}$$

$$\begin{array}{c}
 4 \quad 16 \rightarrow (4)^2 \\
 \swarrow \quad \searrow \\
 36 \rightarrow (6)^2
 \end{array}$$

Similarly,

$$\begin{array}{c}
 9 \quad 25 \rightarrow (5)^2 \\
 \swarrow \quad \searrow \\
 ? \rightarrow (7)^2 = 49
 \end{array}$$

2. (2) First figure
 $\Rightarrow 15 + 12 = 27$
 and, $27 \div 9 = 3$
 Second figure $\Rightarrow 44 + 28 = 72$
 and, $72 \div 9 = 8$
 Similarly, in third figure
 $64 + 53 = 117$
 and, $117 \div 9 = \boxed{13}$

3. (2) First of all take the product of all the numbers given outside the square and divide it by 10 to get the number inside the square.

1st Figure

$$5 \times 3 \times 4 \times 2 = 120$$

$$120 \div 10 = 12$$

2nd Figure

$$5 \times 6 \times 2 \times 3 = 180$$

$$180 \div 10 = 18$$

3rd Figure

$$5 \times 2 \times 2 \times 9 = 180$$

$$180 \div 10 = \boxed{18}$$

4. (1) First figure

$$15 + 16 = 22 + 9$$

$$\text{or, } 31 = 31$$

Second figure

$$13 + 7 = 11 + 9$$

$$\text{or, } 20 = 20$$

Third figure

$$21 + 15 = ? + 13$$

$$\text{or, } ? = 36 - 13 = \boxed{23}$$

5. (1) First arrangement

$$(56 + 15) - (22 + 8)$$

$$\Rightarrow 71 - 30 = 41$$

Second arrangement

$$(46 + 9) - (10 + 6)$$

$$\Rightarrow 55 - 16 = 39$$

Third arrangement

$$(34 + 11) - (14 + 6)$$

$$\Rightarrow 45 - 20 = \boxed{25}$$

6. (2) $5 \times 6 + 3 \times 3 = 39$
 and, $7 \times 5 + 4 \times 4 = 51$

$$\therefore 5 \times 5 + 3 \times 4 = \boxed{37}$$

7. (1) The numbers located outside the circle are perfect squares and the number inside the circle is decreasing by one.

8. (3) The central number is the sum of square roots of all the four numbers located at the corners in the given arrangement.

1st Figure

$$\sqrt{4} = 2; \quad \sqrt{16} = 4,$$

$$\sqrt{9} = 3; \quad \sqrt{25} = 5$$

$$\text{Now, } 2 + 4 + 3 + 5 = 14$$

2nd Figure

$$\sqrt{9} = 3; \quad \sqrt{49} = 7$$

$$\sqrt{36} = 6; \quad \sqrt{1} = 1$$

$$\text{Now, } 3 + 7 + 6 + 1 = \boxed{17}$$

| FINDING THE MISSING NUMBER |

9. (2) The lower number is twice the difference of the first two numbers.

1st Figure

$$466 - 341 = 125$$

$$\text{and } 125 \times 2 = 250$$

2nd Figure

$$398 - 282 = 116$$

$$\text{and } 116 \times 2 = 232$$

10. (2) Take the sum of all the four numbers to get the central number.

1st Figure

$$17 + 12 + 8 + 52 = 89$$

2nd Figure

$$24 + 33 + 49 + 61 = 167$$

Similarly,

$$61 + 33 + 15 + 121 = \boxed{230}$$

11. (3) $(1)^2 + (2)^2 + (3)^2 + (4)^2$
 $= 1 + 4 + 9 + 16 = 30$

Therefore,

$$(3)^2 + (6)^2 + (5)^2 + (4)^2$$

$$= 9 + 36 + 25 + 16 = \boxed{86}$$

12. (2) $9 \times 3 = 27$ and $9 \times 6 = 54$
 $14 \times 3 = 42$ and $14 \times 6 = 81$
 Similarly,

$$7 \times 3 = \boxed{21} \text{ and } 7 \times 6 = 42$$

13. (3) $\frac{13+19}{8} = \frac{32}{8} = 4 \rightarrow$ Lower number

$$\frac{71+9}{8} = 10$$

Therefore,

$$\frac{128+32}{8} = \frac{160}{8} = \boxed{20}$$

14. (2) **First figure**

$$(6 \times 5) + (3 \times 3) = 30 + 9 = 39$$

Second figure

$$(7 \times 5) + (4 \times 4) = 35 + 16 = 51$$

Third figure

$$(5 \times 5) + (3 \times 4) = 25 + 12 = \boxed{37}$$

15. (3) $22 + 42 = 64$

$$27 + 52 = 79$$

$$\text{Therefore, } ? = 91 - 18 = 73$$

16. (2) Align $3 + 18 = 21$

$$4 + 23 = 27$$

$$? + 27 = 33$$

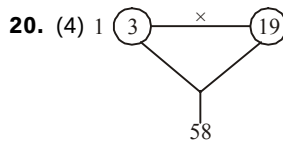
$$\therefore ? = 33 - 27 = 6$$

17. (3) $(5 \times 4) + (3 \times 1) = 23$
 $(7 \times 6) + (3 \times 4) = 54$
 $(11 \times 2) + (? \times 9) = 40$
 or, $? \times 9 = 40 - 22 = 18$

$$\therefore ? = \frac{18}{9} = 2$$

18. (2) The upper numbers are multiples of the lower number.

19. (2) $(7 + 9 + 5 + 4) \times 2 - 10 = 40$
 $(17 + 8 + 3 + 6) \times 2 - 14 = 54$
 $(10 + 21 + 6 + 3) \times 2 - 18 = 62$



$$\Rightarrow 3 \times 19 = 57 \text{ and } 57 + 1 = 58$$

$$81 \times 7 + 1 = 568$$

Similarly,

$$93 \times 1 + 7 = \boxed{100}$$

21. (4) $(6 \times 4) + (10 \times 8)$
 $= 24 + 80 = 104$ $(7 \times 8) + (9 \times 11)$
 $= 56 + 99 = 155$

Similarly,

$$(8 \times 9) + (5 \times 14) = 72 + 70 = \boxed{142}$$

22. (2) $4 + 2 + 3 + 7 = 16$ and
 $1 + 6 = 7$

$$1 + 7 = 8 \text{ and } 8 - 7 = 1$$

$$1 + 1 + 8 + 4 = 14 \text{ and}$$

$$1 + 4 = 5$$

$$1 + 3 = 4 \text{ and } 5 - 4 = 1$$

$$7 + 4 + 9 + 3 = 23 \text{ and}$$

$$2 + 3 = 5$$

$$4 + 2 = 6 \text{ and } 6 - 5 = 1$$

23. (4) $(0)^2 + (3)^2 + (1)^2 + (2)^2$

$$= 0 + 9 + 1 + 4 = 14$$

$$(3)^2 + (5)^2 + (2)^2 + (4)^2$$

$$= 9 + 25 + 4 + 16 = 54$$

$$(3)^2 + (1)^2 + (6)^2 + (5)^2$$

$$= 9 + 1 + 36 + 25 = \boxed{71}$$

24. (2) $8 \times 2 = 16$ and $8 \times 4 = 32$

$$9 \times 2 = 18 \text{ and } 9 \times 4 = 36$$

$$10 \times 2 = 20 \text{ and } 10 \times 4 = 40$$

25. (4) $(18)^2 + (15)^2$

$$= 324 + 225 = 549$$

$$(17)^2 + (19)^2 = 289 + 361 = 650$$

Similarly,

$$(15)^2 + (14)^2 = 225 + 196 = 421$$

26. (3) $13 \times 17 = 221$

$$12 \times 19 = 228$$

$$13 \times 18 = 234$$

27. (2) $5 \times 3 + 1 = 16$

$$16 \times 3 + 1 = 49$$

$$9 \times 3 + 2 = 29$$

$$29 \times 3 + 2 = 89$$

Therefore,

$$15 \times 3 + 3 = \boxed{48}$$

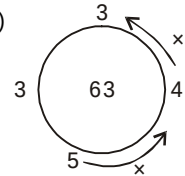
$$48 \times 3 + 3 = 147$$

28. (2) $4 \times 2 \times 3 \times 3 = 75$

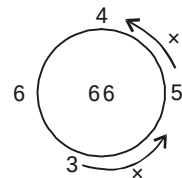
$$9 \times 4 \times 2 \times 10 = 720$$

$$6 \times 20 \times 1 \times 6 = \boxed{720}$$

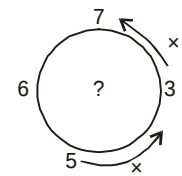
29. (4)



$$5 \times 4 \times 3 = 60; 60 + 3 = 63$$



$$3 \times 5 \times 4 = 60; 60 + 6 = 66$$



$$5 \times 3 \times 7 = 105;$$

$$105 + 6 = 111$$

30. (4) The lower number is the sum of the upper two numbers. Thus,
 $21 + 31 = 52$

$$39 + 46 = 85$$

Therefore,

$$16 + 83 = \boxed{99}$$

31. (3) First of all add the numbers situated at the periphery and then subtract 2 from the sum so obtained get the central number.

1st Figure

$$0 + 6 + 4 + 2 = 12$$

$$\text{and } 12 - 2 = 10$$

2nd Figure

$$2 + 10 + 8 + 6 = 26$$

$$\text{and } 26 - 2 = 24$$

3rd Figure

$$4 + 14 + 12 + 10 = 40$$

$$\text{and } 40 - 2 = 38$$

- 32.** (3) The sum of upper two numbers is equal to the lower number.

$$22 + 10 = 32$$

$$14 + 74 = 88$$

Similarly,

$$33 + 26 = \boxed{59}$$

- 33.** (3) The product of the two diagonally opposite numbers is equal to the central number.

$$12 \times 5 = 60 \text{ and } 15 \times 4$$

$$= 60$$

$$3 \times 14 = 42 \text{ and } 7 \times 6 = 42$$

Similarly,

$$13 \times 6 = \boxed{78} \text{ and } 26 \times 3 = \boxed{78}$$

- 34.** (3) In each arrangement the product of the upper two numbers is equal to the bottom number.

1st arrangement

$$5 \times 4 = 20$$

2nd arrangement

$$8 \times 8 = 64$$

3rd arrangement

$$6 \times 6 = \boxed{36}$$

- 35.** (4) In each arrangement the central number is equal to the sum of the products of diagonally opposite numbers.

1st arrangement

$$8 \times 8 = 64 \text{ and } 7 \times 5 = 35$$

$$\text{Now, } 64 + 35 = 99$$

2nd arrangement

$$3 \times 3 = 9 \text{ and } 6 \times 9 = 54$$

$$\text{Now, } 9 + 54 = 63$$

3rd arrangement

$$17 \times 7 = 119 \text{ and } 9 \times 9 = 81$$

$$\text{Now, } 119 + 81 = \boxed{200}$$

- 36.** (3) Subtract the upper left number from the upper right number to get the lower number.

$$30 - 24 = 6$$

$$43 - 30 = 13$$

Similarly,

$$62 - 43 = \boxed{19}$$

- 37.** (4) $13 \times 2 = 26$

$$26 \times 2 = \boxed{52}$$

$$24 \times 2 = 48$$

$$48 \times 2 = 96$$

$$16 \times 2 = 32$$

$$32 \times 2 = 64$$

- 38.** (2) Take the sum of all the four numbers located at the four corners to get the central number.

1st arrangement :

$$3 + 6 + 4 + 7 = 20$$

2nd arrangement :

$$5 + 8 + 6 + 4 = 23$$

3rd arrangement :

$$11 + 7 + 9 + 9 = \boxed{36}$$

- 39.** (4)
$$\begin{array}{rcl} 2 & - & 1 = \boxed{1} \\ 6 & - & 3 = \boxed{3} \\ 5 & - & 4 = \boxed{1} \end{array}$$

$$4 - 2 = \boxed{2}$$

$$6 - 2 = \boxed{4}$$

$$8 - 0 = \boxed{8}$$

Similarly,

$$7 - 5 = \boxed{2}$$

$$9 - 3 = \boxed{6}$$

$$3 - 1 = \boxed{2}$$

$$\therefore ? = 262$$

- 40.** (2) The sum of the top numbers is equal to the bottom number.

1st Figure

$$42 + 39 = 81$$

2nd Figure

$$22 + 36 = 58$$

3rd Figure

$$17 + 43 = \boxed{60}$$

- 41.** (2) Add the four numbers located at the four corners to get the central number in each figure.

1st Figure

$$9 + 5 + 92 + 15 = 121$$

2nd Figure

$$16 + 19 + 20 + 24 = 79$$

3rd Figure

$$56 + 7 + 9 + 8 = \boxed{80}$$

- 42.** (1) **1st Figure**

$$18 \times 2 = 36; 18 \times 5 = 90$$

2nd Figure

$$13 \times 2 = 26; 13 \times 5 = 65$$

3rd Figure

$$15 \times 2 = \boxed{30}; 15 \times 5 = 75$$

- 43.** (2) The sum of the upper two number is equal to the bottom number.

1st Arrangement

$$15 + 19 = 34$$

2nd Arrangement

$$49 + 13 = 62$$

3rd Arrangement

$$36 + 26 = 62$$

- 44.** (2) The central number is equal to the sum of the products of diagonally opposite numbers.

1st Arrangement

$$3 \times 2 = 6 \text{ and } 5 \times 4 = 20$$

$$\text{And } 6 + 20 = 26$$

2nd Arrangement

$$4 \times 3 = 12 \text{ and } 6 \times 5 = 30$$

$$\text{And, } 12 + 30 = 42$$

3rd Arrangement

$$4 \times 5 = 20 \text{ and } 7 \times 6 = 42$$

$$\text{And, } 20 + 42 = 62$$

- 45.** (2) Take the sum of the products of the diagonally opposite numbers to get the central number.

1st Figure

$$(5 \times 4) + (3 \times 2)$$

$$\text{or, } 20 + 6 = 26$$

2nd Figure

$$(8 \times 4) + (6 \times 5)$$

$$\text{or, } 32 + 30 = 62$$

3rd Figure

$$(3 \times 8) + (7 \times 5)$$

$$\text{or, } 24 + 35 = \boxed{59}$$

- 46.** (3) **1st Figure**

$$4 \times 2 = 8$$

$$4 \times 4 = 16$$

$$4 \times 8 = 32$$

$$4 \times 16 = 64$$

2nd Figure

$$5 \times 2 = 10$$

$$5 \times 4 = 20$$

$$5 \times 8 = 40$$

$$5 \times 16 = 80$$

3rd Figure

$$2.5 \times 2 = 5$$

$$2.5 \times 4 = 10$$

$$2.5 \times 8 = 20$$

$$2.5 \times 16 = 40$$

- 47.** (1) $3 \times \frac{2}{2} = 3 \times 1 = 3$

$$5 \times \frac{4}{2} = 5 \times 2 = 10$$

$$7 \times \frac{6}{2} = 7 \times 3 = 21$$

48. (3) The product of all the four numbers located at the periphery is equal to the central number.

First Arrangement

$$1 \times 4 \times 3 \times 2 = 24$$

Second Arrangement

$$2 \times 5 \times 4 \times 3 = 120$$

Third Arrangement

$$1 \times 6 \times 2 \times 4 = \boxed{48}$$

49. (1) The upper left number is the sum of the other three numbers.

First Arrangement

$$8 + 1 + 3 + 4 = 16$$

Second Arrangement

$$4 + 8 + 5 + 3 = 20$$

Third Arrangement

$$6 + ? + 5 + 4 = 18$$

$$\therefore ? = 18 - 15 = \boxed{3}$$

50. (2) The product of the upper two numbers is equal to the lower number.

First Figure : $13 \times 17 = 221$

Secod Figure : $12 \times 19 = 228$

Third Figure : $13 \times 18 = \boxed{234}$

51. (2) $\sqrt{(3)^2 + (4)^2}$

$$= \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\sqrt{(5)^2 + (12)^2}$$

$$= \sqrt{25 + 144} = \sqrt{169} = 13$$

$$\Rightarrow (?)^2 + (15)^2 = (17)^2$$

$$\Rightarrow (?)^2 = 289 - 225 = 64$$

$$\therefore ? = \sqrt{64} = 8$$

52. (3) The sum of the four numbers located in the four arms is equal to the central number.

Ist Figure

$$9 + 11 + 23 + 6 = 49$$

2nd Figure

$$7 + 12 + 8 + 13 = 40$$

Third Figure

$$16 + 9 + 26 + 8 = 59$$

53. (2) The sum of the four numbers located in the four arms is equal to the central number.

1st Figure

$$1 + 44 + 33 + 22 = 110$$

2nd Figure

$$16 + 40 + 32 + 24 = 112$$

3rd Figure

$$? + 12 + 34 + 23 = 114$$

$$\Rightarrow ? = 114 - 59 = 45$$

54. (3) The sum of the squares of the upper two numbers is equal to the lower number in each figure.

First Figure

$$(2)^2 + (4)^2 = 4 + 16 = 20$$

Second Figure

$$(3)^2 + (9)^2 = 9 + 81 = 90$$

Third Figure

$$(1)^2 + (7)^2 = 1 + 49 = \boxed{50}$$

55. (4) $\sqrt{1} + \sqrt{16} + \sqrt{9} + \sqrt{4}$
 $= 1 + 4 + 3 + 2 = 10$

$$\sqrt{25} + \sqrt{64} + \sqrt{100} + \sqrt{16}$$

$$= 5 + 8 + 10 + 4 = 27$$

Therefore,

$$\sqrt{36} + \sqrt{256} + \sqrt{144} + \sqrt{64}$$

$$= 6 + 16 + 12 + 8 = \boxed{42}$$

56. (2) Multiply the upper two numbers and then take the square of the product to get the lower number.

$$4 \times 3 = 12 \text{ and } (12)^2 = 144$$

$$11 \times 9 = 99 \text{ and } (99)^2 = 9801$$

$$\text{Therefore, } 15 \times 6 = 90$$

$$\text{and } (90)^2 = \boxed{8100}$$

57. (1) $(7 \times 4) - (5 \times 3)$

$$= \text{Central number}$$

$$\text{or, } 28 - 15 = 13$$

$$(8 \times 4) - (9 \times 3)$$

$$\text{or, } 32 - 27 = 5$$

Therefore,

$$(9 \times 4) - (8 \times 3)$$

$$\text{or, } 36 - 24 = \boxed{12}$$

58. (3) Take the sum of all the four numbers located at the four corners to get the central number.

$$3 + 5 + 7 + 9 = 24$$

$$17 + 13 + 11 + 9 = 50$$

Therefore,

$$? + 7 + 10 + 21 = 47$$

$$\therefore ? = 47 - 38 = \boxed{9}$$

59. (1) Take the sum of the products of the upper and lower numbers respectively to get the central number.

$$5 \times 3 = 15 \text{ and } 6 \times 8 = 48$$

$$15 + 48 = 63$$

$$2 \times 7 = 14 \text{ and } 3 \times 9 = 27$$

$$14 + 27 = 41$$

Therefore,

$$6 \times 7 = 42 \text{ and } 8 \times 5 = 40$$

$$42 + 40 = \boxed{82}$$

60. (3) $\frac{27}{3} = 9$ and $\frac{16}{4} = 4$

$$\Rightarrow 9 + 4 = 13 \Rightarrow \text{Central number}$$

$$\frac{42}{7} = 6 \text{ and } \frac{65}{13} = 5$$

$$\Rightarrow 6 + 5 = 11$$

Therefore,

$$\frac{27}{9} = 3 \text{ and } \frac{72}{8} = 9$$

$$\Rightarrow 3 + 9 = \boxed{12}$$

61. (1) Take the sum of cube roots of all the four numbers outside the oval to get the central number.

$$\sqrt[3]{1} = 1; \sqrt[3]{64} = 4;$$

$$\sqrt[3]{27} = 3; \sqrt[3]{8} = 2$$

$$1 + 4 + 3 + 2 = 10$$

$$\sqrt[3]{8} = 2; \sqrt[3]{125} = 5;$$

$$\sqrt[3]{64} = 4; \sqrt[3]{27} = 3$$

$$2 + 5 + 4 + 3 = 14$$

$$\sqrt[3]{27} = 3; \sqrt[3]{216} = 6;$$

$$\sqrt[3]{125} = 5; \sqrt[3]{64} = 4$$

$$3 + 6 + 5 + 4 = \boxed{18}$$

62. (1) $\sqrt{6 + 8 + 4 + 7} = \sqrt{25} = 5$

$$\sqrt{9 + 12 + 8 + 7} = \sqrt{36} = 6$$

Similarly,

$$\sqrt{10 + 9 + 16 + 14} = \sqrt{49} = 7$$

63. (1) $4 + 1 = 5$ and $5 + 1 = 6$

$$1 + 1 = 2 \text{ and } 2 + 1 = 3$$

$$2 + 1 = 3 \text{ and } 3 + 1 = 4$$

$$3 + 1 = 4 \text{ and } 4 + 1 = 5$$

$$\textcircled{30} \xrightarrow{+12} \textcircled{42} \xrightarrow{+12} \textcircled{54}$$

64. (2) **Upper numbers**

$$4 + 1 = 5 \text{ and } 5 + 1 = 6$$

Left numbers

$$1 + 1 = 2 \text{ and } 2 + 1 = 3$$

Right numbers

$$3 + 1 = 4 \text{ and } 4 + 1 = 5$$

Bottom numbers

$$2 + 1 = 3 \text{ and } 3 + 1 = 4$$

Central numbers

$$4 = 4 = 4$$

FINDING THE MISSING NUMBER

65. (3) First figure : $6 \times 6 = 4 \times 9$

Second figure : $9 \times 8 = 24 \times 3$

Third figure

$$15 \times 6 = 9 \times ?$$

$$\therefore ? = \frac{90}{9} = \boxed{10}$$

66. (2) $5 + 0 = 5$ $4 + 2 = 6$ $4 + 3 = 7$
 $3 + 1 = 4$ $2 + 1 = 3$ $1 + 1 = 2$
 9 7 5

67. (2) The sum of squares of the outside numbers is equal to central number.

$$(3)^2 + (2)^2 + (2)^2 + (4)^2$$

$$= 9 + 4 + 4 + 16 = 33$$

$$(3)^2 + (2)^2 + (5)^2 + (4)^2$$

$$= 9 + 4 + 25 + 16 = 54$$

Similarly,

$$(6)^2 + (5)^2 + (4)^2 + (3)^2$$

$$= 36 + 25 + 16 + 9 = 86$$

68. (2) $7 \times 4 = 28$; $19 + 9 = 28$
 $8 \times 5 = 40$; $28 + 12 = 40$
 $9 \times 3 = 27$; $27 - 6 = 21$

69. (3) $12 + 13 = 25$

$$27 + 24 = 51$$

$$\therefore ? = 64 - 27 = 37$$

70. (2) $25 + 45 + 35 + 30 = 135$

$$\frac{135}{5} = 27$$

$$60 + 20 + 40 + 30 = 150$$

$$\frac{150}{5} = 30$$

$$25 + 40 + 35 + 65 = 165$$

$$\frac{165}{5} = 33$$

71. (2) $24 + 22 = 46$

$$27 + 42 = 69$$

$$\therefore ? = 79 - 38 = 41$$

72. (2) $4 \times 3 = 12$ and $(12)^2 = 144$

$$11 \times 9 = 99 \text{ and } (99)^2 = 9801$$

$$15 \times 6 = 90 \text{ and } (90)^2 = 8100$$

73. (2) $12 \times 4 = 48$

$$\sqrt{25} = 5$$

$$16 \times 4 = 64$$

$$\sqrt{81} = 9$$

Similarly,

$$15 \times 4 = \boxed{60}$$

$$\sqrt{49} = 7$$

74. (2) $(27 + 18) - (12 + 13)$

$$\Rightarrow 45 - 25 = 20$$

$$(16 + 12) - (6 + 9)$$

$$\Rightarrow 28 - 15 = 13$$

$$(10 + 11) - (5 + 4)$$

$$\Rightarrow 21 - 9 = \boxed{12}$$

75. (3) $27 = 9 \times (2 + 1)$

$$35 = 7 \times (3 + 2)$$

$$36 = 4 \times (4 + 5)$$

76. (4) First figure

$$12 - 6 = 6 ;$$

$$15 - 9 = 6$$

Second figure

$$12 - 4 = 8 ;$$

$$16 - 8 = 8$$

Third figure

$$11 - 5 = 6 ;$$

$$14 - 8 = 6$$

77. (3) First Figure

$$3 + 5 + 2 + 4 = 14$$

$$\Rightarrow 14 + 13 = 27$$

Second Figure

$$6 + 2 + 3 + 5 = 16$$

$$\Rightarrow 16 + 21 = 37$$

Third Figure

$$2 + 2 + 9 + 5 = 18$$

$$\Rightarrow 18 + 29 = \boxed{47}$$

78. (2) Figure (1)

$$2 \times 3 \times 5 = 30$$

Figure (2)

$$5 \times 1 \times (-1) = -5$$

Figure (3)

$$4 \times 3 \times (-1) = -12$$

79. (3) In each figure the lower left number is the square of the upper right number.

Again, First figure

$$3 + 9 = 7 + 5$$

Second figure

$$2 + 8 = 4 + 6$$

Third figure

$$4 + 7 = 5 + ?$$

$$\Rightarrow ? = 11 - 5 = 6$$

80. (3) The sum of cross products of the numbers gives the central number.

First Figure

$$(4 \times 6) + (6 \times 7) = 24 + 42 = 66$$

Second Figure

$$(8 \times 9) + (5 \times 14)$$

$$= 72 + 70 = 142$$

Third Figure

$$(11 \times 7) + (9 \times 6) = 77 + 54$$

$$= 131$$

81. (3) First figure

$$9 \times 9 = 81$$

$$81 \times 9 = 729$$

Second figure

$$8 \times 8 = 64$$

$$64 \times 8 = 512$$

Third figure

$$7 \times 7 = 49$$

$$49 \times 7 = \boxed{343}$$

82. (3) First arrangement

$$2 \times 2 \times 2 \times 2 = 16$$

Second arrangement

$$3 \times 3 \times 3 \times 3 = 81$$

Third arrangement

$$? = 5 \times 5 \times 5 \times 5 = 625$$

83. (1) The sum of upper two numbers gives the lower right number while their product is equal to the lower left number.

First arrangement

$$5 + 4 = 9 \text{ and } 5 \times 4 = 20$$

Second arrangement

$$3 + 8 = 11 \text{ and } 3 \times 8 = 24$$

Third arrangement

$$9 + 4 = 13 \text{ and } 9 \times 4 = \boxed{36}$$

84. (2) The sum of the upper right number and the lower left number is equal to the lower right number while their product gives the upper left number.

First arrangement

$$7 + 4 = 11 \text{ and } 7 \times 4 = 28$$

Second arrangement

$$5 + 5 = 10 \text{ and } 5 \times 5 = 25$$

Third arrangement

$$8 + 3 = 11 \text{ and } 8 \times 3 = \boxed{24}$$

$$85. (4) \frac{7 \times 8}{9} = 3$$

$$\Rightarrow \sqrt{9} = 3$$

$$\frac{8 \times 9}{144} = 12$$

$$\Rightarrow \sqrt{144} = 12$$

$$\frac{10 \times 11}{169} = 13$$

$$\Rightarrow \sqrt{169} = 13$$

Therefore,

$$\frac{12 \times 13}{441} \Rightarrow \sqrt{441} = 21$$

86. (2) First Figure

$$(3)^3 + (2)^3$$

$$= 27 + 8 = 35$$

Second Figure

$$(1)^3 + (5)^3$$

$$= 1 + 125 = 126$$

Third Figure

$$(3)^3 + (4)^3$$

$$= 27 + 64 = \boxed{91}$$

87. (2) First Figure

$$4 \times 3 = 12$$

$$12 \times 3 = 36$$

Second Figure

$$13 \times 1 = 13$$

$$13 \times 1 = 13$$

Third Figure

$$8 \times 2 = 16$$

$$16 \times 2 = 32$$

88. (2) First Figure

$$(6 \times 7) + (8 + 4) = 42 + 12 = 54$$

Second Figure

$$(8 \times 4) + (12 + 7) = 32 + 19 = 51$$

Third Figure

$$(9 \times 5) + (14 + 9) = 45 + 23 = 68$$

89. (4) First Figure

$$4 \times 3 \times 2 = 24$$

Second Figure

$$(-2) \times 2 (-1) = 4$$

Third Figure

$$0 \times 6 \times 5 = 0$$

90. (4) First Figure

$$14 + 12 + 7 + 3 = 36$$

Second Figure

$$9 + 11 + ? + 16 = 54$$

$$\Rightarrow ? = 54 - 36 = 18$$

91. (3) First figure

$$\sqrt{64} + \sqrt{36} + \sqrt{49}$$

$$\Rightarrow 8 + 6 + 7 = 21$$

Second figure

$$\sqrt{121} + \sqrt{81} + \sqrt{100}$$

$$\Rightarrow 11 + 9 + 10 = 30$$

92. (1) From first figure to second figure the digits of central part are written in the reverse order.

$$3456 \Rightarrow 6543$$

93. (1) The product of outer four numbers is equal to the central number.

First Figure

$$1 \times 3 \times 4 \times 5 = 60$$

Second Figure

$$6 \times 7 \times 1 \times 2 = 84$$

Third Figure

$$3 \times 10 \times 3 \times 1 = \boxed{90}$$

94. (3) First arrangement

$$6 + 4 + 3 + 1 + 5 + 0 = 19$$

$$\Rightarrow 19 \times 7 = 133$$

Second arrangement

$$2 + 5 + 3 + 4 + 6 + 8 = 28$$

$$\Rightarrow 28 \times 7 = 196$$

Third arrangement

$$1 + 5 + 7 + 3 + 4 + 2 = 22$$

$$\Rightarrow 22 \times 7 = 154$$

95. (1) First Figure

$$2 \times 2 \times 3 = 12$$

Second Figure

$$2 \times 3 \times 5 = 30$$

Third Figure

$$5 \times 1 \times (-1) = -5$$

Fourth Figure

$$4 \times 3 \times (-1) = -12$$

96. (1) First Figure

$$(11 \times 12) - (6 \times 9)$$

$$\Rightarrow 132 - 54 = 78$$

Second Figure

$$(14 \times 10) - (7 \times 8)$$

$$\Rightarrow 140 - 56 = 84$$

97. (2) $\sqrt{625} + \sqrt{676} + \sqrt{729}$

$$\Rightarrow 25 + 26 + 27 = 78$$

Similarly,

$$\sqrt{289} + \sqrt{324} + \sqrt{361}$$

$$\Rightarrow 17 + 18 + 19 = 54$$

98. (4) First figure

$$3 + 9 = 8 + 4$$

Second figure

$$4 + 7 = 6 + 5$$

Third figure

$$5 + ? = 9 + 3$$

$$\Rightarrow ? = 12 - 5 = 7$$

99. (4) First figure

$$(4)^2 + (3)^2 = 16 + 9 = 25$$

Second figure

$$(9)^2 + (11)^2 = 81 + 121 = 202$$

Third figure

$$(1)^2 + (7)^2 = 1 + 49 = 50$$

100. (4) First Figure

$$\sqrt{25} + \sqrt{36} + \sqrt{49} + \sqrt{16}$$

$$= 5 + 6 + 7 + 4 = 22$$

Second Figure

$$\sqrt{9} + \sqrt{64} + \sqrt{1} + \sqrt{X} = 16$$

$$= 3 + 8 + 1 + \sqrt{X} = 16$$

$$= \sqrt{X} = 16 - 12 = 4$$

$$\therefore X = (4)^2 = 16$$

Third Figure

$$\sqrt{4} + \sqrt{9} + \sqrt{1} + \sqrt{81} = 15$$

$$= 2 + 3 + 1 + 9 = 15$$

101. (3) First Figure

$$(4 \times 2) + (7 \times 1) + 3$$

$$= 8 + 7 + 3 = 18$$

$$18 \Rightarrow 1 + 8 = 9$$

Second Figure

$$(3 \times 3) + (5 \times 2) + 4$$

$$= 9 + 10 + 4 = 23$$

$$23 \Rightarrow 2 + 3 = 5$$

Third Figure

$$(6 \times 9) + (2 \times 3) + 4$$

$$= 54 + 6 + 4 = 64$$

$$64 \Rightarrow 6 + 4 = 10$$

102. (1) First Figure

$$5 \times 4 = 20 (= 10 \times 2)$$

$$6 \times 2 = 12$$

Second Figure

$$6 \times 5 = 30 (= 10 \times 3)$$

$$7 \times 3 = 21$$

Third Figure

$$4 \times 10 = 40 (= 10 \times 4)$$

$$8 \times 4 = 32$$

103. (4) First Figure

$$(5 \times 9) + (4 \times 8)$$

$$\Rightarrow 45 + 32 = 77$$

Second Figure

$$(9 \times 6) + (3 \times 7)$$

$$\Rightarrow 54 + 21 = 75$$

- 104.** (3) First Figure
 $(8 + 7)^2 = 225$
 Second Figure
 $(3 + 4)^2 = 49$
 Third Figure
 $\sqrt{121} = 11$
 $6 + ? = 11$
 $\therefore ? = 11 - 6 = 5$

- 105.** (1) First Figure
 $6 \times 2 = 12$
 $6 \times 3 = 18$
 $6 \times 5 = 30$
 Second Figure
 $8 \times 2 = 16$
 $8 \times 4 = 32$
 $8 \times 5 = 40$
 Third Figure
 $9 \times 4 = 36$
 $9 \times 2 = 18$
 $9 \times 3 = 27$

- 106.** (3) First Figure
 $4 \times 8 = 32; \frac{32}{2} = 16$
 Second Figure
 $6 \times 4 = 24; \frac{24}{2} = 12$
 Third Figure
 $8 \times 9 = 72; \frac{72}{2} = 36$

- 107.** (1) First figure
 $7 + 9 - 6 = 10$
 Second figure
 $5 + 8 - 3 = 10$
 Third figure
 $9 + 6 - 4 = \boxed{11}$

- 108.** (2) First Figure
 $(1 \times 2) + (3 \times 4)$
 $= 2 + 12 = 14$
 Second Figure
 $(6 \times 5) + (7 \times 8)$
 $= 30 + 56 = 86$
 Third Figure
 $(10 \times 9) + (11 \times 12)$
 $= 90 + 132 = 222$

- 109.** (3) First arrangement
 $(1)^3 = 1$
 $(2)^3 = 8$
 $(3)^3 = 27$

Second arrangement

$$(4)^3 = 64$$

$$(5)^3 = 125$$

$$(6)^3 = 216$$

Third arrangement

$$(7)^3 = 343$$

$$(8)^3 = \boxed{512}$$

$$(9)^3 = 729$$

- 110.** (4) First Figure
 $(2)^2 = 4; (3)^2 = 9; (4)^2 = 16$
 Second Figure
 $(1)^2 = 1; (2)^2 = 4; (3)^2 = 9$
 Third Figure
 $(3)^2 = 9; (4)^2 = 16; (5)^2 = 25$

- 111.** (2) First Figure
 $25 + 17 = 6 \times 7 = 42 = 42$
 Second Figure
 $38 + 18 = 8 \times 7 = 56 = 56$
 Third Figure
 $89 + 16 = 105$

$$\Rightarrow \frac{105}{7} = 15$$

- 112.** (3) First Figure
 $(20 - 9)^2 = 121$
 $\Rightarrow (11)^2 = 121$
 Second Figure
 $(24 - 11)^2 = 169$
 $\Rightarrow (13)^2 = 169$
 Third Figure
 $(32 - 17)^2$
 $\Rightarrow (15)^2 = 225$

- 113.** (4) First Figure
 $\frac{14 \times 24}{8} = 42$

Second Figure

$$\frac{64 \times 12}{8} = 96$$

Third Figure

$$\frac{32 \times 18}{8} = 72$$

TYPE-II

- 1.** (2) $9 + 2^2 = 13$
 $13 + 3^2 = 22$
 $22 + 4^2 = \boxed{38}$
- 2.** (2) The lowermost number in each column is the product of the other numbers.
 $3 \times 5 \times 4 = 60$
 $5 \times 7 \times 4 = 140$
 Therefore,
 $? = \frac{96}{4 \times 4} = \boxed{6}$
- 3.** (2) The sum of numbers of each column is 15.
 First column
 $= 4 + 3 + 8 = 15$
 Second column
 $= 9 + 5 + 1 = 15$
 Similarly,
 Third column
 $= 2 + 7 + ? = 15$
 or, $? = 15 - 9 = \boxed{6}$
- 4.** (2) The number in the sector is perfect cube.
 $(1)^3 = 1$
 $(2)^3 = 8$
 $(3)^3 = 27$
 $(4)^3 = \boxed{64}$
- 5.** (3) The numbers in the given arrangement follow the rule given below :
 $4 \times 1 = 4$
 $4 \times 2 = 8$
 $8 \times 3 = 24$
 $24 \times 4 = 96$
 $96 \times 5 = 480$
 $480 \times 6 = \boxed{2880}$
 $2880 \times 7 = 20160$
- 6.** (2) First column
 $9 \times 6 \times 5 = 270$
 Second column
 $8 \times 7 \times 4 = 224$
 Third column
 $8 \times 7 \times ? = 336$
 $\therefore ? = \frac{336}{56} = \boxed{6}$
- 7.** (2) First column
 $9 - 1 = 8, 8 - 1 = 7$
 Second column
 $4 + 1 = 5; 5 + 1 = 6$

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Third column

$$2 + 0 = 2; 2 + 1 = 3 (1 + 2);$$

$$3 + 1 = \boxed{4}$$

8. (4) a = 23 (529) (1024)

The second number is the square of the first number and the third number is the square of that number obtained by interchanging the digits of the first number.

$$\text{Thus, } (23)^2 = 23 \times 23 = 529$$

$$(32)^2 = 32 \times 32 = 1024$$

$$\Rightarrow b = 21 (441) (144)$$

$$(21)^2 = 21 \times 21 = 441$$

$$(12)^2 = 12 \times 12 = 144$$

Similarly,

$$c = 19 (361) (?)$$

$$(19)^2 = 19 \times 19 = 361$$

$$(91)^2 = 91 \times 91 = 8281$$

9. (1) Proceed column wise. In each column the Second Number is equal to First Number + 1 and the Third Number is equal to First Number + 2 (or Second Number + 1).

1st Column

$$0 + 1 = 1 \text{ and } 1 + 1 = 2$$

2nd Column

$$-1 + 1 = 0 \text{ and } 0 + 1 = \boxed{1}$$

3rd Column

$$-2 + 1 = -1 \text{ and } -1 + 1 = 0$$

10. (3) Divide the second number by the first number and then add the third number to get the lowest number.

First Column

$$25 \div 5 = 5 \text{ and } 5 + 2 = 7$$

Second Column

$$42 \div 6 = 7 \text{ and } 7 + 10 = 17$$

Third Column

$$21 \div 3 = 7 \text{ and } 7 + 20 = \boxed{27}$$

11. (4) **First column**

$$(7 + 6) \times 5 = 13 \times 5 = 65$$

Second column

$$(2 + 3) \times 4 = 5 \times 4 = 20$$

Third column

$$(4 + ?) \times 9 = 45$$

$$\text{or, } 4 + ? = \frac{45}{9} = 5$$

$$\therefore ? = 1$$

12. (3) **First Column**

$$9 \times 3 \times 3 = 81$$

Second Column

$$11 \times 4 \times 4 = 176$$

Third Column

$$13 \times 7 \times 5 = \boxed{455}$$

13. (3) $9 + 7 - 5 = 11$

$$\text{and, } 6 + 4 - 8 = 2$$

$$\therefore 8 + ? - 4 = 7$$

$$\text{or, } ? = 11 - 8 = \boxed{3}$$

14. (3) $8 + 7 = 15$

$$15 + 14 = 29$$

$$29 + 28 = 57$$

$$57 + 56 = 113$$

$$113 + 112 = \boxed{225}$$

15. (4) $4 + 12 = 11 + 5$

$$6 + 7 = 10 + 3$$

$$8 + 9 = 10 + 7$$

$$7 + 5 = \boxed{8} + 4$$

16. (3) $6 \div 3 = 2$ and $2 \times 4 = 8$

$$18 \div 2 = 9 \text{ and } 9 \times 3 = 27$$

$$15 \div 5 = 3 \text{ and } 3 \times \boxed{3} = 9$$

17. (4) $9 + 3 = 12$, $12 + 6 = 18$

$$18 + 9 = \boxed{27}$$

18. (1) $\frac{8 \times 5 \times 2}{2} = 40$

$$\frac{5 \times 4 \times 3}{2} = 30$$

Therefore,

$$? = \frac{2 \times 1 \times 10}{2} = 10$$

19. (4) $4 \times 4 = 16$

$$16 \times 4 = 64$$

$$64 \times 4 = \boxed{256}$$

20. (1) **First Column**

$$2 (2 + 50 + 10)$$

$$\text{or, } 2 \times 62 = 124$$

Second Column

$$2 (12 + 300 + 60)$$

$$\text{or, } 2 \times 372 = 744$$

Third Column

$$2 (? + 550 + 110) = 1364$$

$$\text{or, } ? + 660 = \frac{1364}{2}$$

$$\text{or, } ? = 682 - 660 = \boxed{22}$$

$$\begin{array}{cccc} 21. (1) & 8 & 7 & 10 & 12 \\ & +5\downarrow & +5\downarrow & +5\downarrow & +5\downarrow \\ & 13 & 12 & 15 & 17 \\ & -3\downarrow & -3\downarrow & -3\downarrow & -3\downarrow \\ & 10 & 9 & \boxed{12} & 14 \end{array}$$

22. (3) Proceed clockwise

$$9 + 3 = 12$$

$$12 + (3 \times 2) = 18$$

$$18 + (6 \times 2) = \boxed{30}$$

$$30 + (12 \times 2) = 54$$

$$54 + (24 \times 2) = 102$$

23. (1) $8 + 2 = 10$, $10 + 4 = 14$

$$14 + 6 = \boxed{20}$$

24. (4) The product of the first three numbers is the lowest number in each column.

First Column

$$2 \times 5 \times 1 = 10$$

Second Column

$$7 \times 2 \times ? = 42$$

$$\therefore ? = \frac{42}{14} = \boxed{3}$$

25. (3) Subtract 12 from the preceding number and proceed anti-clockwise.

$$58 - 12 = 46; 46 - 12 = 34$$

$$34 - 12 = 22$$

26. (3) The square of the sum of the first three numbers in each column is equal to the lowermost number.

First Column

$$(1 + 4 + 2)^2 = (7)^2 = 49$$

Second Column

$$(4 + 2 + 2)^2 = (8)^2 = 64$$

Third Column

$$\sqrt{169} = 13$$

$$\therefore ? + 5 + 3 = 13$$

$$\text{or, } ? = 13 - 8 = 5$$

27. (2) The product of the first three numbers in each column is equal to the lowermost number.

First Column

$$9 \times 12 \times 13 = 1404$$

Second Column

$$3 \times 2 \times 5 = 30$$

Third Column

$$7 \times 9 \times ? = 504$$

$$\therefore ? = \frac{504}{7 \times 9} = 8$$

FINDING THE MISSING NUMBER

28. (2) Proceed clockwise starting with the lowest number in the following manner :

$$\begin{aligned} 7 \times 2 + 1 &= 15 \\ 15 \times 2 + 1 &= 31 \\ 31 \times 2 + 1 &= 63 \\ 63 \times 2 + 1 &= 127 \\ 127 \times 2 + 1 &= 255 \end{aligned}$$

29. (4) $25 + 27 = 52$, $23 + 30 = 53$, $33 + 21 = 54$,

$$\text{Therefore, } ? + 36 = 55$$

$$\therefore ? = 55 - 36 = 19$$

30. (4) Subtract the upper number from the sum of the second and the third number to get the lowest number in each column.

$$49 + 98 - 7 = 147 - 7 = 140$$

$$441 + 882 - 21$$

$$= 1323 - 21 = 1302$$

$$225 + 450 - 15$$

$$= 675 - 15 = 660$$

$$\begin{array}{ccc} \text{N} & \text{E} & \text{T} \end{array}$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$14 + 5 + 20 = 39$$

$$\text{Therefore,}$$

$$\begin{array}{ccc} \text{N} & \text{U} & \text{T} \end{array}$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$14 + 21 + 20 = 55$$

31. (3) $25 + 11 - 17 = 19$

$$34 + 6 - 12 = 28$$

$$19 + 8 - ? = 11$$

$$\therefore ? = 27 - 11 = \boxed{16}$$

32. (3) First column

$$3 \times 2 \times 1 = 6$$

$$\text{Second column}$$

$$4 \times 3 \times 2 = 24$$

$$\text{Third column}$$

$$2 \times 1 \times ? = 0$$

$$2 \times 1 \times 0 = 0$$

33. (3) The product of three numbers in each column is equal to the lowermost number.

$$3 \times 4 \times 4 = 48$$

$$4 \times 5 \times 3 = 60$$

$$5 \times 3 \times ? = 105$$

$$\text{or, } ? = \frac{105}{15} = 7$$

34. (4) $5 + 2 = 7$, $7 + 4 = 11$, $11 + 6 = 17$

35. (1) The second number in each column is square root of the first number. Again, the sum of second and third numbers is equal to the lowermost number in each column.

$$\sqrt{81} = 9 \quad \text{or} \quad 20 - 11 = 9$$

36. (1) $\frac{12}{3} = 4$ and $4 + 9 = 13$

$$\frac{6}{2} = 3 \quad \text{and} \quad 3 + 10 = 13$$

Similarly,

$$\frac{8}{1} = 8, \quad \text{and} \quad 8 + 5 = 13$$

37. (3) $836 + 112 = 948$,

$$\frac{948}{3} = 316$$

Similarly,

$$213 + 420 = 633, \quad \frac{633}{3} = 211$$

38. (3) $18 \times 6 \times 3 = 324$

$$15 \times 5 \times 4 = 300$$

$$? = \frac{528}{11 \times 8} = 6$$

39. (2) $5 + 20 + 6 + 9 = 40$

$$4 + 8 + 15 + 3 = 30$$

$$9 + 25 + 7 + 9 = 50$$

$$22 + 7 + 8 + 8 = 45$$

40. (2) $(3 + 4 + 1) + (5 + 2 + 1)$

$$= 8 + 8 = 16$$

$$(6 + 1 + 3) + (8 + 1 + 6)$$

$$= 10 + 15 = 25$$

$$(4 + 5 + 2) + (3 + 2 + 6)$$

$$= 11 + 11 = 22$$

41. (3) $4 + 9 + 17 + 6 = 36$

$$20 + 5 + 8 + 9 = 42$$

$$7 + 23 + 9 + 9 = 48$$

$$9 + 4 + 19 + ? = 40$$

$$\rightarrow ? = 40 - 32 = 8$$

42. (1) $5 \times 6 \times 4 = 120$

$$\frac{120}{2} = 60$$

$$8 \times 9 \times 12 = 864$$

$$\frac{864}{2} = 432$$

$$\frac{13 \times 15 \times ?}{2} = 1560$$

$$\therefore ? = \frac{1560 \times 2}{13 \times 15} = 16$$

43. (4) $4 - 3 = 1$, $1 - 1 = 0$
 $9 - 6 = 3$, $6 - 2 = 4$
 $12 - 8 = 4$, $6 - 3 = 3$
 $12 - 10 = 2$, $4 - 1 = 3$

44. (2) $3 \times 3 = 9$, $9 \times 3 = 27$, $27 \times 3 = 81$
 $6 \times 3 = 18$, $18 \times 3 = 54$, $54 \times 3 = 162$

$$7 \times 3 = 21, \quad 21 \times 3 = 63, \quad 63 \times 3 = 189$$

45. (4) Different quadrants in question:

$$\begin{array}{|c|c|} \hline 5 & 5 \\ \hline 5 & 0 \\ \hline \end{array} \Rightarrow 5 - 5 = 0 \rightarrow 0^2 \times 4 = 0$$

$$\begin{array}{|c|c|} \hline 7 & 3 \\ \hline 6 & 4 \\ \hline \end{array} \Rightarrow 7 - 3 = 4 \rightarrow 4^2 \times 4 = 64$$

$$\begin{array}{|c|c|} \hline 8 & 36 \\ \hline 11 & 11 \\ \hline \end{array} \Rightarrow 11 - 8 = 3 \rightarrow 3^2 \times 4 = 36$$

$$\begin{array}{|c|c|} \hline ? & 8 \\ \hline 2 & 2 \\ \hline \end{array} \Rightarrow 8 - 2 = 6$$

$$\rightarrow ? = 6^2 \times 4 = \boxed{144}$$

46. (1) $1 + 7 + 3 + 5 + 2 + 6 = 24$

$$4 + 3 + 1 + 3 + 2 + 5 = 18$$

Therefore,

$$2 + 5 + 3 + 4 + 7 + 1 = 22$$

47. (4) $5 + 4 = 9$ and $9 \times 2 = 18$

$$6 + 3 = 9 \quad \text{and} \quad 9 \times 3 = 27$$

$$12 + 4 = 16 \quad \text{and} \quad ?$$

$$= \frac{96}{16} = \boxed{6}$$

48. (2) $3 \times 4 \times 5 = 60$

$$7 \times 2 \times 4 = 56$$

$$5 \times ? \times 6 = 90$$

$$\therefore ? = \frac{90}{5 \times 6} = 3$$

49. (2) $8 \times 5 - 28 = 40 - 28 = 12$

$$10 \times 3 - 16 = 30 - 16 = 14$$

$$9 \times 4 = ? = 25$$

$$\Rightarrow 36 - ? = 25$$

$$\therefore ? = 36 - 25 = 11$$

50. (2) $1 + 2 + 4 + 3 = 10$;

$$10 \times 5 = 50$$

$$3 + 4 + 5 + 2 = 14$$

$$14 \times 5 = 70$$

$$7 + 4 + 9 + 3 = 23$$

$$23 \times 5 = \boxed{115}$$

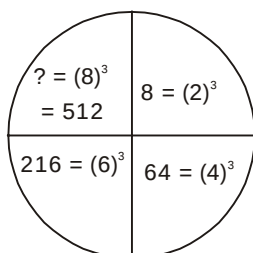
51. (3) $(5)^3 + 1 = 125 + 1 = 126$

$$(6)^3 + 1 = 216 + 1 = 217$$

52. (1) $(21 + 1) - 2 = 22 - 2 = 20$
 $(22 + 2) - 1 = 24 - 1 = 23$
 $(? + 5) - 2 = 43$
 $\Rightarrow ? = (43 + 2) - 5$
 $\Rightarrow ? = 45 - 5 = 40$

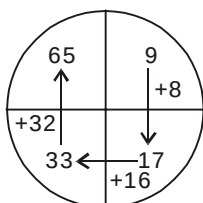
53. (3) **Columnwise**
 First Column
 $(7)^2 + (4)^2 + (2)^2 = 49 + 16 + 4 = 69$
 Second Column
 $(3)^2 + (9)^2 + (1)^2 = 9 + 81 + 1 = 91$
 Third Column
 $(2)^2 + (6)^2 + (5)^2 = 4 + 36 + 25 = 65$

54. (2)



55. (3) **Columnwise**
 I. $2 \times 7 \times 9 = 126$
 II. $7 \times 3 \times 8 = 168$
 III. $9 \times 4 \times ? = 216$
 $\therefore ? = \frac{216}{9 \times 4} = 6$

56. (4)



57. (1) **First Row**
 $\Rightarrow \sqrt{169} + \sqrt{64} + \sqrt{81}$
 $= 13 + 8 + 9 = 30$
Second Row
 $\Rightarrow \sqrt{625} + \sqrt{?} + \sqrt{49} = 50$
 $\Rightarrow 25 + \sqrt{?} + 7 = 50$
 $\Rightarrow \sqrt{?} = 50 - 32 = 18$
 $\therefore (?) = (18)^2 = 324$
Third Row
 $\Rightarrow \sqrt{1296} + \sqrt{576} + \sqrt{100}$
 $= 36 + 24 + 10 = 70$

58. (3) **First Column**
 $= 4 \times 7 - 1 = 28 - 1 = 27$
Second Column
 $= 5 \times 8 - 2 = 40 - 2 = 38$

Third Column
 $= 6 \times 9 - 3 = 54 - 3 = 51$

59. (1) **Columnwise**
 $6 \times 6 = 36$
 $6 \times (6 - 2) = 6 \times 4 = 24$
 $9 \times 9 = 81$
 $9 \times (9 - 2) = 9 \times 7 = 63$
 $12 \times 12 = 144$
 $12 \times (12 - 2) = 12 \times 10 = 120$
 60. (2) $18 \times 3 + 6 = 54 + 6 = 60$
 $60 \times 3 + 6 = 180 + 6 = 186$
 $186 \times 3 + 6 = 558 + 6 = 564$
 $564 \times 3 + 6 = 1692 + 6 = 1698$
 $1698 \times 3 + 6 = 5094 + 6 =$

5100

61. (2) The sum of all the three numbers in each column is equal to 57.

First Column
 $18 + 17 + 22 = 57$

Second Column
 $23 + 19 + ? = 57$

$\Rightarrow ? = 57 - 42 =$ 15

Third Column
 $16 + ? + ? = 57$
 $\Rightarrow ? + ? = 57 - 16 = 41$
 $\Rightarrow 20 + 21 = 41$

62. (3) $3 \times 2 - 1 = 6 - 1 = 5$
 $5 \times 2 - 2 = 10 - 2 = 8$
 $8 \times 2 - 3 = 16 - 3 = 13$
 $13 \times 2 - 4 = 26 - 4 = 22$
 $22 \times 2 - 5 = 44 - 5 = 39$

63. (2) Starting from the lowest number, when we proceed clockwise the successive number is the square of the previous number.

$(2)^2 =$ 4

$(4)^2 = 16$

$(16)^2 = 256$

64. (2) Subtract the third number from the sum of the first two numbers to get the lowermost number in each column.

1st Column
 $16 + 13 = 29$
 and $29 - 14 = 15$

2nd Column
 $28 + 12 = 40$
 and $40 - 10 = 30$

3rd Column
 $29 + 16 = 45$
 and $45 - 15 = 30$

65. (1) The lowermost number in each column is the product of the other three numbers.

1st Column

$5 \times 8 \times 10 = 400$

2nd Column

$6 \times 9 \times 7 = 378$

3rd Column

$5 \times 7 \times ? = 315$

or, $? = \frac{315}{5 \times 7} =$ 9

66. (3) If we proceed in clockwise direction the next number is 6 less than the previous number.

$27 - 6 = 21$

$21 - 6 = 15$

Therefore,

$15 - 6 =$ 9

67. (2) Multiply the first number by the second number and then add the third number to the product so obtained in each column to get the result.

$6 \times 8 = 48$ and $48 + 3 = 51$

$15 \times 4 = 60$ and $60 + 5 = 65$

$20 \times 5 = 100$ and $100 + 20 =$

120

68. (2) The product of the first three numbers is equal to the fourth in each column.

$9 \times 3 \times 3 = 81$

$11 \times 4 \times 4 = 176$

Therefore,

$13 \times 7 \times ? = 455$

or, $? = \frac{455}{13 \times 7} = 5$

69. (2) The lowermost number in each column is the product of the first three numbers.

1st Column

$1 \times 8 \times 9 = 72$

2nd Column

$3 \times 6 \times 5 = 90$

3rd Column

$2 \times 7 \times ? = 56$

or, $? = \frac{56}{2 \times 7} =$ 4

70. (2) **First column**

$4 = (2)^2$

$9 = (2 + 1)^3 = (3)^3$

$16 = (3 + 1)^2 = (4)^2$

$25 = (4 + 1)^2 = (5)^2$

Second column

$$1 = (1)^2$$

$$4 = (1 + 1)^2 = (2)^2$$

$$9 = (2 + 1)^2 = (3)^2$$

$$16 = (3 + 1)^2 = (4)^2$$

Third column

$$25 = (5)^2$$

$$36 = (5 + 1)^2 = (6)^2$$

$$49 = (6 + 1)^2 = (7)^2$$

$$\boxed{64} = (7 + 1)^2 = (8)^2$$

- 71. (2)** The lower most number in each column is the product of the first three numbers.

First column

$$4 \times 5 \times 6 = 120$$

Second column

$$5 \times 4 \times 5 = 100$$

Third column

$$2 \times 2 \times ? = 24$$

$$\text{or, } ? = \frac{24}{4} = 6$$

- 72. (4) First Column**

$$\begin{array}{ccc} \times 2 & \times 2 & \times 2 \\ 7 \rightarrow 14 \rightarrow 28 \rightarrow 56 \end{array}$$

Second Column

$$\begin{array}{ccc} \times 2 & \times 2 & \times 2 \\ 4 \rightarrow 8 \rightarrow 16 \rightarrow 32 \end{array}$$

Third Column

$$9 \xrightarrow{\times 2} \boxed{18} \xrightarrow{\times 2} 36 \xrightarrow{\times 2} 72$$

- 73. (4) First Column**

$$5 \times 4 \times 3 = 60$$

Second Column

$$6 \times 5 \times 4 = 120$$

Third Column

$$7 \times ? \times 5 = 140$$

$$\text{or, } ? = \frac{140}{35} = 4$$

- 74. (1)** Out of two opposite numbers, one is the perfect square of the other.

$$(3)^3 = 3$$

$$(2)^2 = 4$$

$$(5)^2 = 25$$

$$\therefore (9)^2 = \boxed{81}$$

- 75. (1)** 408 (169) 395

$$408 - \sqrt{169} = 408 - 13 = 395$$

Therefore,

$$129 - \sqrt{x} = 122$$

$$\text{or, } \sqrt{x} = 129 - 122 = 7$$

$$\therefore x = (7)^2 = 49$$

- 76. (2)** The product of first three number is equal to the fourth number in each column

1st Column

$$3 \times 2 \times 6 = 36$$

2nd Column

$$4 \times 5 \times 7 = 140$$

3rd Column

$$9 \times 2 \times ? = 18$$

$$\text{or, } 18 \times ? = 18$$

$$\therefore ? = \frac{18}{18} = 1$$

- 77. (2)** Divide the product of the first two numbers by the third number to get the lowermost number in each column.

1st Column

$$5 \times 4 = 20 \text{ and } 20 \div 2 = 10$$

2nd Column

$$7 \times 6 = 42 \text{ and } 42 \div 3 = 14$$

3rd Column

$$8 \times 6 = 48$$

$$\text{or, } \frac{48}{?} = 12$$

$$\therefore ? = \frac{48}{12} = 4$$

- 78. (1)** The product of the first three numbers is equal to the lowermost number in each column.

First Column

$$4 \times 2 \times 3 = 24$$

Second Column

$$3 \times 5 \times 7 = 105$$

Third Column

$$6 \times 4 \times ? = 120$$

$$\therefore ? = \frac{120}{24} = 5$$

- 79. (2)** The product of the first three numbers in each column is equal to the lowermost number :

1st Column

$$8 \times 8 \times 4 = 256$$

2nd Column

$$6 \times 10 \times 2 = 120$$

3rd Column

$$4 \times 12 \times ? = 192$$

$$\Rightarrow ? = \frac{192}{48} = 4$$

- 80. (1)** $43 - 16 = 27$

$$56 - 29 = \boxed{27}$$

$$47 - 25 = 22$$

- 81. (3)** $1 + 2 = 3$, $2 + 3 = 5$,
 $3 + 5 = 8$,

$$\therefore ? = 5 + 8 = \boxed{13}$$

- 82. (1)** $19 \times 6 \Rightarrow \frac{19 \times 6}{3} = 38$

$$32 \times 12 \Rightarrow \frac{32 \times 12}{3} = 128$$

$$17 \times 9 \Rightarrow \frac{17 \times 9}{3} = 51$$

$$9 \times 3 \Rightarrow \frac{9 \times 3}{3} = \boxed{9}$$

- 83. (2)** $3 \times 2 - 1 = 5$

$$5 \times 2 - 2 = 8$$

$$8 \times 2 - 3 = 13$$

$$13 \times 2 - 4 = 22$$

$$22 \times 2 - 5 = \boxed{39}$$

- 84. (4)** The sum of the first two numbers is equal to the lowermost number in each column.

$$6 + 36 = 42 ; 7 + 49 = 56$$

$$\text{Therefore, } 8 + 64 = 72$$

- 85. (2)** $7 \times 4 = 28$

$$9 \times 4 = 36$$

$$3 \times 4 = 12$$

- 86. (2)** The sum of the second, the third and the fourth numbers in each column is equal to the first number.

1st column

$$11 + 5 + 10 = 26$$

2nd column

$$9 + 4 + 5 = 18$$

3rd column

$$7 + 1 + ? = 10$$

$$\therefore ? = 10 - 8 = \boxed{2}$$

- 87. (1)** $1 \times 2 = 2$; $2 \times 2 = 4$;

$$2 \times 4 = 8$$

$$\text{And, } 4 \times 8 = \boxed{32}$$

- 88. (2)** $7 + 9 - 5 = 11$

$$4 + 15 - 12 = 7$$

Therefore,

$$13 + 8 - 11 = \boxed{10}$$

- 89. (2)** $8 = (2)^3$

$$27 = (2 + 1)^3$$

$$64 = (2 + 2)^3$$

$$125 = (2 + 3)^3$$

$$216 = (2 + 4)^3$$

$$\therefore ? = (2 + 5)^3 = 7 \times 7 = 343$$

FINDING THE MISSING NUMBER

90. (3) The lowermost number in each column is the sum of the squares of the upper three numbers.

1st Column

$$(3)^2 + (5)^2 + (2)^2 = 9 + 25 + 4 = 38$$

2nd Column

$$(1)^2 + (4)^2 + (8)^2$$

$$= 1 + 16 + 64 = 81$$

3rd Column

$$(4)^2 + (7)^2 + (?)^2 = 74$$

$$= 16 + 49 + (?)^2 = 74$$

$$= (?)^2 = 74 - 65 = 9$$

$$\therefore ? = \sqrt{9} = 3$$

91. (2) Subtract 5 from the product of two numbers outside the bracket to get the number under the bracket.

$$a = 12 (175) 15$$

$$\text{or, } 12 \times 15 - 5 = 175$$

$$b = 14 (219) 16$$

$$\text{or, } 14 \times 16 - 5 = 219$$

Therefore,

$$e = 17 (?) 14$$

$$\text{or, } 17 \times 14 - 5$$

$$\text{or, } 238 - 5 = 233$$

92. (3) Proceed clockwise

$$2 + 1 = 3$$

$$3 + 2 = 5$$

$$5 + 4 = 9$$

$$9 + 8 = 17$$

$$17 + 16 = \boxed{33}$$

93. (3) **First Column**

$$24 \div 4 = 6$$

$$6 \div 3 = 2$$

$$2 \div 2 = 1$$

Second Column

$$144 \div 4 = 36$$

$$36 \div 3 = 12$$

$$12 \div 2 = 6$$

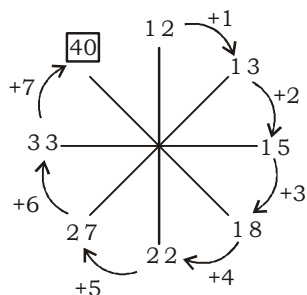
Third Column

$$384 \div 4 = \boxed{96}$$

$$96 \div 3 = 32$$

$$32 \div 2 = 16$$

94. (3)



$$95. (2) \begin{matrix} 6 & +7 & 13 & +2 & 15 \\ F & \rightarrow & M & \rightarrow & O \end{matrix}$$

$$\begin{matrix} 10 & +7 & 17 & +2 & 19 \\ J & \rightarrow & Q & \rightarrow & S \end{matrix}$$

$$\begin{matrix} 14 & +7 & 21 & +2 & \boxed{23} \\ N & \rightarrow & U & \rightarrow & W \end{matrix}$$

96. (4) **First Column**

$$(6)^2 = 36 \text{ and, } 6 \times (6-2) = 24$$

Second Column

$$(8)^2 = 64 \text{ and } 8 \times (8-2) = 48$$

Third Column

$$(7)^2 = 49 \text{ and } 7 \times (7-2) = \boxed{35}$$

97. (4) **First Column**

$$(5 + 4) \times 2 = 18$$

Second Column

$$(2 + 4) \times 5 = 30$$

Third Column

$$(4 + 7) \times 3 = \boxed{33}$$

$$98. (4) \text{ First Row : } \frac{3}{2} - 1 = \frac{1}{2}$$

$$\text{Second Row : } \frac{8}{3} - 2 = \frac{2}{3}$$

$$\text{Third Row : } \frac{19}{5} - 3 = \frac{4}{5}$$

99. (1) $(10 + 5 + 6) \times 2 = 42$

$$(17 + 3 + 14) \times 2 = 68$$

$$(8 + 15 + ?) \times 2 = 92$$

$$\Rightarrow 23 + ? = \frac{92}{2}$$

$$\Rightarrow ? = 46 - 23 = 23$$

100. (2) $(3 + 6 + 2)^2 = 121$

$$(10 + 2 + 3)^2 = 225$$

$$(? + 6 + 1)^2 = 100$$

$$\text{or, } ? + 6 + 1 = 10$$

$$\therefore ? = 10 - 7 = 3$$

101. (2) $5 + 7 + 9 = 21$

$$8 + 6 + 7 = 21$$

$$9 + 6 + ? = 21$$

$$\therefore ? = 21 - 15 = 6$$

102. (4) $7 \times 6 \times 12 = 504$

$$8 \times 5 \times 13 = 520$$

$$6 \times 9 \times ? = 486$$

$$\therefore ? = \frac{486}{9 \times 6} = 9$$

103. (3) $14 + 1 = 15$; $15 + 2 = 17$

$$17 + 4 = 21$$
; $21 + 8 = 29$

$$29 + 16 = 45$$

104. (2) $12 \times 5 - 10 = 60 - 10 = 50$

$$14 \times 9 - 16 = 126 - 16 = 110$$

$$16 \times 6 - ? = 84$$

$$\Rightarrow 96 - ? = 84$$

$$\therefore ? = 96 - 84 = 12$$

105. (2) $\sqrt{144} = 12$, $\sqrt{121} = 11$

$$12 \times 11 = 132$$

$$\sqrt{64} = 8$$
, $\sqrt{100} = 10$

$$8 \times 10 = 80$$

106. (3) $6 \times 12 \times 10 = 720$

$$8 \times 9 \times 5 = 360$$

$$\therefore ? = \frac{336}{12 \times 4} = 7$$

107. (2) $7 \times 8 + 6 = 56 + 6 = 62$

$$8 \times 9 + 7 = 72 + 7 = 79$$

$$6 \times ? + 5 = 47$$

$$\Rightarrow 6 \times ? = 47 - 5$$

$$\therefore ? = \frac{42}{6} = 7$$

108. (2) LCM of 9, 12, 15 = 180

$$\text{LCM of } 4, 16, 20 = 80$$

$$\text{LCM of } 5, 15, 25 = 75$$

109. (3) $5 \times 6 - 7 = 30 - 7 = 23$

$$4 \times 5 - 6 = 20 - 6 = 14$$

$$3 \times 4 - 5 = 12 - 5 = \boxed{7}$$

110. (4) $5 \times 2 \times 6 = 60$

$$8 \times 4 \times 2 = 64$$

$$\therefore ? = 7 \times 6 \times 3 = 126$$

111. (2) $6 \times 7 \times 11 = 462$

$$5 \times 8 \times 12 = 480$$

$$7 \times 4 \times ? = 224$$

$$\Rightarrow ? = \frac{224}{28} = 8$$

112. (4) $8 \times 6 - 9 = 48 - 9 = 39$

$$9 \times 7 - 11 = 63 - 11 = 52$$

$$9 \times 8 - ? = 59$$

$$\Rightarrow ? = 72 - 59 = \boxed{13}$$

113. (2) $8 \times 4 \times 2 = 64$

$$9 \times 5 \times 4 = 180$$

$$7 \times 6 \times ? = 294$$

$$\therefore ? = \frac{294}{42} = \boxed{7}$$

114. (4) First Row $\Rightarrow 3 \times 5 = 15$

$$\text{Second Row } \Rightarrow 5 \times 7 = 35$$

$$\text{Third Row } \Rightarrow 9 \times 5 = \boxed{45}$$

FINDING THE MISSING NUMBER

115. (2) First Column

$$\sqrt{36} = 6; 6 \times 2 = 12$$

Second Column

$$\sqrt{64} = 8; 8 \times 2 = 16$$

Third Column

$$\sqrt{100} = 10; 10 \times 2 = 20$$

116. (2) First Column

$$2 \times 7 \times 3 = 42$$

Second Column

$$7 \times 5 \times 8 = 280$$

Third Column

$$8 \times 3 \times ? = 120$$

$$\Rightarrow ? = \frac{120}{24} = 5$$

117. (3) First Column

$$14 \times 4 + 5 = 56 + 5 = 61$$

Second Column

$$22 \times 6 + 3 = 132 + 3 = 135$$

Third Column

$$37 \times 3 + 5 = 111 + 5 = 116$$

118. (1) First Column

$$\sqrt{64} + \sqrt{36} = 8 + 6 = 14$$

Second Column

$$\sqrt{25} + \sqrt{49} = 5 + 7 = 12$$

Third Column

$$\sqrt{81} + \sqrt{16} = 9 + 4 = 13$$

119. (1) First Row

$$27 = (3)^3; 9 = (3)^2$$

Second Row

$$16 = (4)^2; 64 = (4)^3$$

Third Row

$$512 = (8)^3; 64 = (8)^2$$

120. (3) First Column

$$5 + 7 = 4 \times 3$$

Second Column

$$3 + 5 = 4 \times 2$$

Third Column

$$7 + 9 = 4 \times ?$$

$$\Rightarrow ? = \frac{16}{4} = 4$$

Fourth Column

$$1 + 3 = 4 \times 1$$

121. (4) The product of the first number and the third number in each row is equal to the middle number.

First Row

$$8 \times 9 = 72$$

Second Row

$$6 \times 11 = 66$$

Third Row

$$9 \times 6 = \boxed{54}$$

122. (2) First Column

$$14 = 2 + (3 \times 4) = 2 + 12$$

Second Column

$$25 = 4 + (3 \times 7) = 4 + 21$$

Third Column

$$42 = 6 + (? \times 9) =$$

$$? = \frac{42 - 6}{9} = \frac{36}{9} = 4$$

123. (4) Row -wise

First Row

$$(5)^2 + (4)^2 = 25 + 16 = 41$$

Second Row

$$(7)^2 + (3)^2 = 49 + 9 = 58$$

Third Row

$$(10)^2 + (2)^2 = 100 + 4 = 104$$

124. (1) The lowermost number in each column is equal to the product of the first three numbers.

First Column

$$5 \times 6 \times 7 = 210$$

Second Column

$$6 \times 5 \times 4 = 120$$

Third Column

$$3 \times 4 \times 5 = 60$$

Fourth Column

$$? \times 8 \times 7 \times 6 = 336$$

125. (3) Row -wise

First Number = Second Number $\times$ Third Number

$$16 = 4 \times 4$$

$$81 = 3 \times 27$$

$$? = 25 \times 5 = 125$$

126. (1) The product of the first three numbers is equal to the lowermost number in each column.

$$6 \times 3 \times 2 = 36$$

$$4 \times 6 \times 3 = 72$$

$$5 \times 5 \times 4 = 100$$

127. (3) First column

$$5 \times 7 \times 1 = 35$$

Second column

$$4 \times 5 \times 3 = 60$$

Third column

$$9 \times 2 \times 4 = 72$$

Fourth column

$$6 \times 4 \times 5 = 120$$

128. (2) First Row

$$\sqrt{64} - \sqrt{36} = 8 - 6 = 2$$

Second Row

$$\sqrt{81} - \sqrt{25} = 9 - 5 = 4$$

Third Row

$$\sqrt{144} - \sqrt{16} = 12 - 4 = \boxed{8}$$

129. (2) First Row

$$10 \times 8 + 5 = 85$$

Second Row

$$7 \times 7 + 5 = 54$$

Third Row

$$8 \times 9 + 5 = \boxed{77}$$

130. (1) First Column

$$12 \times 3 \times 8 = 288$$

Second Column

$$13 \times 5 \times 8 = 520$$

Third Column

$$16 \times 6 \times ? = 384$$

$$\Rightarrow ? = \frac{384}{96} = \boxed{4}$$

131. (2) **Column**

$$(Ist)^2 - (IIrd) = (IIIrd)$$

$$Ist \text{ Row } 4^2 - 10 = 16 - 10 = 6$$

$$IIrd \text{ Row } 12^2 - 96 = 144 - 96 = 48$$

$$IIIrd \text{ Row } 16^2 - 152 = 256 - 152 = \boxed{104}$$

132. (2) First Column

$$9 \times 7 \times 4 = 252$$

Second Column

$$5 \times 6 \times 8 = 240$$

Third Column

$$6 \times 7 \times ? = 210$$

$$? = \frac{210}{42} = 5$$

133. (2) First Column

$$13 + 6 + 1 = 20$$

Second Column

$$6 + 8 + 6 = 20$$

Third Column

$$4 + 6 + ? = 20$$

$$\Rightarrow ? = 20 - 10 = \boxed{10}$$

134. (2) $16 + 32 = 48$

$$12 + 24 = 36$$

$$13 + 26 = 39$$

135. (1) First number $\times$ Third number + 3 = Middle number in each row.

FINDING THE MISSING NUMBER

First Row
 $3 \times 4 + 3 = 12 + 3 = 15$
 Second Row
 $7 \times 5 + 3 = 35 + 3 = 38$
 Third Row

$$3 \times 5 + 3 = 15 + 3 = \boxed{18}$$

- 136.** (3) In each column the lowermost number is 5 more than the sum of the first two numbers.

First Column
 $9 + 5 + 5 = 19$
 Second Column
 $10 + 7 + 5 = 22$
 Third Column
 $11 + 8 + 5 = 24$

- 137.** (3) The product of the first three numbers in each column is equal to the lowermost number.

First Column
 $7 \times 8 \times 3 = 168$
 Second Column
 $9 \times 6 \times 5 = 270$
 Third Column
 $6 \times 7 \times ? = 126$

$$\therefore ? = \frac{126}{42} = \boxed{3}$$

- 138.** (3) The product of the first three numbers in each column is equal to the lowermost number.

First Column
 $15 \times 7 \times 8 = 840$
 Second Column
 $11 \times 9 \times 6 = 594$
 Third Column
 $7 \times 12 \times ? = 420$

$$\therefore ? = \frac{420}{84} = 5$$

- 139.** (4) $(1)^2 + (5)^2 = 1 + 25 = 26$
 $(3)^2 + (2)^2 = 9 + 4 = 13$

$$(4)^2 + (8)^2 = 16 + 64 = \boxed{80}$$

- 140.** (2) $(10)^2 + 5 = 100 + 5 = 105$
 $(11)^2 + 5 = 121 + 5 = 126$

$$(12)^2 + 5 = 144 + 5 = \boxed{149}$$

- 141.** (2) $7 = 1 + 2 \times 3 = 1 + 6$
 $23 = 3 + 5 \times 4 = 3 + 20$
 $21 = 0 + 7 \times 3 = 0 + 21$

- 142.** (1) $(11 - 1) \div 2 = 5$
 $(2)^3 = 8$
 $(7 - 1) \div 2 = 3 + 20$
 $(3)^3 = 27$

$$\text{Therefore, } ? = \frac{3}{27}$$

- 143.** (4) First column

$$14 + 26 = 40 \text{ and } \frac{40}{4} = 10$$

Second Column

$$16 + 32 = 48 \text{ and } \frac{48}{4} = 12$$

Third Column

$$18 + 18 = 36 \text{ and } \frac{36}{4} = \boxed{9}$$

- 144.** (4) $\begin{array}{ccc} 9 \times 5 = 45 & 9 \downarrow & 9 \downarrow \\ 9 \times 5 = 45 & 5 \downarrow & 5 \downarrow \\ 17 \times 4 = 68 & & \\ 17 \times 4 = 68 & & \\ 8 \times 8 = 64 & & \end{array}$
 Now, $16 \times ? = 64$

$$\therefore ? = \frac{64}{16} = 4$$

- 145.** (2) $29 + 9 = 38$
 $38 - (7 + 8) = 38 - 15 = 23$
 $38 + 12 = 50$
 $50 - (10 + 11) = 50 - 21 = 29$
 $23 + 6 = 29$
 $(? + 15) - (13 + 14) = 29 + 6$
 $\Rightarrow ? + 15 - 27 = 35$
 $\Rightarrow ? = 35 + 12 = 47$

- 146.** (2) First Column
 $7 \times 2 \times 5 = 70$
 Second Column
 $4 \times 3 \times 9 = 108$
 Third Column
 $5 \times 2 \times 6 = 240$

$$\Rightarrow ? = \frac{240}{30} = 8$$

- 147.** (1) First Column
 $5 \times 6 + 4 = 30 + 4 = 34$
 Second Column
 $4 \times 7 + 2 = 28 + 2 = 30$
 Third Column
 $3 \times 8 + ? = 30$
 $\Rightarrow ? = 30 - 24 = 6$

- 148.** (2) Columnwise :

First Number $\times$ Third Number -
 Second Number = Lowermost
 Number

First Column

$$5 \times 4 - ? = 15 \Rightarrow 20 - ? = 15$$

$$\therefore ? = 20 - 15 = \boxed{5}$$

Second Column

$$2 \times 5 - 3 = 10 - 3 = 7$$

Third Column

$$7 \times 2 - 1 = 14 - 1 = 13$$

- 149.** (2) First Row

$$4 \times 8 \times 10 = 320$$

Second Row

$$2 \times ? \times 3 = 180$$

$$\Rightarrow ? = \frac{180}{6} = \boxed{30}$$

Third Row

$$9 \times 6 \times 4 = 216$$

- 150.** (2) First Column

$$18 - 12 = 6; 6 \times 6 = 36$$

Second Column

$$13 - 11 = 2; 2 \times 2 = 4$$

Third Column

$$19 - 16 = 3; 3 \times 3 = \boxed{9}$$

- 151.** (4) Columnwise

First Column

$$8 + 21 = 29; 29 + 21 = 50$$

Second Column

$$15 + 21 = \boxed{36}; 36 + 21 = 57$$

Third Column

$$22 + 21 = 43; 43 + 21 = 64$$

- 152.** (3) Columnwise

$$\text{First Column } \% 7 \times 8 \times 2 = 112$$

Second Column

$$\% 9 \times 3 \times 5 = 135$$

Third Column

$$\% 3 \times 5 \times ? = 900$$

$$\therefore ? = \frac{900}{15} = 60$$

- 153.** (1) First Column

$$7 + 4 - 2 = 9$$

Second Column

$$9 + 5 - 3 = 11$$

Third Column

$$6 + 7 - ? = 9$$

$$\Rightarrow ? = 13 - 9 = \boxed{4}$$

- 154.** (1) First Column

$$9 \times 12 = 18; 18 \times 2 = 36$$

Second Column

$$8 \times 2 = 16; 16 \times 2 = 32$$

Third Column

$$7 \times 2 = 14; 14 \times 2 = \boxed{28}$$

- 155.** (2) $3 + 2 = 5$

$$5 + 4 = 9$$

$$9 + 6 = \boxed{15}$$

FINDING THE MISSING NUMBER

156. (2) Columnwise

$$\frac{7 \times 10}{2} = 35$$

$$\frac{6 \times ?}{2} = 12$$

$$\Rightarrow ? = \frac{12}{3} = \boxed{4}$$

$$\frac{15 \times 12}{2} = 90$$

157. (1) First arrangement

$$18 - 6 + 17 + 9$$

$$\Rightarrow 12 + 17 + 9 = 38$$

Second arrangement

$$11 - 9 + 11 + 19$$

$$\Rightarrow 2 + 11 + 19 = 32$$

Third arrangement

$$6 - 3 + 15 + 26$$

$$\Rightarrow 3 + 15 + 26 = 44$$

Fourth arrangement

$$12 - 20 + 8 + 9$$

$$\Rightarrow 29 - 20 = \boxed{9}$$

158. (3) $2 + 3 + 5 = 30$

$$\Rightarrow 2 \times 3 \times 5 = 30$$

$$3 + 4 + 6 = 72$$

$$\Rightarrow 3 \times 4 \times 6 = 72$$

$$5 + 6 + 2 = 60$$

$$\Rightarrow 5 \times 6 \times 2 = 60$$

$$5 + 4 + 0 = 0$$

$$\Rightarrow 5 \times 4 \times 0 = 0$$

159. (1) $20 \times 5 = 4 \Rightarrow 20 \div 5 = 4$

$$48 \times 12 = 4 \Rightarrow 48 \div 12 = 4$$

Therefore, 77×11

$$\Rightarrow 77 \div 11 = 7$$

160. (2) $53 \div 31 = 2$

$$\Rightarrow (5 + 3) \div (3 + 1) = 2$$

$$\Rightarrow 8 \div 4 = 2$$

$$45 \div 27 = 1$$

$$\Rightarrow (4 + 5) \div (2 + 7) = 1$$

$$\Rightarrow 9 \div 9 = 1$$

$$69 \div 32 = 3$$

$$\Rightarrow (6 + 9) \div (3 + 2) = 3$$

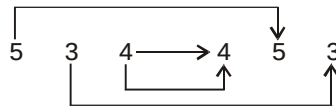
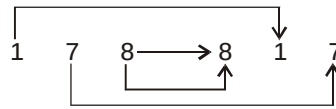
$$\Rightarrow 15 \div 5 = 3$$

So, $97 \div 26$

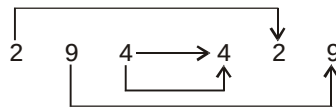
$$\Rightarrow (9 + 7) \div (2 + 6)$$

$$\Rightarrow 16 \div 8 = 2$$

161. (1)



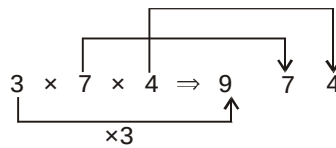
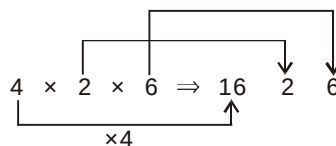
So,



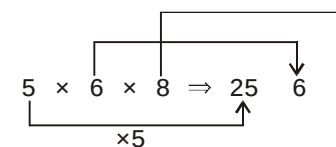
162. (4) $3 \times 2 = 6$; $6 \times 2 = 12$;

$$12 \times 2 = 24; 24 \times 2 = 48$$

163. (1)



Similarly,



164. (3) $(3)^3 = 27$

$$(2)^3 = 8$$

$$(4)^3 = 64$$

165. (2) $72 - 9 = 63$

$$37 - 9 = 28$$

$$\Rightarrow 72 + 37 \Rightarrow 6328$$

$$54 - 9 = 45$$

$$13 - 9 = 04$$

$$\Rightarrow 54 + 13 \Rightarrow 4504$$

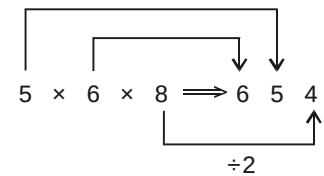
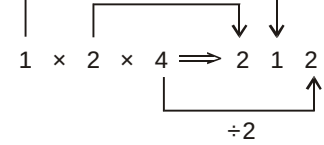
Similarly,

$$61 - 9 = 52$$

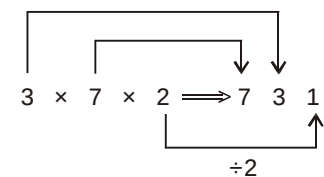
$$53 - 9 = 44$$

$$\Rightarrow 61 + 53 \Rightarrow 5244$$

166. (3)



Similarly,



167. (1) First Column

$$6 \times 8 + 3 = 48 + 3 = 51$$

Second Column

$$15 \times 4 + 5 = 60 + 5 = 65$$

Third Column

$$20 \times 5 + 20 = 100 + 20 = \boxed{120}$$

168. (1) First Column

$$\sqrt{16} + \sqrt{36}$$

$$\Rightarrow 4 + 6 = 10$$

Second Column

$$\sqrt{25} + \sqrt{64}$$

$$\Rightarrow 5 + 8 = 13$$

Third Column

$$\sqrt{9} + \sqrt{81}$$

$$\Rightarrow 3 + 9 = \boxed{12}$$

169. (3) First number + Second number = Third number in each row

Second number - First number = Fourth number in each row.

First Row

$$2 + 9 = 11; 9 - 2 = 7$$

Second Row

$$8 + 5 = 13; 5 - 8 = -3$$

Third Row

$$7 + ? = 10$$

$$\Rightarrow 10 - 7 = \boxed{3}$$

$$3 - 7 = -4$$

Fourth Row

$$6 + 4 = 10; 4 - 6 = \boxed{-2}$$

170. (1) $55 + 66 \Rightarrow 5 + 6 = 11$

$11 \times 3 = 33$

$22 + 99 \Rightarrow 2 + 9 = 11$

$11 \times 3 \Rightarrow 33$

Similarly,

$44 + 88 \Rightarrow 4 + 8 = 12$

$12 \times 3 = 36$

171. (2) **First Column**

$\sqrt{16} + \sqrt{25}$

$\Rightarrow 4 + 5 = 9$

Second Column

$\sqrt{49} + \sqrt{36}$

$\Rightarrow 7 + 6 = 13$

Third Column

$\sqrt{64} + \sqrt{81}$

$\Rightarrow 8 + 9 = \boxed{17}$

172. (1) **First Column**

$(3)^2 + (5)^2 + (1)^2$

$\Rightarrow 9 + 25 + 1 = 35$

Second Column

$(4)^2 + (7)^2 + (2)^2$

$\Rightarrow 16 + 49 + 4 = 69$

Third Column

$(6)^2 + (3)^2 + (7)^2$

$\Rightarrow 36 + 9 + 49 = \boxed{94}$

173. (1) $19 - 18 = 1; 1 \times 2 = 2$

$34 - 32 = 2; 2 \times 2 = 4$

$44 - 41 = 3; 3 \times 2 = 6$

174. (1)

$$\begin{array}{ccc}
 & 4 & \\
 & \swarrow & \searrow \\
 36 & & 2 \\
 (4 + 2)^2 = (6)^2 = 36 & & \\
 & 3 & \\
 & \swarrow & \searrow \\
 100 & & 7 \\
 (3 + 7)^2 = (10)^2 = 100 & & \\
 & 2 & \\
 & \swarrow & \searrow \\
 ? & & 5 \\
 (2 + 5)^2 = (7)^2 = 49
 \end{array}$$

175. (1) **First Column**

$\sqrt{49} - \sqrt{4} + \sqrt{25}$

$\Rightarrow 7 - 2 + 5 = 10$

Second Column

$\sqrt{81} - \sqrt{49} + \sqrt{16}$

$\Rightarrow 9 - 7 + 4 = \boxed{6}$

Third Column

$\sqrt{64} - \sqrt{9} + \sqrt{36}$

$\Rightarrow 8 - 3 + 6 = 11$

176. (2) **First Column**

$2 + 1 = 3; 3 + 1 = 4$

Second Column

$14 + 7 = 21; 21 + 7 = \boxed{28}$

Third Column

$21 + 7 = 28; 28 + 7 = 35$

Fourth Column

$28 + 7 = 35; 35 + 7 = \boxed{42}$

177. (4) **First Column**

$2 \times 12 = 24$

$2 \times 10 = 20$

Second Column

$3 \times 13 = 39$

$3 \times 10 = 40$

Third Column

$4 \times 14 = \boxed{56}$

$4 \times 10 = 40$

178. (1) **First Column**

$7 + 4 + 2 = 13$

$13 \times 3 = 39$

Second Column

$3 + 9 + 1 = 13$

$13 \times 5 = 65$

Third Column

$2 + 6 + 5 = 13$

$13 \times 7 = \boxed{91}$

179. (2) **First Column**

$\sqrt{81} \times \sqrt{4} \times \sqrt{36}$

$\Rightarrow 9 \times 2 \times 6 = 108$

Second Column

$\sqrt{64} \times \sqrt{9} \times \sqrt{16}$

$\Rightarrow 8 \times 3 \times 4 = 94$

Third Column

$\sqrt{16} \times \sqrt{49} \times \sqrt{25}$

$\Rightarrow 4 \times 7 \times 5 = \boxed{140}$

180. (4) First number in each row is the product of the second and third numbers.

First Row

$25 = 5 \times 5$

Second Row

$30 = 5 \times 6$

Third Row

$35 = ? \times 5$

$\therefore ? = \frac{35}{5} = \boxed{7}$

181. (4) **First Column**

$24 \times 2 + 5 = 48 + 5 = 53$

Second Column

$51 \times 4 + 7 = 204 + 7 = 211$

Third Column

$67 \times 6 + 5 = 402 + 5 = \boxed{407}$

182. (4)

$$\begin{array}{ccccccc}
 & & & & & & \\
 & & & & & & \\
 5 & \times & 6 & \times & 3 & \longrightarrow & 3 & 5 & 6 \\
 & & & & & & \uparrow & & \uparrow \\
 & & & & & & & &
 \end{array}$$

$$\begin{array}{ccccccc}
 & & & & & & \\
 & & & & & & \\
 1 & \times & 0 & \times & 5 & \longrightarrow & 5 & 1 & 0 \\
 & & & & & & \uparrow & & \uparrow \\
 & & & & & & & &
 \end{array}$$

Therefore,

$$\begin{array}{ccccccc}
 & & & & & & \\
 & & & & & & \\
 5 & \times & 6 & \times & 7 & \longrightarrow & 7 & 5 & 6 \\
 & & & & & & \uparrow & & \uparrow \\
 & & & & & & & &
 \end{array}$$

183. (1) Consecutive square numbers are given :

$(1)^2 = 1; (2)^2 = 4; (3)^2 = 9;$

$(4)^2 = 16; (5)^2 = 25; (6)^2 = 36;$

$(7)^2 = 49; (8)^2 = \boxed{64}$

$(9)^2 = 81; (10)^2 = 100;$

$(11)^2 = \boxed{121}; (12)^2 = 144$

184. (2) $(4)^2 = 16; (16)^2 = 256$ and $(256)^2 = 65536$

$(3)^2 = 9; (9)^2 = 81$ and

$(81)^2 = 6561$

$(2)^2 = 4; (4)^2 = 16$ and

$(16)^2 = 256$

185. (2) **First Row**

$16 \times 2 = 32; 16 \times 3; 16 \times 4 = 64$

Second Row

$17 \times 2 = 34;$

$17 \times 3 = 51; 17 \times 4 = 68$

Third Row

$18 \times 2 = 36; 18 \times 3 = 54$

$18 \times 4 = 72$

186. (1)

$$\begin{array}{ccc}
 4 & & \\
 & \searrow & \\
 & & 2 \\
 36 & \longleftarrow &
 \end{array}$$

FINDING THE MISSING NUMBER

$$(4 + 2)^2 = (6)^2 = 36$$

$$\begin{array}{c} 5 \\ \searrow \\ 49 \leftarrow 2 \\ (5 + 2)^2 = (7)^2 = 49 \end{array}$$

$$\begin{array}{c} 10 \\ \searrow \\ ? \leftarrow 2 \\ (10 + 2)^2 = (12)^2 = 144 \end{array}$$

187. (3) First Row

$$5 \times 5 = 25 \rightarrow \text{Third number}$$

Second Row

$$6 \times 3 = 18 \rightarrow \text{Third number}$$

$$[5 - 2 = 3]$$

Third Row

$$10 \times 2.5 = 25 \rightarrow \text{Third number}$$

$$[3 - 0.5 = 2.5]$$

Fourth Row

$$3 \times 1 = 3 \rightarrow \text{Third number}$$

$$[2.5 - 1.5 = 1]$$

188. (1) 29×48

$$\Rightarrow 2 \times 9 \times 4 \times 8 = 576$$

$$35 \times 16$$

$$\Rightarrow 3 \times 5 \times 1 \times 6 = 90$$

$$22 \times 46$$

$$\Rightarrow 2 \times 2 \times 4 \times 6 = 96$$

Therefore,

$$42 \times 17$$

$$\Rightarrow 4 \times 2 \times 1 \times 7 = \boxed{56}$$

189. (4) $54 \quad 30$

$$\begin{array}{c} 54 \quad 30 \\ \swarrow \quad \searrow \\ 24 \end{array}$$

$$30 + 24 = 54$$

$$\begin{array}{c} 112 \quad 42 \\ \swarrow \quad \searrow \\ 70 \end{array}$$

$$42 + 70 = 112$$

Therefore,

$$\begin{array}{c} ? \quad 28 \\ \swarrow \quad \searrow \\ 38 \end{array}$$

$$? = 28 + 38 = \boxed{66}$$

190. (4) First Row

$$216 - 7 = 209 ; 209 - 7 = 202$$

Second Row

$$522 - 7 = 515 ; 515 - 7 = 508$$

Third Row

$$633 - 7 = 626 ; 626 - 7 = \boxed{619}$$

191. (2) $15 + 29 = 44$

$$44 + (29 - 9) = 64$$

$$12 + 16 = 28$$

$$28 + (16 + 9) = 53$$

$$3 + 5 = 8 ; 8 + 5 = \boxed{13}$$

192. (1) First Column

$$(3 \times 4 \times 5) - 2 = 58$$

Second Column

$$(5 \times 6 \times 2) - 2 = 58$$

Third Column

$$(8 \times 4 \times 2) - 2 = 62$$

Fourth Column

$$(7 \times 6 \times 3) - 2 = \boxed{124}$$

193. (4) $\sqrt{\text{First number} \times \text{Third number}}$
= Second number in each column.

First Column

$$\sqrt{4 \times 9} = \sqrt{36} = 6$$

Second Column

$$\sqrt{3 \times 27} = \sqrt{81} = 9$$

Third Column

$$\sqrt{2 \times ?} = 10$$

$$\Rightarrow 2 \times ? = 100$$

$$\therefore ? = \frac{100}{2} = \boxed{50}$$

194. (1) $84 \div 12 = 7$ and $7 \times 2 = 14$

$$81 \div 9 = 9$$
 and $9 \times 2 = 18$

$$88 \div 11 = 8$$
 and $8 \times 2 = \boxed{16}$

195. (4) $235 \Rightarrow (2)^2 + (3)^2 + (5)^2 = 38$

$$452 \Rightarrow (4)^2 + (5)^2 + (2)^2 = 45$$

$$345 \Rightarrow (3)^2 + (4)^2 + (5)^2 = \boxed{50}$$

196. (2) $\begin{array}{c} 2 \times 3 \\ \times 2 \downarrow \quad \times 3 \downarrow \\ 4 \quad 9 \end{array}$

$$\begin{array}{c} 5 \times 6 \\ \times 5 \downarrow \quad \times 6 \downarrow \\ 25 \quad 36 \end{array}$$

$$\begin{array}{c} 1 \times 9 \\ \times 1 \downarrow \quad \times 9 \downarrow \\ 1 \quad 81 \end{array}$$

$$\begin{array}{c} 4 \times 7 \\ \times 4 \downarrow \quad \times 7 \downarrow \\ 16 \quad 49 \end{array}$$

197. (2) $6 \times \frac{4}{1} = 24$

$$24 \times \frac{5}{2} = 60$$

$$60 \times \frac{6}{3} = 120$$

$$120 \times \frac{7}{4} = 210$$

$$210 \times \frac{8}{5} = 336$$

So, correct answer is (2) 336.

Ind Method : Number of given figure :

$$\begin{array}{c} 6 \\ \downarrow +18 \\ 24 \\ \downarrow +36 \\ 60 \\ \downarrow +60 \\ 120 \\ \downarrow +90 \\ 210 \\ \downarrow +\square \\ \square \end{array} \begin{array}{l} +3 \times 6 \\ +4 \times 6 \\ +5 \times 6 \\ +6 \times 6 \end{array}$$

$$\text{So } ? = 210 + [90 + (6 \times 6)] = 336$$

198. (4) First Column

$$12 \times 3 + 4 = 40$$

Second Column

$$15 \times 4 + 6 = 66$$

Third Column

$$16 \times 5 + 4 = \boxed{84}$$

199. (1) $52 + 26 + 38 = 46$

$$\Rightarrow (5 \times 2) + (2 \times 6) + (3 \times 8) = 46$$

$$\Rightarrow 10 + 12 + 24 = 46$$

$$24 + 36 + 52 = 36$$

$$\Rightarrow (2 \times 4) + (3 \times 6) + (5 \times 2) = 36$$

$$\Rightarrow 8 + 18 + 10 = 36$$

$$79 + 55 + 28$$

$$\Rightarrow (7 \times 9) + (5 \times 5) + (2 \times 8)$$

$$\Rightarrow 63 + 25 + 16 = \boxed{104}$$

200. (1) The required answer would be 5.

201. (3) First Column

$$10 \times 3 + 6 = 36$$

Second Column

$$12 \times 4 + 6 = 54$$

Third Column

$$14 \times 4 + 6 = \boxed{62}$$

- 202.** (3) In each row the second number is twice of the first number and the third number is 5 more than that of the second number.

First Row

$$7 \times 2 = 14; 14 + 5 = 19$$

Second Row

$$6 \times 2 = 12; 12 + 5 = 17$$

Third Row

$$5 \times 2 = 10; 10 + 5 = \boxed{15}$$

- 203.** (3) First Column

$$92 + 64 + 52 = 208$$

Second Column

$$70 + 53 + 45 = 168$$

The difference between the sums of the three numbers of the first column and second column is 40.

Third Column

Therefore, the sum of the three numbers of the third column would be $168 - 40 = 128$

Thus,

$$? = 128 - (48 + 42)$$

$$\Rightarrow 128 - 90 = 38$$

- 204.** (1) First Column

$$\sqrt{25} + \sqrt{36} \Rightarrow 5 + 6 = 11$$

Second Column

$$\sqrt{49} + \sqrt{81} \Rightarrow 7 + 9 = 16$$

Third Column

$$\sqrt{16} + \sqrt{64} \Rightarrow 4 + 8 = 12$$

- 205.** (2) Subtract the first number from the product of the second and the third numbers to get the lowermost number in each column.

First Column

$$6 \times 9 - 3 = 54 - 3 = \boxed{51}$$

Second Column

$$7 \times 10 - 5 = 70 - 5 = 65$$

Third Column

$$8 \times 4 - 2 = 32 - 2 = 30$$

- 206.** (3) Column

$$(IIInd)^2 - (Ist) = (IIIrd)$$

$$Ist \text{ Row } 4^2 - 3 = 13$$

$$IIInd \text{ Row } 8^2 - 8 = 56$$

$$IIIrd \text{ Row } 3^2 - 5 = \boxed{4}$$

- 207.** (4) The sum of three numbers in each column is 83.

First Column

$$25 + 32 + 26 = 83$$

Second Column

$$\Rightarrow 83 - (17 + 40)$$

$$\Rightarrow 83 - 57 = 26$$

Third Column

$$41 + 11 + 31 = 83$$

- 208.** (3) The sum of all the three numbers in each column is 81.

First column

$$24 + 31 + 26 = 81$$

Second column

$$20 + 25 + 36 = 81$$

Third column

$$37 + ? + 19 = 81$$

$$\therefore ? = 81 - 56 = 25$$

- 209.** (4) Subtract the third number from the product of the first and the second numbers to get the lowermost number in each column.

First column

$$6 \times 7 - 5 = 42 - 5 = 37$$

Second column

$$5 \times 6 - 7 = 30 - 7 = 23$$

Third column

$$4 \times 5 - 6 = 20 - 6 = \boxed{14}$$

- 210.** (1) Subtract the second number from the first number and add the resultant to the third number to get the lowermost number in each column.

First column

$$8 - 6 + 2 = 2 + 2 = 4$$

Second column

$$12 - 5 + 8 = 7 + 8 = 15$$

Third column

$$13 - 10 + ? = 18$$

$$\Rightarrow ? = 18 - 3 = \boxed{15}$$

- 211.** (1) $\boxed{12} + 8 = 20$

$$20 + 16 = 36$$

$$36 + 32 = 68$$

$$68 + 64 = 132$$

$$132 + 128 = 260$$

- 212.** (2) First column

$$(7 + 5) + (2 \times 2)$$

$$\Rightarrow 12 + 4 = 16$$

Second column

$$(9 + 7) + (4 \times 4)$$

$$\Rightarrow 16 + 16 = 32$$

Third column

$$(8 + 6) + (3 \times 3)$$

$$\Rightarrow 14 + 9 = 23$$

- 213.** (2) $16 = 4 \times 4$

$$64 = 4 \times 4 \times 4$$

$$36 = 6 \times 6$$

$$216 = 6 \times 6 \times 6$$

$$64 = 8 \times 8$$

$$512 = 8 \times 8 \times 8$$

- 214.** (4) In each column the first number is the product of the second and the third numbers.

The lowermost number in each column can be obtained by subtracting the second number from the sum of the first and the third numbers.

First Column

$$3 \times 27 = 81$$

$$81 + 27 - 3 = 105$$

Second Column

$$5 \times 125 = 625$$

$$625 + 125 - 5 = 745$$

Third Column

$$7 \times ? = 2401$$

$$\therefore ? = \frac{2401}{7} = 343$$

Again,

$$2401 + ? - ? = 2737$$

$$\therefore 2737 + 7 - 2401$$

$$\Rightarrow ? = 2744 - 2401 = 343$$

- 215.** (4) First Row

$$72 + 44 = 116$$

$$116 - 68 = 48$$

Second Row

$$91 + ? - 86 = 48 + 12$$

$$\Rightarrow 91 + ? = 60 + 86$$

$$\Rightarrow ? = 146 - 91 = 55$$

Third Row

$$43 + 66 = 109$$

$$109 - 37 = 60 + 12 = 72$$

- 216.** (1) First Column

$$8 \times 3 + 6 = 24 + 6 = 30$$

Second Column

$$2 \times 9 + 4 = 18 + 4 = 22$$

FINDING THE MISSING NUMBER

Third Column

$$9 \times 6 + 9 = 54 + 9 = \boxed{63}$$

217. (2) First Column

$$8 \times 2 \times 4 = 64$$

$$\sqrt[3]{64} = 4$$

Second Column

$$3 \times 3 \times 3 = 27$$

$$\sqrt[3]{27} = 3$$

Third Column

$$12 \times 6 \times 3 = 216$$

$$\sqrt[3]{216} = 6$$

218. (3) First Column

$$4 \times 8 + 3$$

$$= 32 + 3 = 35$$

Second Column

$$7 \times 6 + 7$$

$$= 42 + 7 = 49$$

Third Column

$$9 \times 8 + 9$$

$$= 72 + 9 = 81$$

219. (1) First Column

$$(7)^2 + (5)^2 + (3)^2$$

$$= 49 + 25 + 9 = 83$$

Second Column

$$(6)^2 + (4)^2 + (2)^2$$

$$= 36 + 16 + 4 = 56$$

Third Column

$$(8)^2 + (9)^2 + (1)^2$$

$$= 64 + 81 + 1 = 146$$

220. (2) First Row

$$\frac{225}{15} = 15$$

$$\Rightarrow 15 \times 2 = 30$$

Second Row

$$\frac{70}{7} = 10$$

$$\Rightarrow 10 \times 2 = 20$$

Third Row

$$\frac{?}{3} = \frac{8}{2}$$

$$\Rightarrow 2 \times ? = 8 \times 3$$

$$\therefore ? = \frac{24}{2} = 12$$

221. (2) First Number $\div 3 =$ Second Number and Second Number $\div 2 =$ The lowest number in each column

First Column

$$96 \div 3 = 32$$

$$32 \div 2 = 16$$

Second Column

$$? \div 3 = 48$$

$$\Rightarrow ? = 48 \times 3 = 144$$

$$48 \div 2 = 24$$

Third Column

$$168 \div 3 = 56$$

$$56 \div 2 = 28$$

222. (4) First Column

$$7 \times 6 \times 2 = 84$$

Second Column

$$8 \times 9 \times 3 = 216$$

Third Column

$$5 \times ? \times 6 = 900$$

$$\Rightarrow ? = \frac{900}{30} = 30$$

223. (2) First Column

$$2 + 3 + 4 + 5 = 14$$

$$\Rightarrow 14 \times 14 = 196$$

Second Column

$$3 + 1 + 2 + 5 = 11$$

$$\Rightarrow 11 \times 11 = 121$$

Third Column

$$4 + 4 + 2 + 2 = 12$$

$$\Rightarrow 12 \times 12 = 144$$

Fourth Column

$$\sqrt{225} = 15$$

$$\Rightarrow ? + 2 + 6 + 2 = 15$$

$$\Rightarrow ? = 15 - 10 = 5$$

224. (4) First Column

$$6 \times 6 = 36$$

$$6 \times 4 = 24$$

$$6 \times 3 = 18$$

Second Column

$$8 \times 8 = 64$$

$$8 \times 6 = 48$$

$$8 \times 3 = 24$$

Third Column

$$7 \times 7 = 49$$

$$7 \times 5 = 35$$

$$7 \times 3 = 21$$

225. (3) First Column

$$5 \times 3 \times 4 = 60$$

Second Column

$$2 \times 6 \times 7 = 84$$

Third Column

$$3 \times 5 \times 2 = 30$$

226. (4) First Column

$$14 \times 2 = 7 \times 4$$

$$28 = 28$$

Second Column

$$10 \times 3 = 3 \times 10$$

$$30 = 30$$

Third Column

$$22 \times 4 = 11 \times ?$$

$$? = \frac{88}{11} = \boxed{8}$$

227. (4) First Column

$$8 \times 6 = 48$$

$$48 \div 3 = 16$$

First Column

$$4 \times 9 = 36$$

$$36 \div 3 = 12$$

First Column

$$7 \times 9 = \boxed{63}$$

$$63 \div 3 = 21$$

228. (1) First Row

$$3 + 8 + 7 = 18$$

Second Row

$$9 + 6 + 3 = 18$$

Third Row

$$2 + ? + 12 = 18$$

$$\Rightarrow ? = 18 - 14 = \boxed{4}$$

229. (1) First Column

$$(2 + 9) \times (9 - 2)$$

$$= 11 \times 7 = 77$$

Second Column

$$(4 + 7) \times (7 - 4)$$

$$= 11 \times 3 = 33$$

Third Column

$$(3 + 6) \times (6 - 3)$$

$$= 9 \times 3 = 27$$

Fourth Column

$$(2 + 5) \times (5 - 2)$$

$$= 7 \times 3 = 21$$

230. (2) Divide the sum of lower numbers by 2 to get the upper number.

First arrangement

$$\frac{16 + 20}{2} = \frac{36}{2} = 18$$

Second arrangement

$$\frac{18 + 22}{2} = \frac{40}{2} = 20$$

Third arrangement

$$\frac{9 + ?}{2} = 8$$

$$\Rightarrow 9 + ? = 2 \times 8$$

$$\Rightarrow ? = 16 - 9 = \boxed{7}$$

231. (3) First arrangement

$$2 + 3 + 1 = 6$$

$$4 + 3 + 2 = 9$$

$$9 - 6 = 3$$

Second arrangement

$$5 + 6 + 1 = 12$$

$$3 + 6 + 6 = 15$$

$$15 - 12 = \boxed{3}$$

232. (2) First column

$$(1 + 2 + 4 + 3) \times 5$$

$$= 10 \times 5 = 50$$

Second column

$$(3 + 4 + 5 + 2) \times 5$$

$$= 14 \times 5 = 70$$

Third column

$$? = (7 + 4 + 9 + 3) \times 5$$

$$\Rightarrow ? = 23 \times 5 = 115$$

233. (1) The sum of all the three numbers in each column is 40.

First column

$$13 + 11 + 16 = 40$$

Second column

$$9 + ? + 20 = 40$$

$$\Rightarrow ? = 40 - 29 = \boxed{11}$$

Third column

$$24 + 6 + 10 = 40$$

234. (1) Subtract the first number from the second number to get the third number in each row.

First Row

$$30 - 9 = 21$$

Second Row

$$? - 6 = 14$$

$$\Rightarrow ? = 14 + 6 = \boxed{20}$$

Third Row

$$40 - 12 = 28$$

235. (4) First column

$$7 \times 8 \times 3 = 168$$

Second column

$$6 \times 6 \times 4 = 144$$

Third column

$$6 \times ? \times 5 = 120$$

$$\Rightarrow ? = \frac{120}{30} = 4$$

236. (1) First column

$$(8)^2 + (3)^2 + (1)^2$$

$$\Rightarrow 64 + 9 + 1 = 74$$

Second column

$$(5)^2 + (7)^2 + (4)^2$$

$$\Rightarrow 25 + 49 + 16 = 90$$

Third column

$$? = (6)^2 + (5)^2 + (2)^2$$

$$\Rightarrow ? = 36 + 25 + 4 = \boxed{65}$$

237. (2) First Row

$$46 - 22 = 24$$

Second Row

$$58 - 27 = 31$$

Third Row

$$68 - 32 = \boxed{36}$$

238. (2) $3 \times 4 \Rightarrow (4)^3 = 64$

$$2 \times 3 \Rightarrow (3)^2 = 9$$

$$3 \times 2 \Rightarrow (2)^3 = 8$$

$$9 \times 2 \Rightarrow (2)^9 = \boxed{512}$$

239. (4) First Column

$$9 \times 4 \times 7 = 252$$

Second Column

$$5 \times 7 \times 8 = 280$$

Third Column

$$7 \times ? \times 3 = 126$$

$$\Rightarrow ? = \frac{126}{21} = \boxed{6}$$

240. (1) First Column

$$14 + 13 + 18 = 45$$

Second Column

$$19 + 15 + ? = 45$$

$$\Rightarrow ? = 45 - 34 = \boxed{11}$$

Third Column

$$12 + ? + 16 = 45$$

$$\Rightarrow ? = 45 - 28 = \boxed{17}$$

241. (3) First Column

$$5 \times 4 \times 8 = 160$$

Second Column

$$7 \times 8 \times 6 = 336$$

Third Column

$$9 \times 2 \times ? = 108$$

$$\Rightarrow ? = \frac{108}{18} = 6$$

242. (3) The sum of all the four numbers in each column is equal to 21.

First Column

$$6 + 5 + 7 + 3 = 21$$

Second Column

$$7 + 3 + ? + 3 = 21$$

$$\Rightarrow ? = 21 - 13 = 8$$

Third Column

$$4 + 5 + 6 + 6 = 21$$

243. (1) **First Column**

$$9 \times 8 - 7 = 72 - 7 = 65$$

Second Column

$$6 \times 5 - 4 = 30 - 4 = 26$$

Third Column

$$? \times 6 - 3 = 39$$

$$= ? \times 6 = 39 + 3 = 42$$

$$= ? = \frac{42}{6} = 7$$

244. (3) $3 \times 8 = 4 \times 6$

Similarly,

$$? + 10 = 20 + 14$$

$$= ? = 34 - 10 = 24$$

245. (3) First Column

$$15 \times 4 = 60$$

$$60 \times 4 = 240$$

Second Column

$$25 \times 4 = 100$$

$$100 \times 4 = 400$$

Third Column

$$30 \times 4 = \boxed{120}$$

$$120 \times 4 = 480$$

246. (4) First Column

$$6 \times 6 = 36$$

$$6 \times 3 = 18$$

Second Column

$$7 \times 7 = 49$$

$$7 \times 4 = 28$$

Third Column

$$8 \times 8 = 64$$

$$8 \times 5 = \boxed{40}$$

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247. (3) First Column

$$7 \times 8 \times 2 = 112$$

Second Column

$$5 \times 4 \times 8 = 162$$

Third Column

$$3 \times 9 \times ? = 162$$

$$\Rightarrow ? = \frac{162}{27} = 6$$

248. (4) $2 \times 8 + 1 = 17$

$$17 \times 8 + 1 = 137$$

$$137 \times 8 + 1 = 1097$$

249. (4) First Column

$$9 + 4 = 13$$

$$13 - 3 = 10$$

$$10 + 4 = 14$$

$$14 - 3 = 11$$

Second Column

$$11 + 4 = 15$$

$$15 - 3 = 12$$

$$12 + 4 = 16$$

$$16 - 3 = 13$$

Third Column

$$13 + 4 = 17$$

$$17 - 3 = 14$$

$$14 + 4 = 18$$

$$18 - 3 = 15$$

250. (3) Subtract the lowest number from the sum of the two largest numbers to get the lowermost number.

First Column

$$18 + 6 - 3 = 21$$

Second Column

$$21 + 9 - 4 = 26$$

Third Column

$$24 + 8 - 3 = 29$$

251. (1) First Row

$$25 + 15 = 40 ; \frac{40}{5} = 8$$

Second Row

$$65 + 25 = 90 ; \frac{90}{5} = 18$$

Third Row

$$45 + 15 = 60 ; \frac{60}{5} = 12$$

252. (2) In each row the product of the first and the third numbers is equal to the second number.

$$\text{First row} \quad 2 \times 2 = 4$$

$$\text{Second row} \quad 3 \times 3 = 9$$

$$\text{Third row} \quad 4 \times 4 = 16$$

$$\text{Fourth row} \quad 8 \times 8 = 64$$

253. (4) $4 \times 4 \times 4 = 64$

$$7 \times 7 \times 7 = 343$$

$$11 \times 11 \times 11 = 1331$$

254. (4) First Row

$$40 + 30 = 72$$

$$\text{And, } \frac{72}{6} = 12$$

Second Row

$$30 + 24 = 54$$

$$\text{And, } \frac{54}{6} = 9$$

Third Row

$$54 + ? = 90$$

$$\Rightarrow ? = 90 - 54 = 36$$

$$\text{And, } \frac{90}{6} = 15$$

255. (2) First Column

$$1 + 8 + 27 = 36$$

$$\Rightarrow 36 - 1^2 = 35$$

Second Column

$$216 + 125 + 64 = 405$$

$$\Rightarrow 405 - 2^2 = 401$$

Third Column

$$343 + 512 + ? = 1575 + 3^2$$

$$\Rightarrow 855 + ? = 1584$$

$$\Rightarrow ? = 1584 - 855 = 729$$

256. (3) First Column

$$(2 \times 4) + (4 \times 6)$$

$$\Rightarrow 8 + 24 = 32$$

Second Column

$$(3 \times 5) + (5 \times 7)$$

$$\Rightarrow 15 + 35 = 50$$

Third Column

$$(8 \times 10) + (10 \times 12)$$

$$\Rightarrow 80 + 120 = 200$$

257. (1) First Row

$$4 \times 3 \times 2 + 8$$

$$\Rightarrow 24 + 8 = 32$$

Second Row

$$5 \times 3 \times 1 + 9$$

$$\Rightarrow 15 + 9 = 24$$

Third Row

$$7 \times 3 \times 3 + 7$$

$$\Rightarrow 63 + 7 = 70$$

Fourth Row

$$2 \times 9 \times 4 + 12$$

$$\Rightarrow 72 + 12 = 84$$

258. (4) $5 = 2^2 + 1$

$$10 = 3^2 + 1$$

$$26 = 5^2 + 1$$

$$50 = 7^2 + 1$$

$$122 = 11^2 + 1$$

259. (4) $2 \times 2 = 4$

$$4 \times 2 = 8$$

$$8 \times 2 = 16$$

$$16 \times 2 = 32$$

$$32 \times 2 = 64$$

$$64 \times 2 = 128$$

$$128 \times 2 = \boxed{256}$$

260. (2) $7 \times 2 - 1 = 13$

$$10 \times 2 - 1 = 19$$

$$20 \times 2 - 1 = 39$$

261. (3) First Column

$$(2 + 4) \times 6 \Rightarrow 6 \times 6 = 36$$

Second Column

$$(3 + 6) \times 9 \Rightarrow 9 \times 9 = 81$$

Third Column

$$(4 + 8) \times 12 \Rightarrow 12 \times 12 = 144$$

262. (1) First Column

$$5 \times 6 + 4 = 34$$

Second Column

$$4 \times 7 + 2 = 30$$

Third Column

$$3 \times 8 + ? = 30$$

$$\Rightarrow ? = 30 - 24 = 6$$

263. (1) First Row

$$\frac{21}{7} = 3$$

Second Row

$$\frac{?}{3} = 4$$

$$\Rightarrow ? = 3 \times 4 = 12$$

Third Row

$$\frac{16}{8} = 2$$

264. (3) $2 \times 6 = 12$

$$3 \times 4 = 12$$

$$1 \times ? = 12$$

$$\Rightarrow ? = 12$$

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265. (3) First Row

$$8 + 5 = 13$$

$$13 - 3 = 10$$

Second Row

$$7 + 5 = 12$$

$$12 - 3 = 9$$

Third Row

$$10 + 5 = 15$$

$$15 - 3 = 12$$

266. (2) $7 \times 6 = 42$

$$6 \times 6 = 36$$

$$9 \times 6 = 54$$

267. (3) First Column

$$5 \times 2 = 10$$

$$5 + 2 = 7$$

Second Column

$$6 \times 3 = 18$$

$$6 + 3 = 9$$

Third Column

$$8 \times 5 = 40$$

$$8 + 5 = 13$$

268. (3) First Column

$$21 + 44 = 65$$

Second Column

$$? + 55 = 77$$

$$\Rightarrow ? = 77 - 55 = 22$$

Third Column

$$21 + 66 = 87$$

269. (1) $9 \times 8 - 1 = 71$

$$6 \times 5 - 4 = 26$$

$$7 \times 6 - 7 = 35$$

270. (1) $10 + 7 = 17$

$$9 + ? + 6 = 17$$

$$\Rightarrow ? = 17 - 15 = 2$$

$$15 + 2 = 17$$

271. (1) First Row

$$874 = 46 \times 19$$

Second Row

$$1 + 3 + 5 = 9$$

Third Row

$$2 + 4 + 6 = 12$$

Fourth Row

$$3 + 1 + 9 = 13$$

$$\text{Now, } 9 + 12 + 13 = 34$$

Fifth Row

$$1 + 7 + ? = 46 - 34 = 12$$

$$\Rightarrow ? = 12 - 8 = 4$$

272. (2) First Row

$$4 \times 2 = 8$$

$$8 \times 2 = 16$$

$$16 \times 2 = 32$$

Second Row

$$5 \times 3 = 15$$

$$15 \times 3 = 45$$

$$45 \times 3 = 135$$

Third Row

$$6 \times 4 = 24$$

$$24 \times 4 = 96$$

$$96 \times 4 = 384$$

273. (1) First Column

$$(6 + 7) \times 5 = 65$$

Second Column

$$(3 + 2) \times 4 = 20$$

Third Column

$$(4 + ?) \times 9 = 45$$

$$\Rightarrow 4 + ? = \frac{45}{9}$$

$$\Rightarrow ? = 5 - 4 = 1$$

274. (4) The sum of three numbers in each column and each row is 132

First Column

$$43 + 42 + 47 = 132$$

Second Column

$$48 + 44 + ? = 132$$

$$\Rightarrow ? = 132 - 92 = 40$$

First Row

$$43 + 48 + 41 = 132$$

Second Row

$$42 + 44 + ? = 132$$

$$\Rightarrow ? = 132 - 86 = \boxed{46}$$

Third Row

$$47 + 40 + ? = 132$$

$$\Rightarrow ? = 132 - 87 = \boxed{45}$$

275. (1) $6 \times 3 + 1 = 19$

$$19 \times 3 + 1 = 58$$

$$58 \times 3 + 1 = \boxed{175}$$

276. (4) $7 \times 3 = 21$

$$14 \times 3 = 42$$

$$18 \times 3 = 54$$

277. (2) $9 + 8 = 17$

$$17 + 16 = 33$$

$$33 + 32 = \boxed{65}$$

278. (3) $8 \times 8 = 64$

$$11 \times 11 = 121$$

$$12 \times 12 = 144$$

279. (4) First Column

$$(3)^2 + (2)^2 + (1)^2$$

$$\Rightarrow 9 + 4 + 1 = 14$$

Second Column

$$(4)^2 + (3)^2 + (2)^2$$

$$\Rightarrow 16 + 9 + 4 = 29$$

Third Column

$$(5)^2 + (4)^2 + (3)^2$$

$$\Rightarrow 25 + 16 + 9 = 50$$

280. (3) The sum of three numbers in each row and each column is equal to 342.

First Row

$$113 + 118 + \boxed{111} = 342$$

Second Row

$$112 + \boxed{114} + 116 = 342$$

Third Row

$$\boxed{117} + 110 + 115 = 342$$

First Column

$$113 + 112 + 117 = 342$$

Second Column

$$118 + 114 + 110 = 342$$

Third Column

$$111 + 116 + 115 = 342$$

281. (2) $30 \times 3 = 90$

$$\text{Similarly, } 8 \times 10 = 80$$

282. (1) $22 \times 3 = 66$

$$66 \times 3 = 198$$

$$198 \times 3 = 594$$

283. (2) First Column

$$(2 + 3) \times 15$$

$$= 5 \times 15 = 75$$

Second Column

$$(7 + 4) \times 15$$

$$= 11 \times 15 = 165$$

Third Column

$$(14 + 5) \times 15$$

$$= 19 \times 15 = 285$$

284. (4) First Row

$$(1)^2 = 1; (2)^2 = 4; (3)^2 = 9;$$

$$(4)^2 = 16$$

Second Row

$$1 + 1 = 2; 2 + 1 = 3;$$

$$3 + 1 = 4$$

Third Row

$$2 + 2 = 4; 4 + 2 = 6;$$

$$6 + 2 = \boxed{8}$$

| FINDING THE MISSING NUMBER |

285. (2) First Column

$$(4 + 7)^2 = 121$$

Second Column

$$(8 + 2)^2 = 100$$

Third Column

$$(9 + 5)^2 = 196$$

Fourth Column

$$(7 + 6)^2 = 169$$

286. (1) First Column

$$43 + 21 + 35 = 99$$

Second Column

$$25 + 40 + 34 = 99$$

Third Column

$$? + 35 + 28 = 99$$

$$\Rightarrow ? = 99 - 63 = \boxed{36}$$

287. (2) First Row

$$7 \times 4 = 28 ; 2 \times 14 = 28$$

Second Row ;

$$4 \times 9 = 36 ; 3 \times 12 = 36$$

Third Row

$$? = \frac{6 \times ?}{24} = 4$$

$$\Rightarrow ? = \frac{4 \times 24}{6} = 16$$

288. (3) First Column

$$= 9 - 3 = 6 ; 6 + 2 + 6 = 14$$

Second Column

$$= 8 - 6 = 2 ; 6 + 2 + ? = 14$$

$$\Rightarrow ? = 14 - 8 = 6$$

Third Column

$$= 5 - 5 = 0 ; 7 + 7 + 0 = 14$$

289. (2) $4 + 5 = 9$

$$5 + 9 = 14$$

$$9 + 14 = 23$$

$$14 + 23 = 37$$

$$23 + 37 = 60$$

$$37 + 60 = \boxed{97}$$

290. (4) First Column

$$4 \times 3 + 2 = 14$$

Second Column

$$4 \times 6 + 2 = 26$$

Third Column

$$3 \times 5 + ? = 17$$

$$\Rightarrow ? = 17 - 15 = \boxed{2}$$

291. (2) $2 \times 2 + 1 = 5$

$$5 \times 2 - 1 = 9$$

$$9 \times 2 + 1 = 19$$

$$19 \times 2 - 1 = 37$$

$$37 \times 2 + 1 = 75$$

$$75 \times 2 - 1 = 149$$

$$149 \times 2 + 1 = \boxed{299}$$

292. (1) First Row

$$(1)^3 + (3)^3 = 28$$

Second Row

$$(3)^2 + (7)^2 = 58$$

Third Row

$$(2)^3 + (5)^3 = 133$$

Fourth Row

$$(4)^2 + (9)^2$$

$$\Rightarrow 16 + 81 = \boxed{97}$$

293. (3) $\sqrt[3]{8} + \sqrt[3]{27} + \sqrt[3]{64} + \sqrt{121}$

$$= 2 + 3 + 4 + 11 = 20$$

294. (1) First Column

$$5 \times 4 + 6 = 26$$

Second Column

$$7 \times 4 + 4 = 32$$

Third Column

$$9 \times 4 + 8 = 44$$

295. (3) First Column

$$9 \times 3 \times 3 = 81$$

Second Column

$$11 \times 4 \times 4 = 176$$

Third Column

$$13 \times 7 \times 5 = 455$$

296. (3) First Column

$$\sqrt{25} + \sqrt{81} + \sqrt{16}$$

$$= 5 + 9 + 4 = 18$$

Second Column

$$\sqrt{36} + \sqrt{9} + \sqrt{49}$$

$$= 6 + 3 + 7 = 16$$

Third Column

$$\sqrt{64} + \sqrt{4} + \sqrt{100}$$

$$= 8 + 2 + 10 = 20$$

297. (3) First Row

$$6 \times 5 = 3 \times 10$$

Second Row

$$2 \times 8 = ? \times 4$$

$$\Rightarrow ? = \frac{2 \times 8}{4} = 4$$

Third Row

$$4 \times 6 = 3 \times 8$$

Fourth Row

$$5 \times 9 = 15 \times 3$$

298. (2) $5 \times 7 = 35 ; 35 + 6 = 41$

$$10 \times 4 = 40 ; 40 + 7 = 47$$

$$\text{And, } 47 - 41 = 6$$

$$8 \times 6 = 48 ; 48 + 7 = 55$$

$$\text{And, } 55 - 47 = 8 (= 6 + 2)$$

Alternatively,

$$5 + 1 = 6 ; 6 + 1 = 7$$

$$4 + 3 = 7 ; 7 + 3 = 10$$

$$6 + 1 = 7 ; 7 + 1 = 8$$

299. (4) First Row

$$\frac{6}{2} \times 11 = 33$$

$$\Rightarrow 33 - 25 = 8$$

Second Row

$$\frac{8}{2} \times 6 = 24$$

$$\Rightarrow 24 - 16 = 8$$

Third Row

$$\frac{12}{2} \times 5 = 30$$

$$\Rightarrow 30 - 22 = 8$$

300. (3) First Column

$$8 \times 7 - 12 = 44$$

Second Column

$$5 \times 6 - 20 = 10$$

Third Column

$$4 \times 8 - 12 = \boxed{20}$$

301. (2) First Row

$$121 + 49 = 170$$

$$\text{and } 81 \times 2 + 8 = 170$$

Second Row

$$100 + 36 = 136$$

$$\text{and, } 64 \times 2 + 8 = 136$$

Third Row

$$25 + 15 = 40$$

$$\text{and, } 16 \times 2 + 8 = 40$$

Fourth Row

$$41 + 29 = 70$$

$$\text{and, } 31 \times 2 + 8 = 70$$

302. (4) First Column

$$2 \times 8 + 6 = 22$$

$$\text{and, } 22 - 10 = 12$$

Second Column

$$3 \times 6 + 3 = 21$$

$$\text{and, } 21 - 9 = 12$$

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Third Column

$$4 \times 9 + 5 = 41$$

$$\text{and, } 41 - 29 = 12$$

303. (2) First Column

$$6 \times 8 + 3 = 51$$

Second Column

$$15 \times 4 + 5 = 65$$

Third Column

$$20 \times 5 + 20 = 120$$

304. (3) First Row

$$(56 - 21) \times 2 = 70$$

$$= 35 \times 2 = 70$$

Second Row

$$(87 - 45) \times 2 = 84$$

$$= 42 \times 2 = 84$$

Third Row

$$(180 - 115) \times 2$$

$$= 65 \times 2 = 130$$

305. (2) First Row

$$\frac{24 + 36}{3} = 20$$

$$\Rightarrow \frac{60}{3} = 20$$

Second Row

$$\frac{15 + 18}{3} = 11$$

$$\Rightarrow \frac{33}{3} = 11$$

Third Row

$$\frac{55 + ?}{3} = 40$$

$$\Rightarrow 55 + ? = 3 \times 40$$

$$\Rightarrow ? = 120 - 55 = 65$$

306. (2) First Row

$$8 \times 4 = 32$$

Second Row

$$7 \times 5 = 35$$

Third Row

$$2 \times 3 = 6$$

307. (3) Proceed clockwise :

$$8 \times 2 = 16; 8 \times 3 = 24;$$

$$8 \times 4 = 32; 8 \times 5 = 40;$$

$$8 \times 6 = 48; 8 \times 7 = 56;$$

$$8 \times 8 = 64$$

308. (3) First Column

$$3 \times 4 \times 3 = 36$$

Second Column

$$5 \times 6 \times 7 = 210$$

Third Column

$$9 \times ? \times 2 = 36$$

$$\Rightarrow ? = \frac{36}{18} = 2$$

309. (3) In each row, first number $\times$ third number = middle number.

First Row

$$7 \times 8 = 56$$

Second Row

$$9 \times ? = 45$$

$$\Rightarrow ? = \frac{45}{9} = 5$$

Third Row

$$11 \times 9 = 99$$

310. (1) The larger number is the sum of the two smaller numbers in each row.

First row

$$8 + 4 = 12$$

Second row

$$8 + 4 = 12$$

Third row

$$4 + 8 = 12$$

Fourth row

$$100 - 44 = 56$$

311. (1) First Column

$$17 - 12 = 5 \text{ and } (5)^2 = 25$$

Second Column

$$13 - 11 = 2 \text{ and } (2)^2 = 4$$

Third Column

$$19 - 16 = 3 \text{ and } (3)^2 = 9$$

312. (2) First Column

$$(11)^2 - (1)^2 \Rightarrow 121 - 1 = 120$$

Second Column

$$(7)^2 - (2)^2 \Rightarrow 49 - 4 = 45$$

Third Column

$$(5)^2 - (3)^2 \Rightarrow 25 - 9 = 16$$

313. (2) $3 + 4 = 7$

$$9 + ? = 21$$

$$\therefore ? = 21 - 9 = 12$$

314. (4) First Row

$$28 = 4 \times 7 \text{ and } 4 \times 5 = 20$$

Second Row

$$84 = 7 \times 12 \text{ and } 7 \times 5 = 35$$

Third Row

$$45 = 5 \times 9 \text{ and } 5 \times 5 = 25$$

315. (4) First Column

$$(4 + 3) \times 2 = 14$$

$$\Rightarrow 7 \times 2 = 14$$

Second Column

$$(3 + 5) \times 2 = 16$$

$$\Rightarrow 8 \times 2 = 16$$

Third Column

$$(6 + 4) \times 3$$

$$\Rightarrow 10 \times 3 = 30$$

316. (2) First Column

$$(7)^2 + (4)^2 + (2)^2$$

$$= 49 + 16 + 4 = 69$$

Second Column

$$(3)^2 + (9)^2 + (1)^2$$

$$= 9 + 81 + 1 = 91$$

Third Column

$$(2)^2 + (6)^2 + (5)^2$$

$$= 4 + 36 + 25 = 65$$

317. (2) First Column

$$(8 + 7) \times 6$$

$$\Rightarrow 15 \times 6 = 90$$

Second Column

$$(7 + 6) \times 5$$

$$\Rightarrow 13 \times 5 = 65$$

Third Column

$$(6 + 5) \times 4$$

$$\Rightarrow 11 \times 4 = 44$$

318. (4) First Column

$$6 \times 8 + 3 = 51$$

Second Column

$$15 \times 4 + 5 = 65$$

Third Column

$$20 \times 5 + 20 = 120$$

319. (3) $(1)^3 = 1$

$$(2)^3 = 8$$

$$(3)^3 = 27$$

$$(4)^3 = 64$$

320. (2) $27 - 8 = 19$

$$53 - 34 = 19$$

$$? = 21 + 19 = 40$$

321. (4) First Column

$$8 + 6 + 7 + 2 + 3 + 6 = 32$$

Second Column

$$4 + 8 + 0 + 5 + 1 + 0 + \boxed{2 + 4 + 0}$$

$$= 24$$

Third Column

$$3 + 8 + 2 + 1 + 1 + 2 = 17$$

Now,

$$32 - 24 = 8$$

$$24 - 17 = 7$$

322. (1) First Row

$$8 + 6 = 7 \times 2$$

Second Row

$$9 + 13 = 11 \times 2$$

Third Row

$$3 + 7 = 5 \times 2$$

323. (1) First Column

$$3 \times 2 \times 6 = 36$$

Second Column

$$4 \times 5 \times 7 = 140$$

Third Column

$$9 \times 2 \times ? = 18$$

$$\Rightarrow ? = \frac{18}{18} = 1$$

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324. (3) First Column

$$\begin{aligned}(5 + 4) \times 2 \\ = 9 \times 2 = 18 \\ \text{Second Column} \\ (2 + 4) \times 5 \\ = 6 \times 5 = 30 \\ \text{Third Column} \\ (4 + 7) \times 3 \\ = 11 \times 3 = 33\end{aligned}$$

325. (4) In each row the sum of the first and the third number is twice of the second number

$$\begin{aligned}\text{First Row} \\ ? + 150 = 120 \times 2 \\ \Rightarrow ? = 240 - 150 = 90 \\ \text{Second Row} \\ 110 + 90 = 100 \times 2 \\ \Rightarrow 200 = 200 \\ \text{Third Row} \\ 80 + 40 = 60 \times 2 \\ \Rightarrow 120 = 120\end{aligned}$$

326. (4) First Column

$$7 + \left(\frac{16}{2}\right) = 7 + 8 = 15$$

Third Column

$$5 + \left(\frac{8}{2}\right) = 5 + 4 = 9$$

Therefore,
Second Column

$$10 + \left(\frac{40}{2}\right) = 10 + 20 = 30$$

327. (2) First Row

$$\begin{aligned}7 \times \sqrt{4} \\ \Rightarrow 7 \times 2 = 14 \\ \text{Second Row}\end{aligned}$$

$$\begin{aligned}4 \times \sqrt{9} \\ \Rightarrow 4 \times 3 = 12 \\ \text{Third Row}\end{aligned}$$

$$\begin{aligned}6 \times 4 = 24 \\ \therefore ? = 4 \times 4 = 16\end{aligned}$$

328. (1) First Column

$$\begin{aligned}(2)^3 + (1)^3 + (3)^3 \\ = 8 + 1 + 27 = 36 \\ \text{Second Column} \\ (4)^3 + (2)^3 + (1)^3 \\ = 64 + 8 + 1 = \boxed{73}\end{aligned}$$

$$\begin{aligned}\text{Third Column} \\ (0)^3 + (4)^3 + (3)^3 \\ = 0 + 64 + 27 = 91\end{aligned}$$

329. (3) In each row, first number + second number is equal to third number + fourth number.

First Row

$$2 + 4 = 2 + 4$$

Second Row

$$5 + 5 = 3 + 7$$

Third Row

$$8 + 6 = 4 + ?$$

$$\Rightarrow ? = 14 - 4 = \boxed{10}$$

330. (2) In each row, first number $\times$ 9 = third number and second number $\times$ 9 = fourth number

First Row

$$8 \times 9 = 72; 13 \times 9 = 117$$

Second Row

$$21 \times 9 = 189; 7 \times 9 = 63$$

Third Row

$$4 \times 9 = 36; 6 \times 9 = 54$$

331. (3) First Column

$$6 \times 8 + 3 = 51$$

Second Column

$$15 \times 4 + 5 = 65$$

Third Column

$$20 \times 5 + 20 = 120$$

332. (2) First Column

$$7 \times 8 + 4 = 60$$

Second Column

$$9 \times 9 + 9 = 90$$

Third Column

$$8 \times ? + 6 = 70$$

$$\Rightarrow 8 \times ? = 70 - 6$$

$$\Rightarrow ? = \frac{64}{8} = 8$$

333. (4) First Column

$$5 \times 6 \times 8 = 240$$

Second Column

$$5 \times 8 \times 12 = 480$$

Third Column

$$15 \times 6 \times ? = 450$$

$$\Rightarrow ? = \frac{450}{90} = \boxed{5}$$

334. (3) In each Column,

First number + Third number =
Second number + Fourth number

First Column

$$10 + 28 = 18 + 20$$

Second Column

$$6 + 34 = 14 + 26$$

Third Column

$$9 + ? = 17 + 32$$

$$\Rightarrow ? = 49 - 9 = 40$$

335. (4) First Column

$$4 \times 8 = 32$$

$$32 \times 8 = 256$$

Second Column

$$8 \times 8 = 64$$

$$64 \times 8 = 512$$

Third Column

$$16 \times 8 = 128$$

$$128 \times 8 = 1024$$

Alternatively,

First Row,

$$4 \times 2 = 8; 8 \times 2 = 16$$

Second Row,

$$32 \times 2 = 64; 64 \times 2 = 128$$

Third Row,

$$256 \times 2 = 512; 512 \times 2 = 1024$$

336. (3) First Column

$$6 \times 4 = 3 \times 8$$

Second Column

$$18 \times 3 = 2 \times 27$$

Third Column

$$15 \times ? = 5 \times 9$$

$$\Rightarrow ? = \frac{5 \times 9}{15} = 3$$

337. (2) First Column

$$24 + 5 + 9 = 38$$

Second Column

$$30 + 7 + 1 = 38$$

Third Column

$$19 + 18 + ? = 38$$

$$\Rightarrow ? = 38 - 37 = \boxed{1}$$

338. (3) First Column

$$2 \times 6 \times 7 + (1)^2 [7 - 6 = 1] \\ = 84 + 1 = 85$$

Second Column

$$2 \times 8 \times 9 + (1)^2 [9 - 8 = 1] \\ 144 + 1 = 145$$

Third Column

$$2 \times 12 \times 14 + (2)^2 [14 - 12 = 2] \\ = 336 + 4 = 340$$

339. (4) First Column

$$7 + \left(\frac{16}{2}\right) = 7 + 8 = 15$$

Third Column

$$5 + \left(\frac{8}{2}\right) = 5 + 4 = 9$$

Therefore,
Second Column

$$10 + \left(\frac{40}{2}\right) = 10 + 20 = 30$$

340. (4) First Column

$$8 \times 4 \times 9 = 288$$

Second Column

$$7 \times 5 \times 6 = 210$$

Third Column

$$9 \times 6 \times ? = 162$$

$$\Rightarrow ? = \frac{162}{54} = \boxed{3}$$